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(12) **United States Patent**
Ghani

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(45) **Date of Patent: Jan. 20, 2015**

(54) **METHOD AND DEVICE FOR PROVIDING LOCATION BASED CONTENT DELIVERY**

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(73) Assignee: **Safelite Group, Inc.**, Columbus, OH (US)

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PCT Pub. Date: **Jul. 26, 2007**

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G06F 17/30 (2006.01)
G06Q 30/02 (2012.01)
G06Q 20/20 (2012.01)

(52) **U.S. Cl.**
CPC ***G06Q 30/02*** (2013.01); ***G06Q 20/204*** (2013.01)
USPC **707/640**

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CPC G06F 11/1461; G06F 11/1464; G06F 11/1469; G06F 11/0727; G06F 11/0793; G06F 11/1402; G06F 11/1435; G06F 11/1443; G06F 11/1448; G06F 11/1451; G06F 11/1456; G06F 17/30067; G06F 2201/83
USPC 707/10, 640; 709/228
See application file for complete search history.

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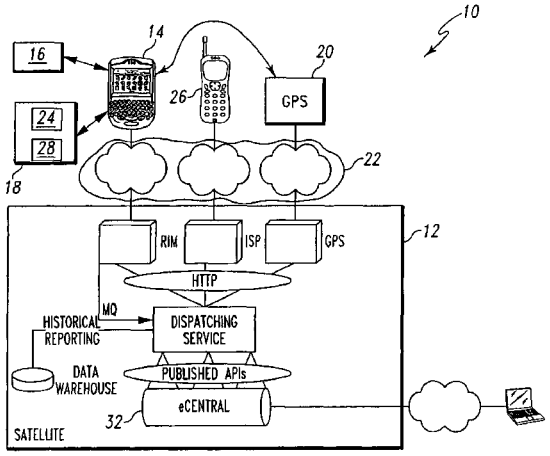
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Primary Examiner — Sheree Brown
(74) *Attorney, Agent, or Firm* — Ice Miller LLP

(57) **ABSTRACT**

A device for providing location based content delivery includes a server located at a first location, a hand-held electronic device configured for communication via a wireless network with the server to receive data relating to the plurality of requests for services and to transmit data regarding the provision of services for updating the data stored in the memory and a processing device communicatively coupled to the hand-held device A method for providing location based content delivery at a job site other than the location of a service provider comprises providing a technician with a hand-held electronic device, clocking the technician in and out remotely utilizing the hand-held electronic device, transferring and viewing work orders to the hand-held electronic device, capturing customer information on the hand-held electronic device at a job site and transferring it to a server accessible at the location of the service provider.

33 Claims, 35 Drawing Sheets



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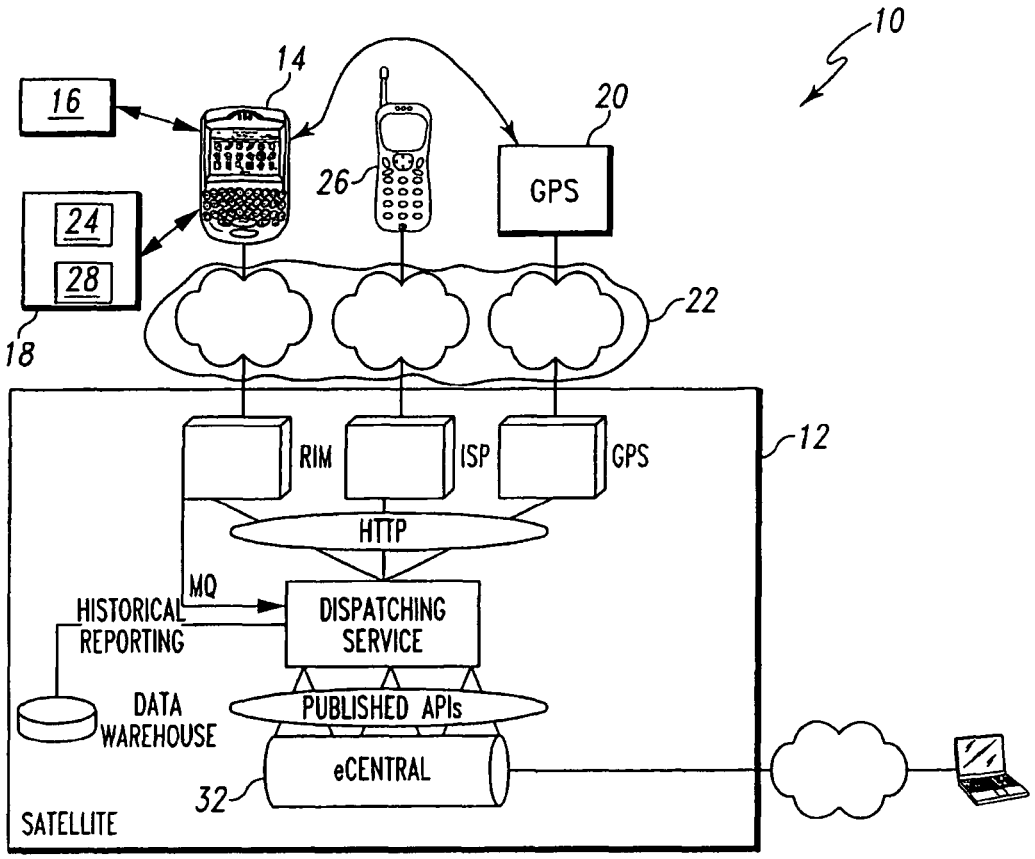

10
14
16
20
26
GPS
24
28
18
22
RIM
ISP
GPS
12
HTTP
MQ
HISTORICAL REPORTING
DISPATCHING SERVICE
DATA WAREHOUSE
PUBLISHED APIs
eCENTRAL
32
SATELLITE

Fig. 1

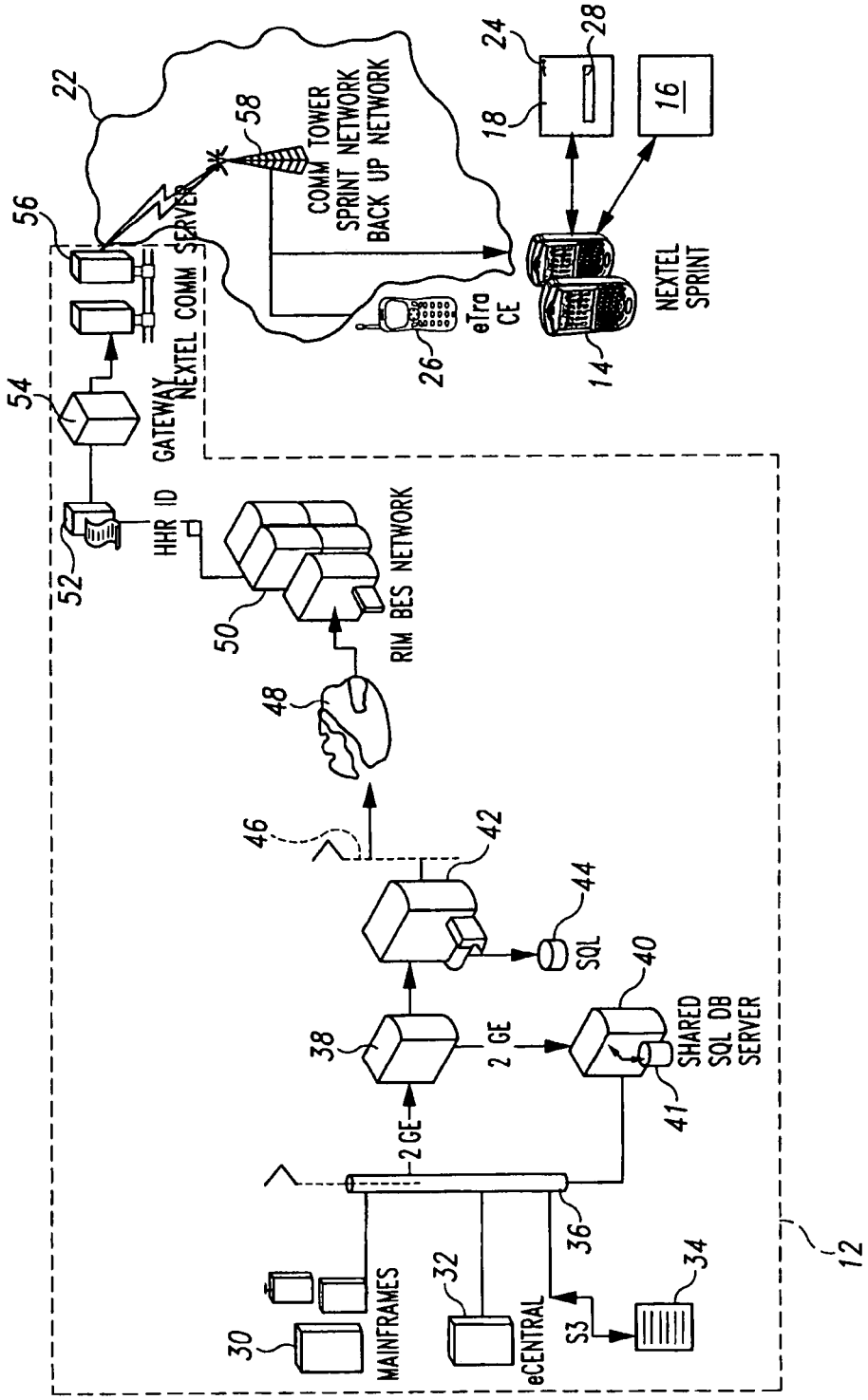

22
58
COMM TOWER
SPRINT NETWORK
BACK UP NETWORK
56
NEXTEL COMM SERVER
24
28
18
16
NEXTEL
SPRINT
26
eTra
CE
14
54
GATEWAY
HHR ID
52
50
RIM BES NETWORK
48
46
42
44
SQL
40
38
2 GE
SHARED
SQL DB
SERVER
41
2 GE
36
30
MAINFRAMES
32
eCENTRAL
S3
34
12

Fig. 2

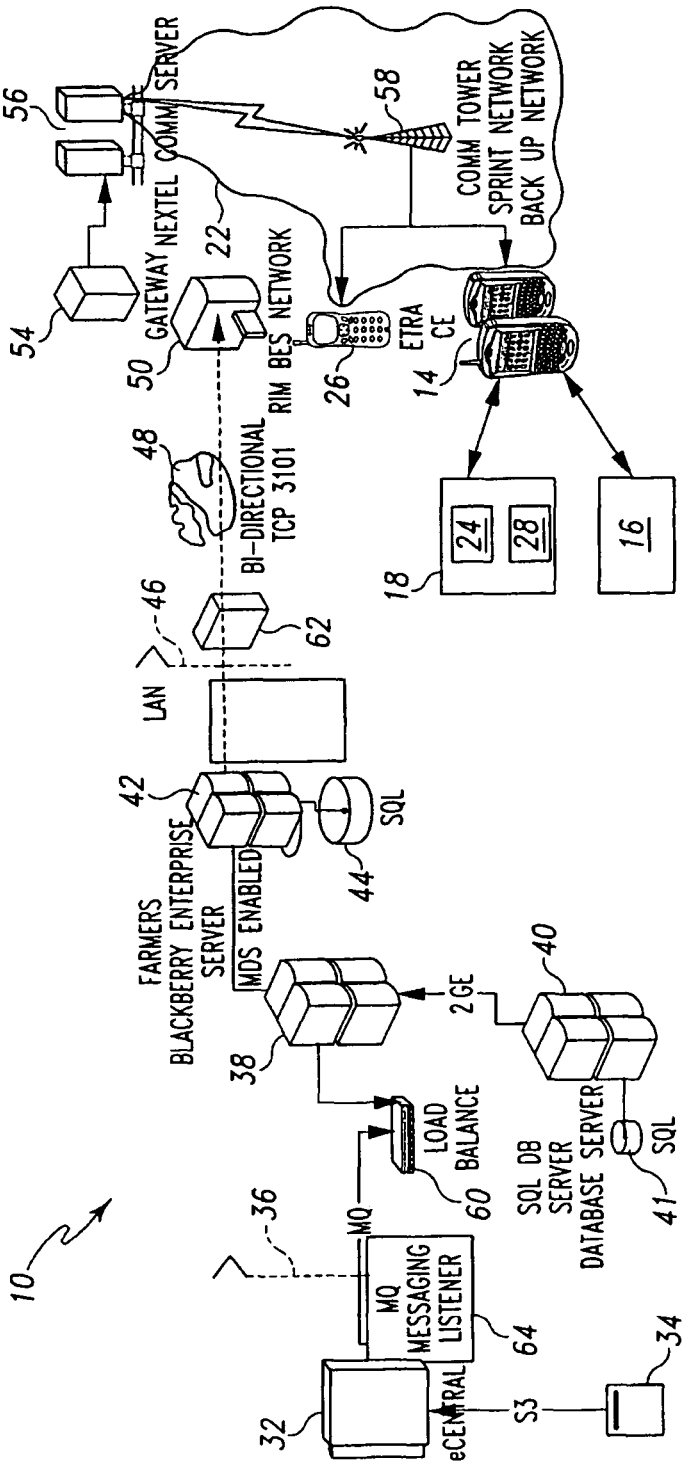

10
56
SERVER
COMM
NEXTEL
58
COMM TOWER
SPRINT NETWORK
BACK UP NETWORK
22
54
GATEWAY
50
RIM BES NETWORK
26
ETRA
CE
14
48
BI-DIRECTIONAL
TCP 3101
18
24
28
16
46
62
LAN
42
SQL
44
FARMERS
BLACKBERRY ENTERPRISE
SERVER
MDS ENABLED
38
2GE
40
LOAD
BALANCE
60
SQL DB
SERVER
DATABASE SERVER
SQL
41
36
MQ
MQ
MESSAGING
LISTENER
64
32
eCENTRAL
S3
34

Fig. 3

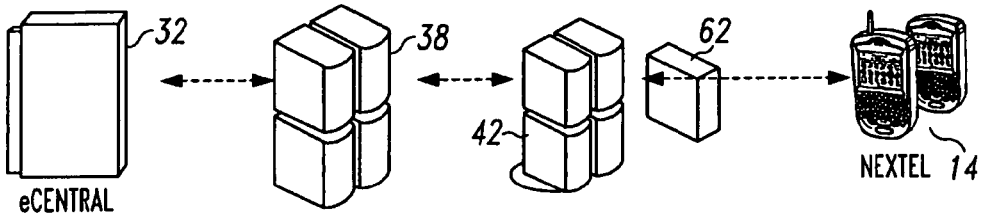


Fig. 4

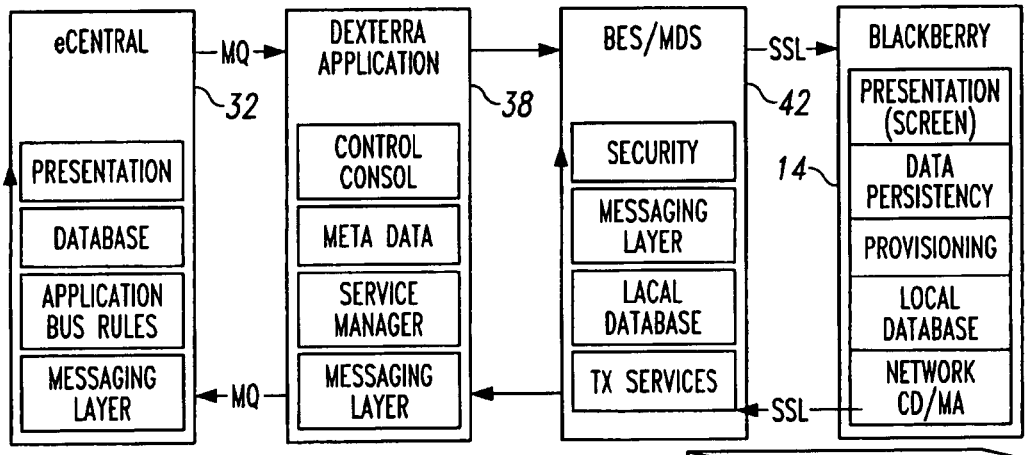


Fig. 5

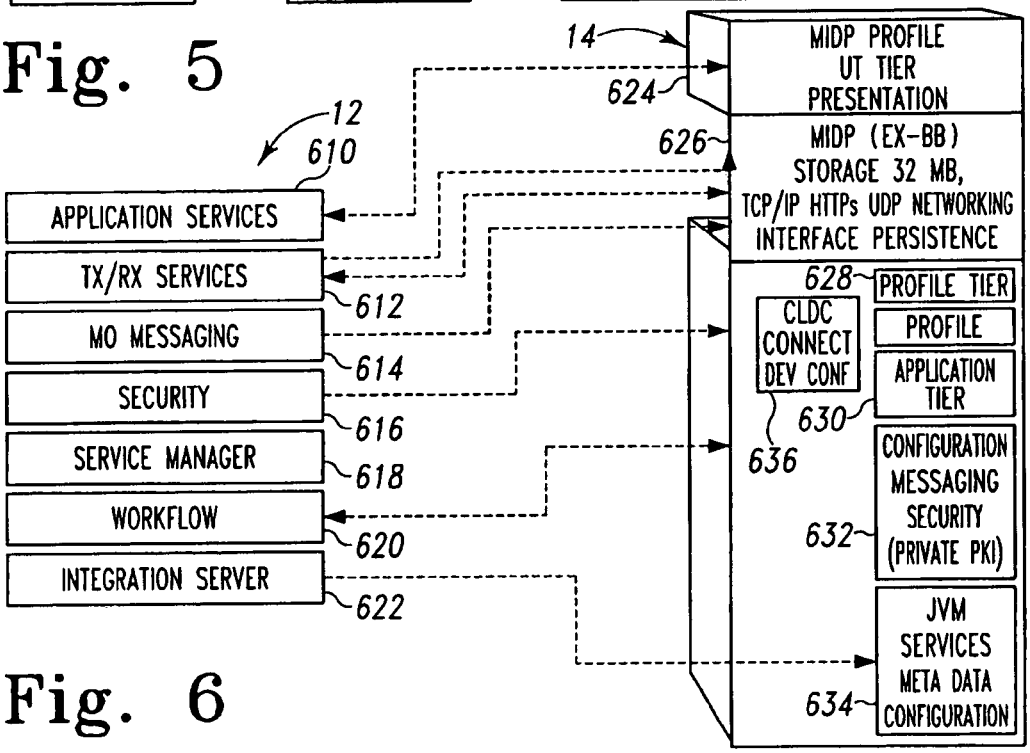


Fig. 6

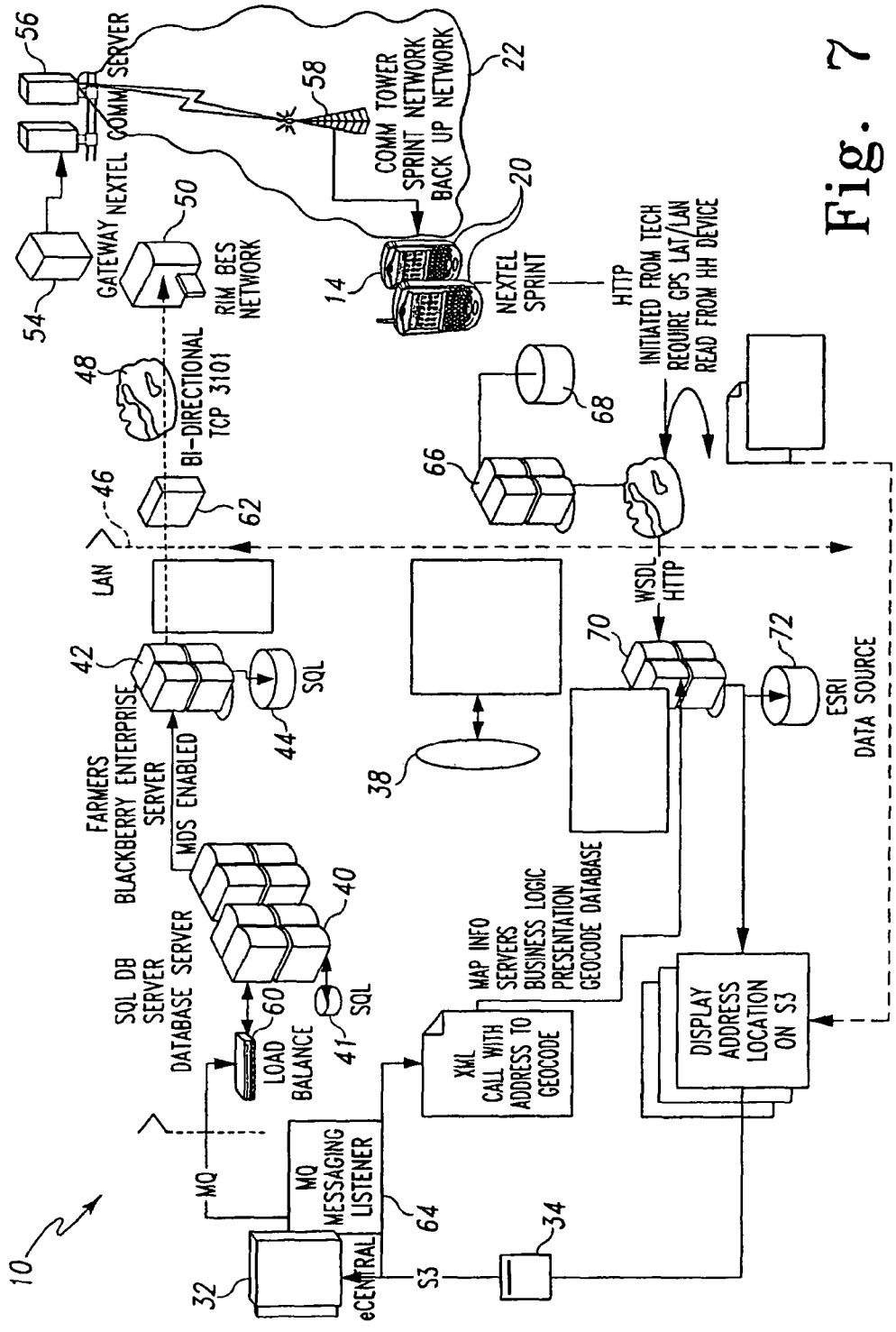

10
32
eCENTRAL
S3
MQ
MQ MESSAGING LISTENER
64
34
LOAD BALANCE
60
SQL DB SERVER
DATABASE SERVER
40
41
SQL
FARMERS BLACKBERRY ENTERPRISE SERVER
MDS ENABLED
42
44
SQL
LAN
46
62
BI-DIRECTIONAL TCP 3101
48
GATEWAY NEXTEL
54
56
SERVER
COMM
58
COMM TOWER SPRINT NETWORK BACK UP NETWORK
22
RIM BES NETWORK
50
14
NEXTEL SPRINT
20
HTTP
INITIATED FROM TECH REQUIRE GPS LAT/LAN READ FROM HH DEVICE
66
68
WSDL HTTP
70
72
ESRI DATA SOURCE
38
MAP INFO SERVERS BUSINESS LOGIC PRESENTATION GEOCODE DATABASE
XML CALL WITH ADDRESS TO GEOCODE
DISPLAY ADDRESS LOCATION ON S3

Fig. 7

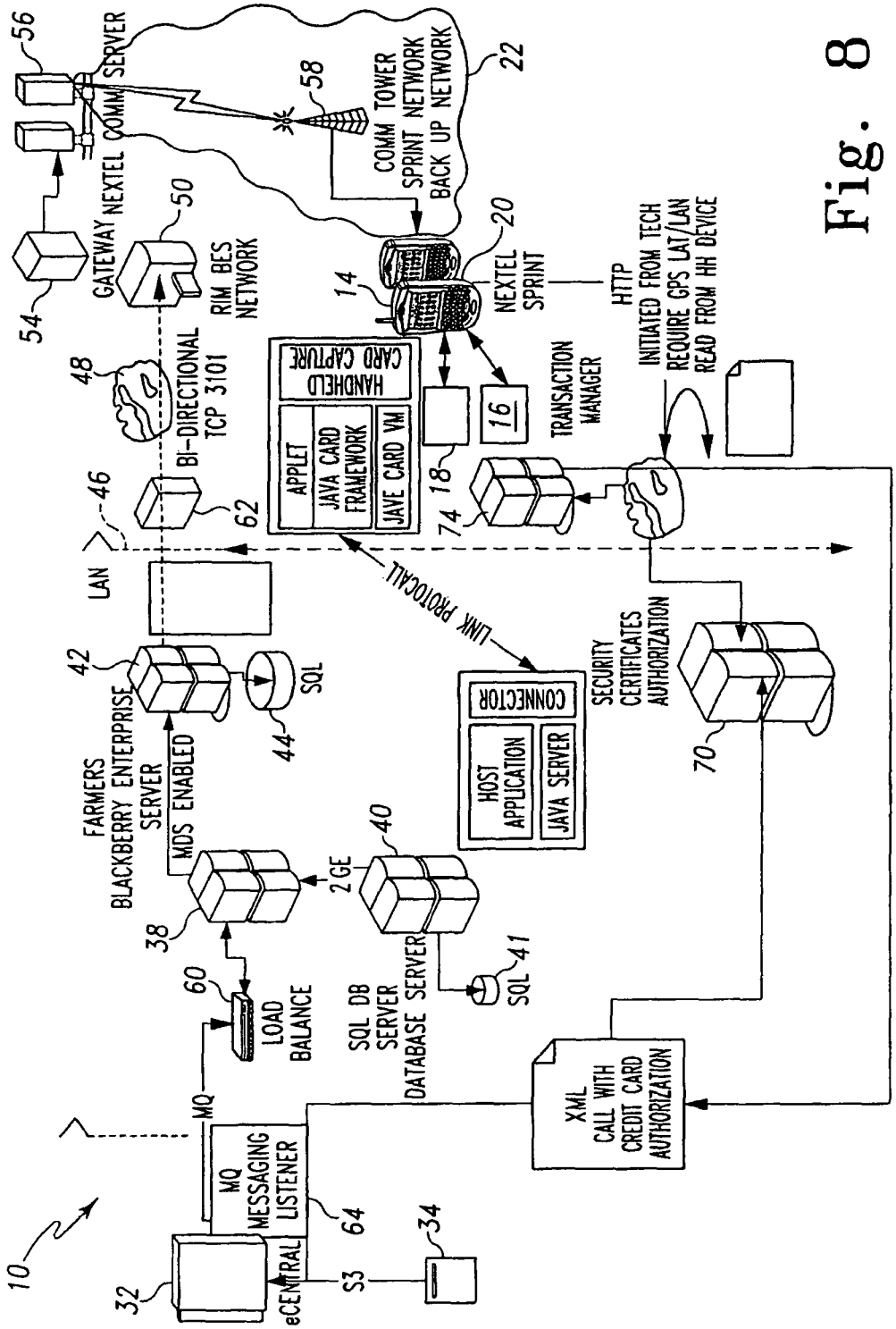

10
32
eCENTRAL
S3
34
MQ MESSAGING LISTENER
64
MQ
LOAD BALANCE
60
SQL DB SERVER
DATABASE SERVER
SQL 41
2GE
40
38
FARMERS BLACKBERRY ENTERPRISE SERVER
MDS ENABLED
42
SQL
44
LAN
46
BI-DIRECTIONAL TCP 3101
48
62
APPLET
JAVA CARD FRAMEWORK
JAVE CARD VM
HANDHELD CARD CAPTURE
16
18
74
TRANSACTION MANAGER
LINK PROTOCALL
CONNECTOR
HOST APPLICATION
JAVA SERVER
SECURITY CERTIFICATES AUTHORIZATION
70
XML CALL WITH CREDIT CARD AUTHORIZATION
HTTP INITIATED FROM TECH REQUIRE GPS LAT/LAN READ FROM HH DEVICE
NEXTEL SPRINT
14
20
RIM BES NETWORK
50
GATEWAY NEXTEL COMM SERVER
54
56
COMM TOWER SPRINT NETWORK BACK UP NETWORK
58
22

Fig. 8

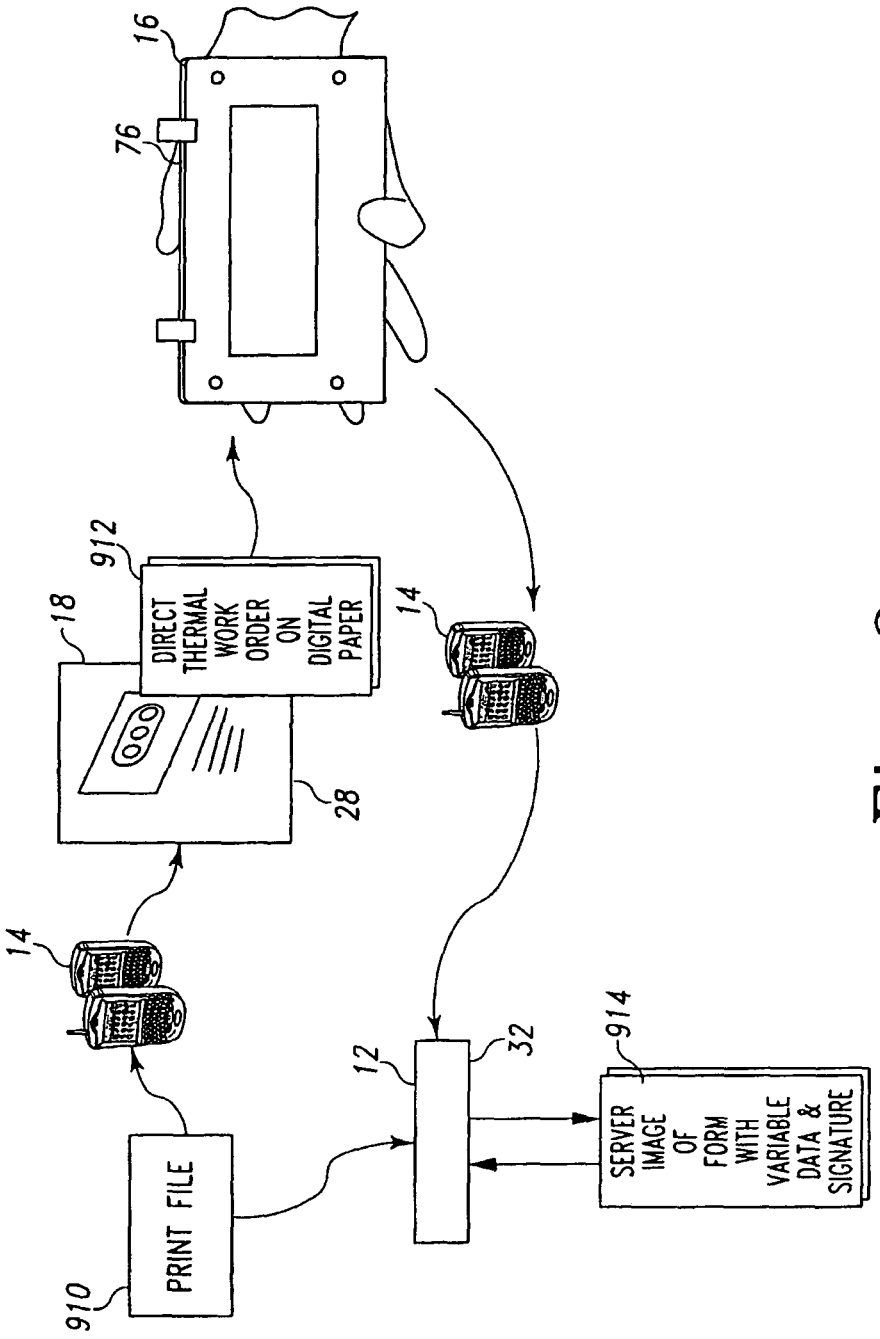

16
76
912
DIRECT THERMAL WORK ORDER ON DIGITAL PAPER
18
14
28
14
12
32
914
SERVER IMAGE OF FORM WITH VARIABLE DATA & SIGNATURE
PRINT FILE
910

Fig. 9

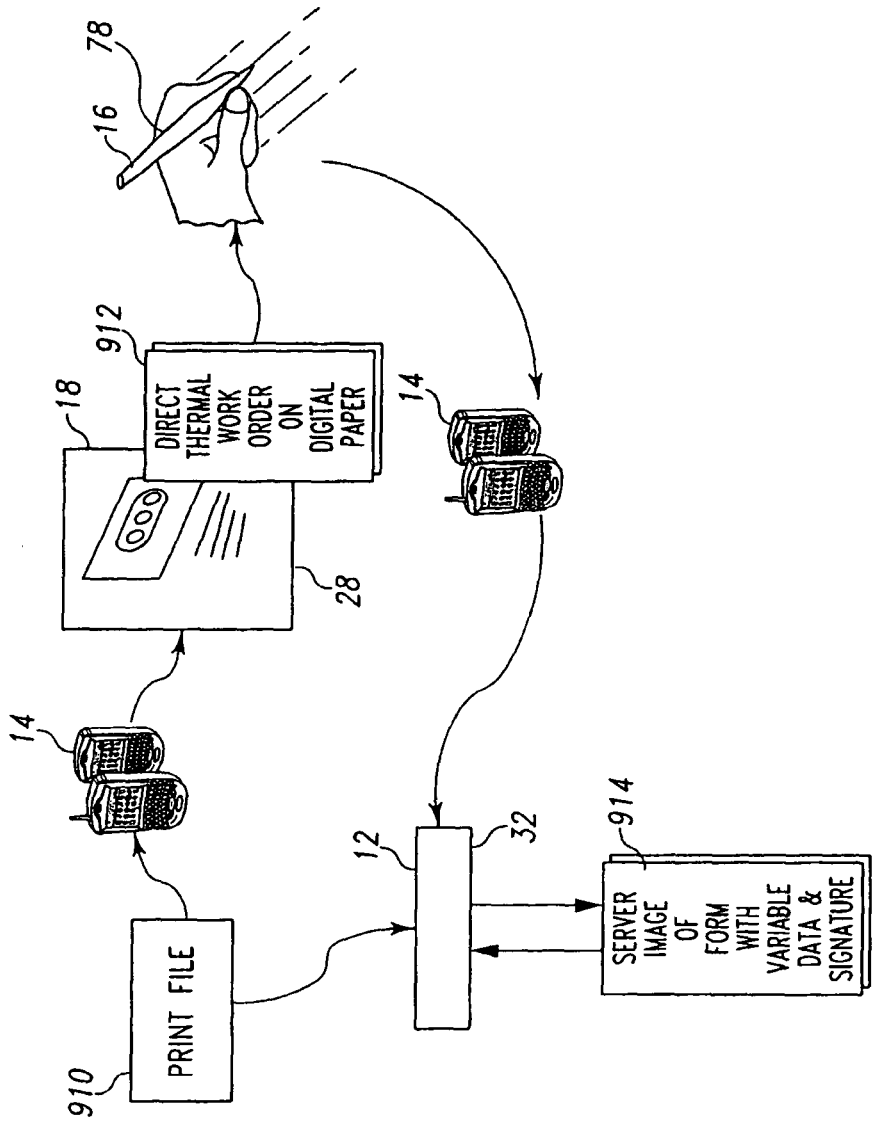

78
16
912
DIRECT THERMAL WORK ORDER ON DIGITAL PAPER
18
14
28
14
12
32
914
SERVER IMAGE OF FORM WITH VARIABLE DATA & SIGNATURE
PRINT FILE
910

Fig. 10

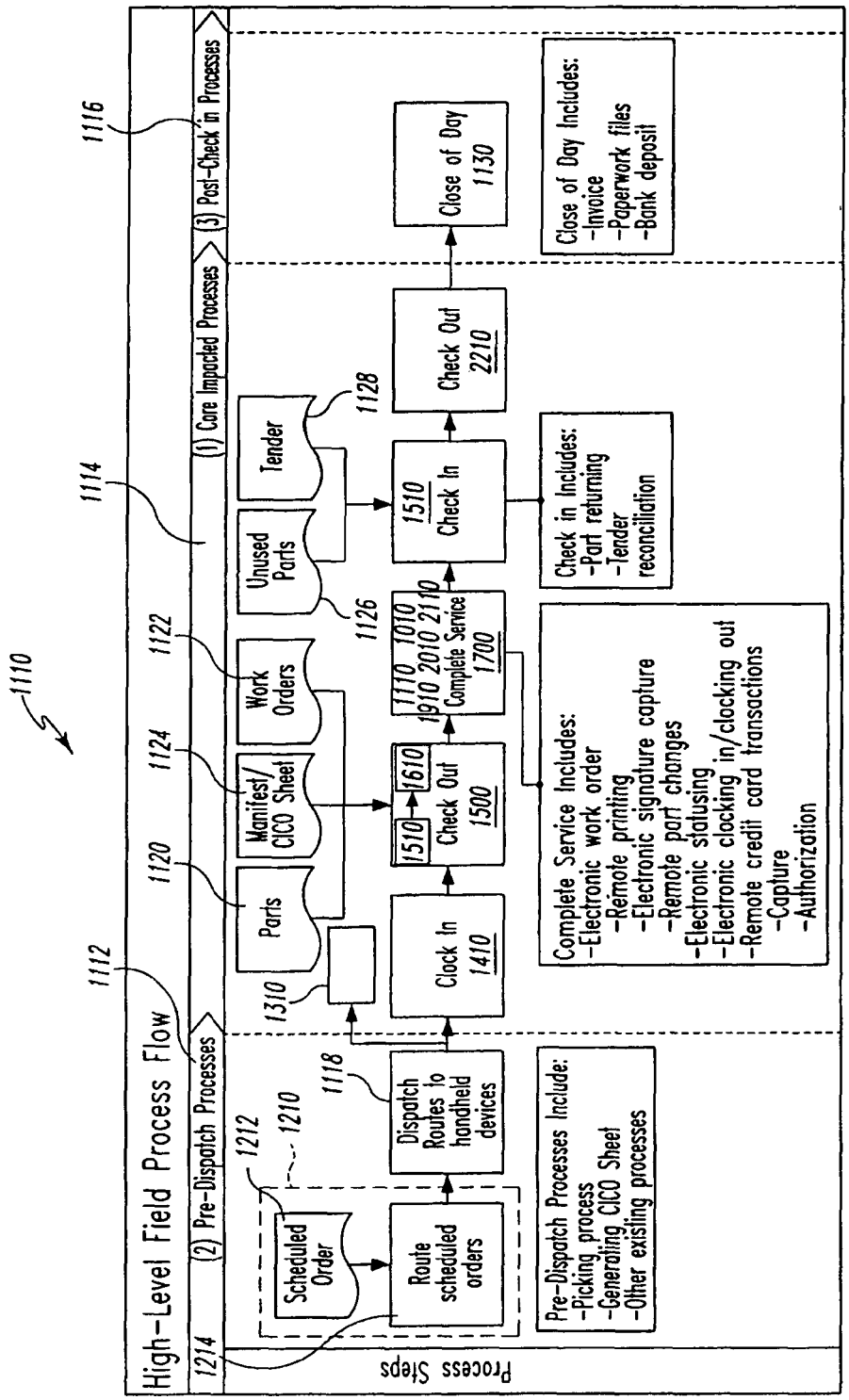

1110
1112
1114
1116
1118
1120
1122
1124
1126
1128
1210
1212
1214
1310
1410
1500
1510
1610
1700
1910
2010
2110
2210
High-Level Field Process Flow
(1) Core Impacted Processes
(2) Pre-Dispatch Processes
(3) Post-Check in Processes
Process Steps
Scheduled Order
Route scheduled orders
Dispatch Routes to handheld devices
Parts
Manifest/ CICO Sheet
Work Orders
Unused Parts
Tender
Clock In 1410
Check Out 1500
Complete Service 1700
Check In 1510
Check Out 2210
Close of Day 1130
Pre-Dispatch Processes Include:
-Picking process
-Generating CICO Sheet
-Other existing processes
Complete Service Includes:
-Electronic work order
-Remote printing
-Electronic signature capture
-Remote part changes
-Electronic statusing
-Electronic clocking in/clocking out
-Remote credit card transactions
-Capture
-Authorization
Check in Includes:
-Part returning
-Tender reconciliation
Close of Day Includes:
-Invoice
-Paperwork files
-Bank deposit

Fig. 11

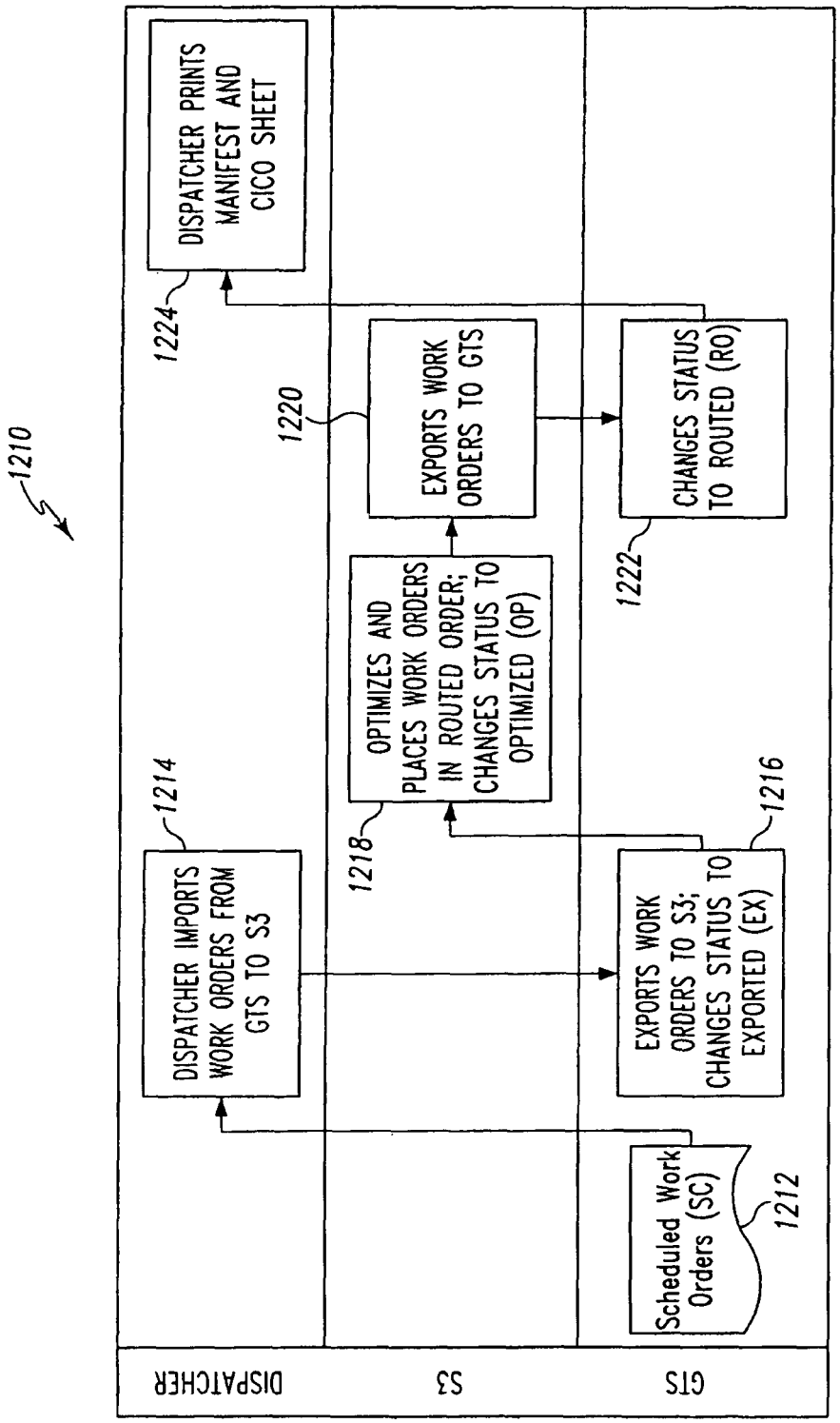

1210
DISPATCHER
S3
GTS
Scheduled Work Orders (SC)
1212
DISPATCHER IMPORTS WORK ORDERS FROM GTS TO S3
1214
OPTIMIZES AND PLACES WORK ORDERS IN ROUTED ORDER; CHANGES STATUS TO OPTIMIZED (OP)
1218
EXPORTS WORK ORDERS TO S3; CHANGES STATUS TO EXPORTED (EX)
1216
EXPORTS WORK ORDERS TO GTS
1220
CHANGES STATUS TO ROUTED (RO)
1222
DISPATCHER PRINTS MANIFEST AND CICO SHEET
1224

Fig. 12

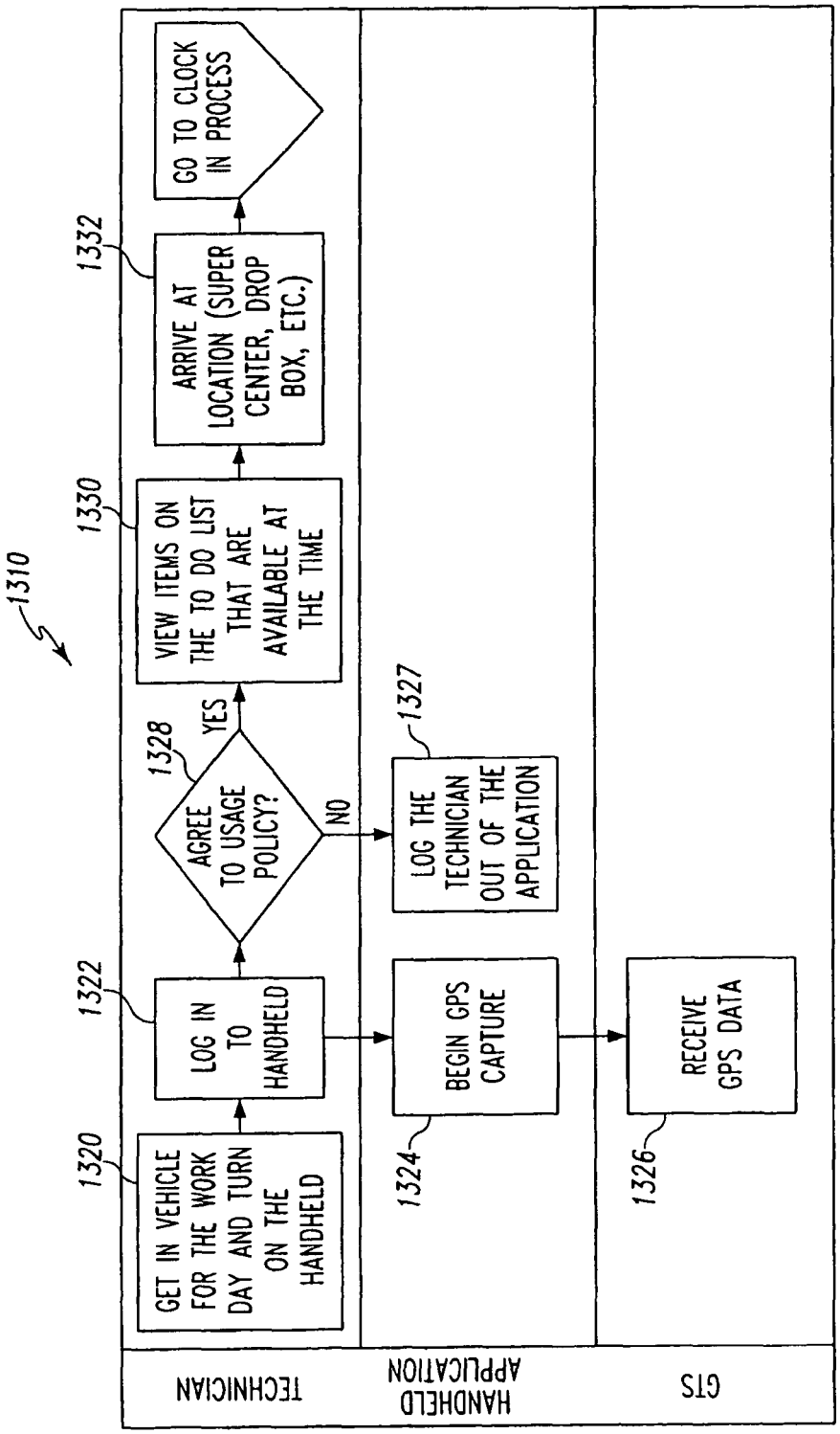

1310
TECHNICIAN
HANDHELD APPLICATION
GIS
1320
GET IN VEHICLE FOR THE WORK DAY AND TURN ON THE HANDHELD
1322
LOG IN TO HANDHELD
1324
BEGIN GPS CAPTURE
1326
RECEIVE GPS DATA
1327
LOG THE TECHNICIAN OUT OF THE APPLICATION
1328
AGREE TO USAGE POLICY?
YES
NO
1330
VIEW ITEMS ON THE TO DO LIST THAT ARE AVAILABLE AT THE TIME
1332
ARRIVE AT LOCATION (SUPER CENTER, DROP BOX, ETC.)
GO TO CLOCK IN PROCESS

Fig. 13

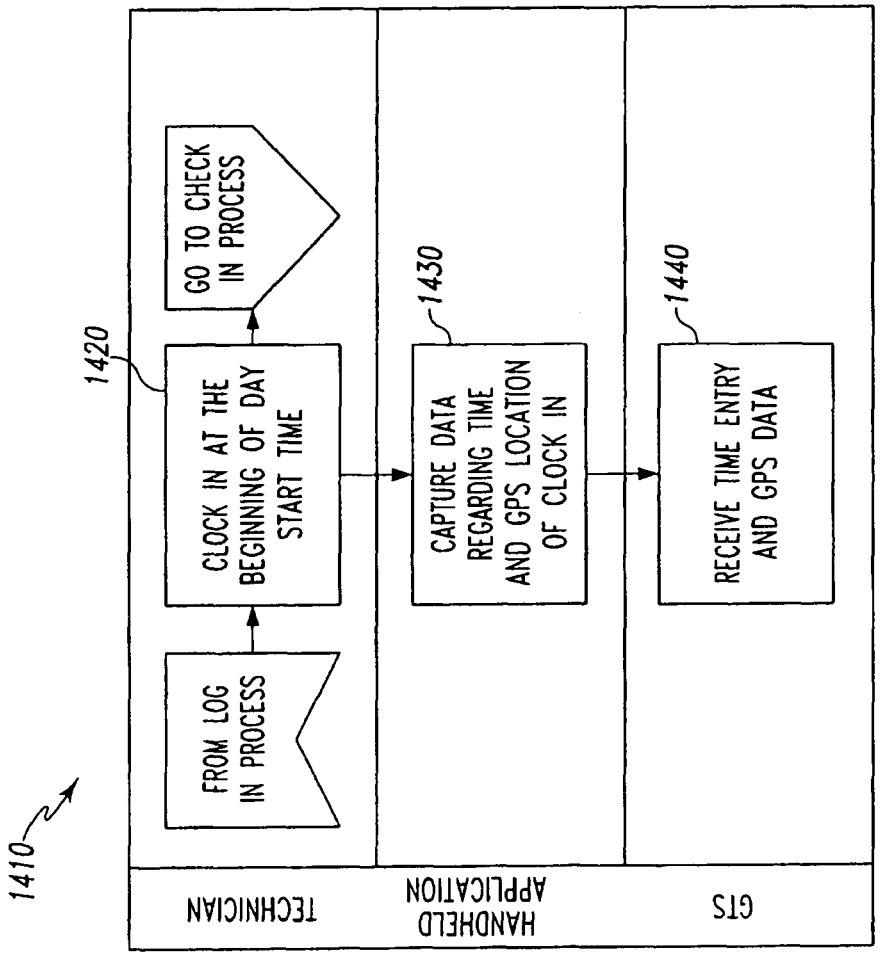

1410
TECHNICIAN
HANDHELD APPLICATION
GIS
FROM LOG IN PROCESS
CLOCK IN AT THE BEGINNING OF DAY START TIME
1420
GO TO CHECK IN PROCESS
CAPTURE DATA REGARDING TIME AND GPS LOCATION OF CLOCK IN
1430
RECEIVE TIME ENTRY AND GPS DATA
1440

Fig. 14

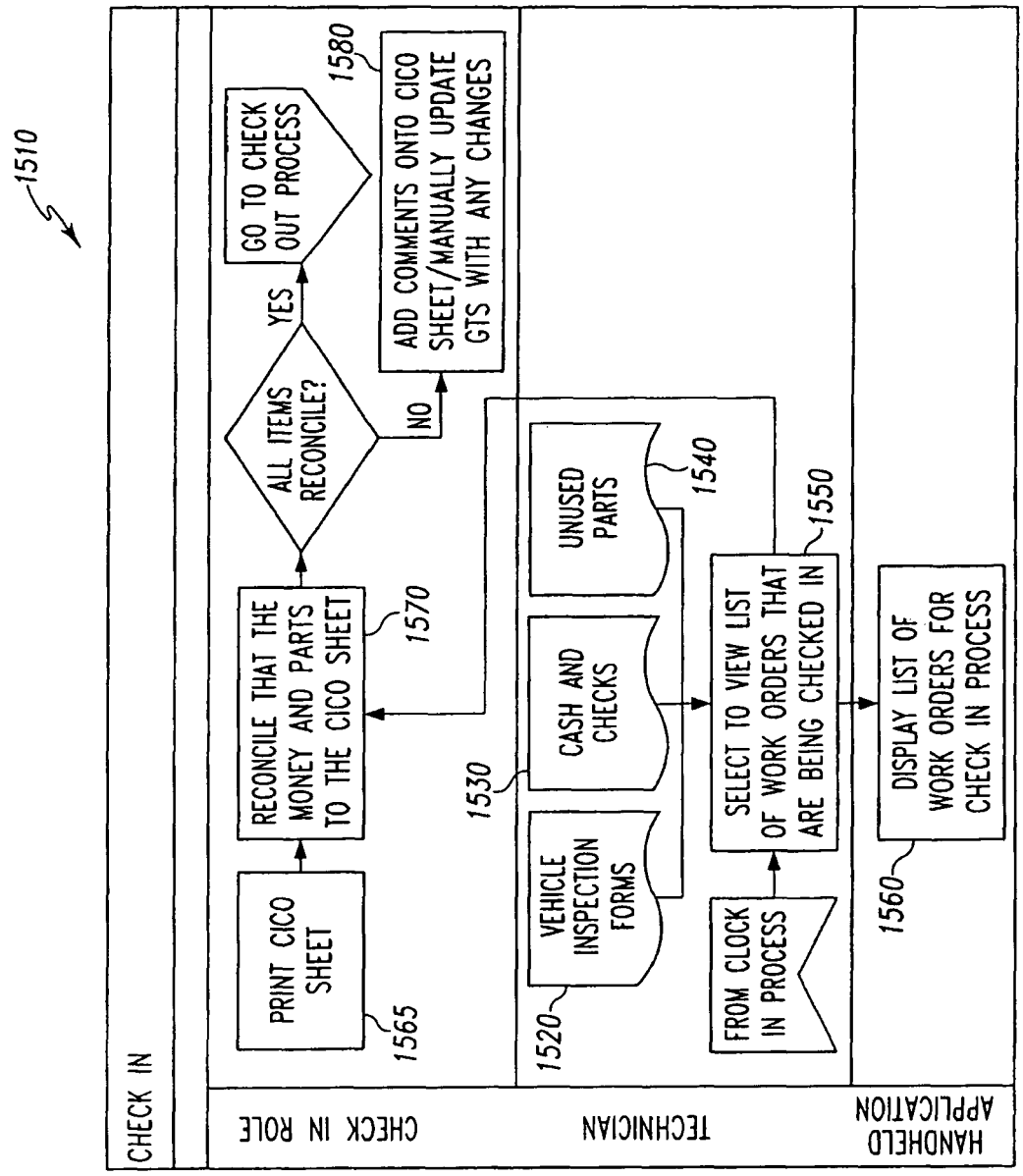

1510
CHECK IN
CHECK IN ROLE
TECHNICIAN
HANDHELD APPLICATION
PRINT CICO SHEET
1565
RECONCILE THAT THE MONEY AND PARTS TO THE CICO SHEET
1570
ALL ITEMS RECONCILE?
YES
NO
GO TO CHECK OUT PROCESS
1580
ADD COMMENTS ONTO CICO SHEET/MANUALLY UPDATE GTS WITH ANY CHANGES
1530
VEHICLE INSPECTION FORMS
1520
CASH AND CHECKS
UNUSED PARTS
1540
FROM CLOCK IN PROCESS
SELECT TO VIEW LIST OF WORK ORDERS THAT ARE BEING CHECKED IN
1550
DISPLAY LIST OF WORK ORDERS FOR CHECK IN PROCESS
1560

Fig. 15

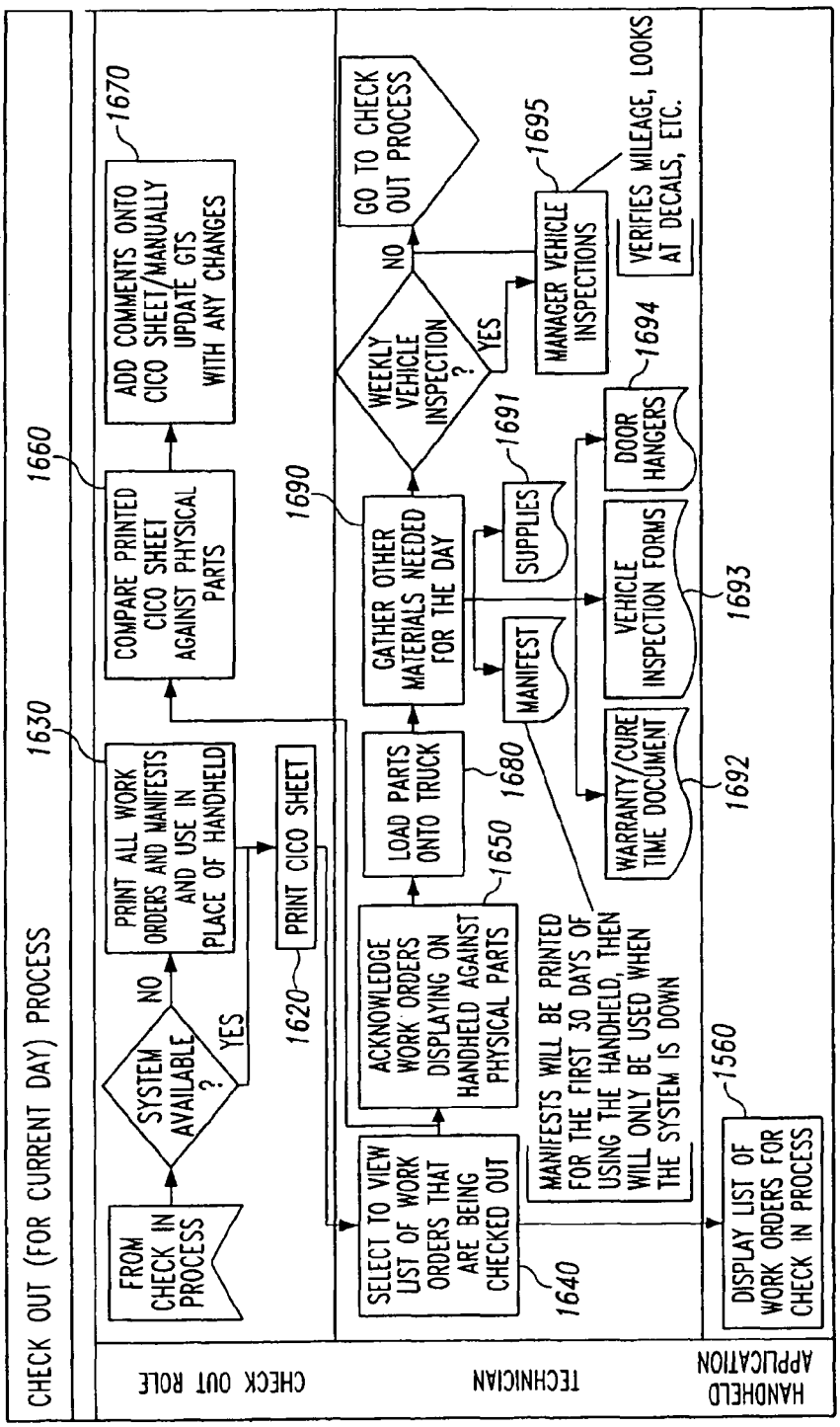

1610
CHECK OUT (FOR CURRENT DAY) PROCESS
CHECK OUT ROLE
TECHNICIAN
HANDHELD APPLICATION
FROM CHECK IN PROCESS
SYSTEM AVAILABLE ?
NO
YES
PRINT ALL WORK ORDERS AND MANIFESTS AND USE IN PLACE OF HANDHELD
PRINT CICO SHEET
1620
1630
COMPARE PRINTED CICO SHEET AGAINST PHYSICAL PARTS
ADD COMMENTS ONTO CICO SHEET/MANUALLY UPDATE GTS WITH ANY CHANGES
1660
1670
SELECT TO VIEW LIST OF WORK ORDERS THAT ARE BEING CHECKED OUT
1640
ACKNOWLEDGE WORK ORDERS DISPLAYING ON HANDHELD AGAINST PHYSICAL PARTS
1650
LOAD PARTS ONTO TRUCK
1680
GATHER OTHER MATERIALS NEEDED FOR THE DAY
1690
MANIFESTS WILL BE PRINTED FOR THE FIRST 30 DAYS OF USING THE HANDHELD, THEN WILL ONLY BE USED WHEN THE SYSTEM IS DOWN
MANIFEST
SUPPLIES
1691
WEEKLY VEHICLE INSPECTION ?
NO
YES
GO TO CHECK OUT PROCESS
MANAGER VEHICLE INSPECTIONS
1694
1695
VERIFIES MILEAGE, LOOKS AT DECALS, ETC.
WARRANTY/CURE TIME DOCUMENT
VEHICLE INSPECTION FORMS
DOOR HANGERS
1692
1693
DISPLAY LIST OF WORK ORDERS FOR CHECK IN PROCESS
1560

Fig. 16

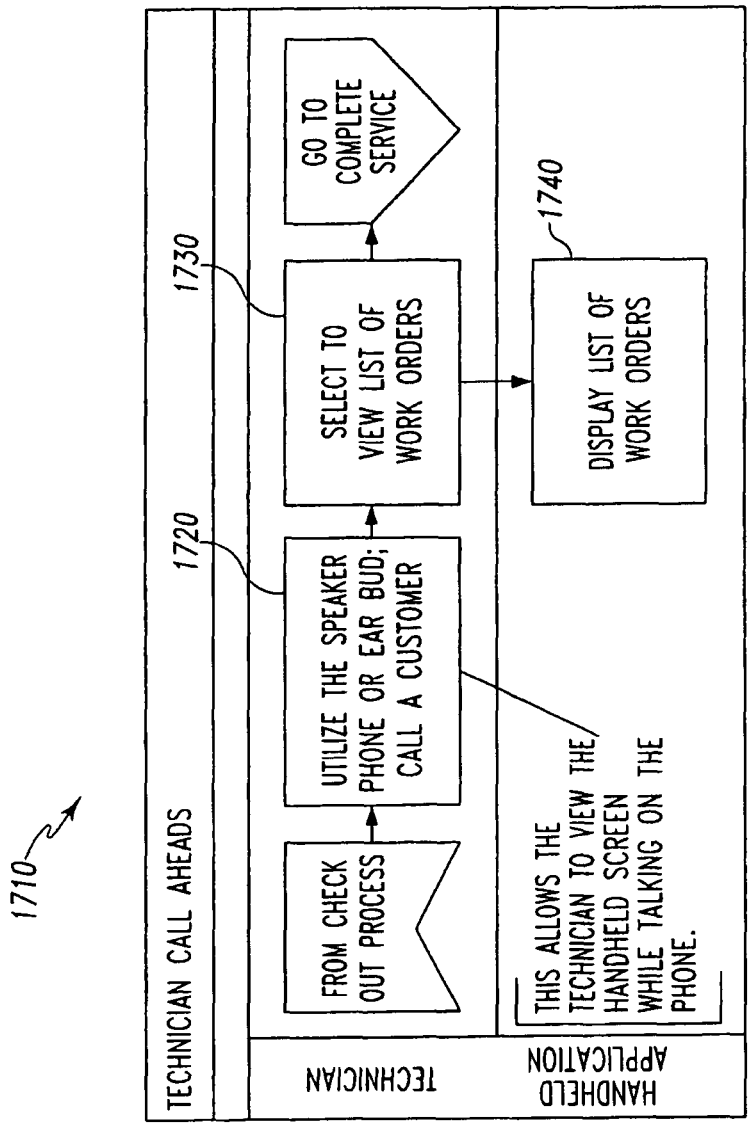

1710
TECHNICIAN CALL AHEADS
TECHNICIAN
HANDHELD APPLICATION
FROM CHECK OUT PROCESS
1720
UTILIZE THE SPEAKER PHONE OR EAR BUD; CALL A CUSTOMER
1730
SELECT TO VIEW LIST OF WORK ORDERS
GO TO COMPLETE SERVICE
1740
DISPLAY LIST OF WORK ORDERS
THIS ALLOWS THE TECHNICIAN TO VIEW THE HANDHELD SCREEN WHILE TALKING ON THE PHONE.

Fig. 17

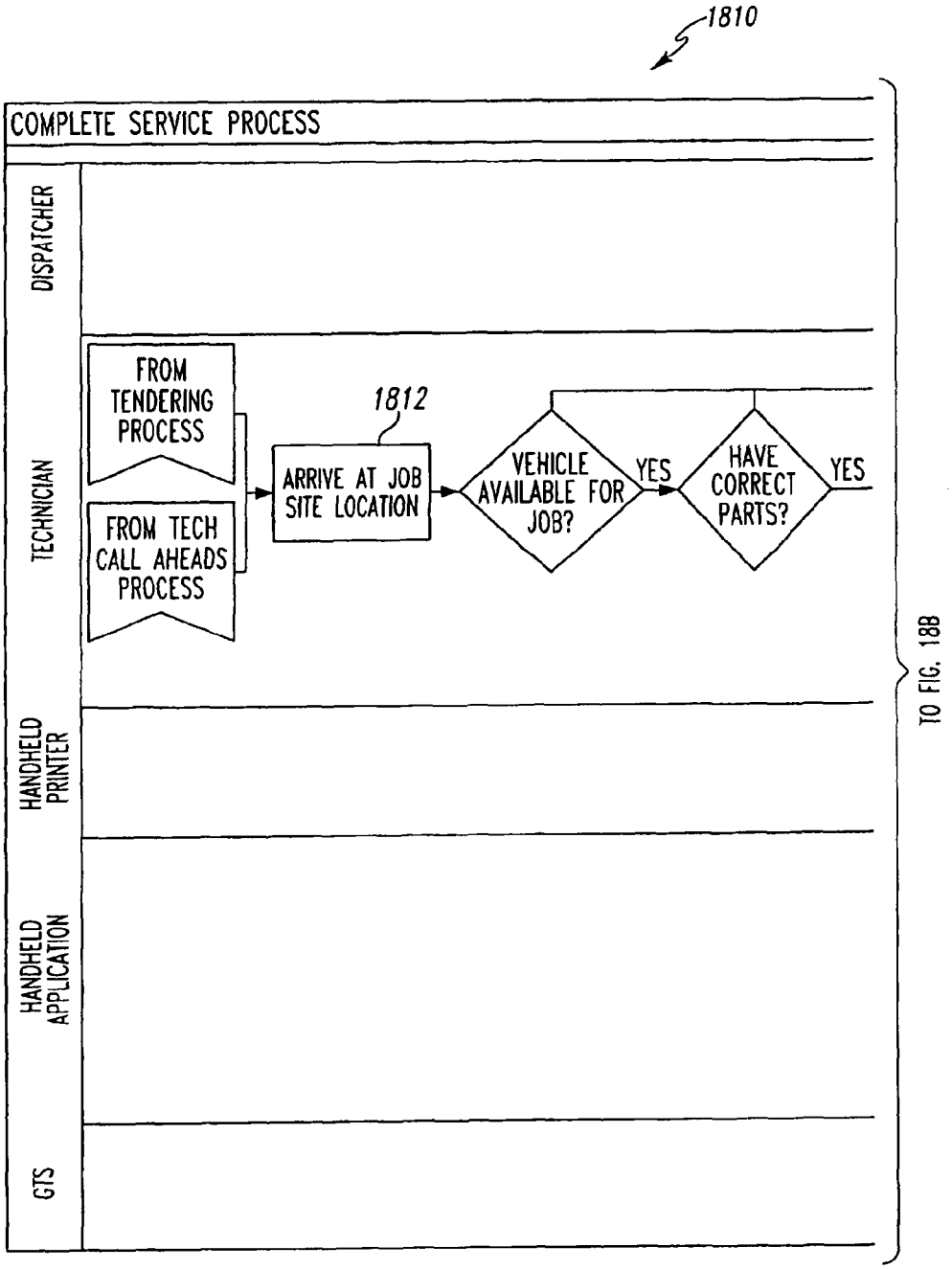

1810
COMPLETE SERVICE PROCESS
DISPATCHER
TECHNICIAN
HANDHELD PRINTER
HANDHELD APPLICATION
GTS
FROM TENDERING PROCESS
FROM TECH CALL AHEADS PROCESS
1812
ARRIVE AT JOB SITE LOCATION
VEHICLE AVAILABLE FOR JOB?
YES
HAVE CORRECT PARTS?
YES
TO FIG. 18B

Fig. 18A

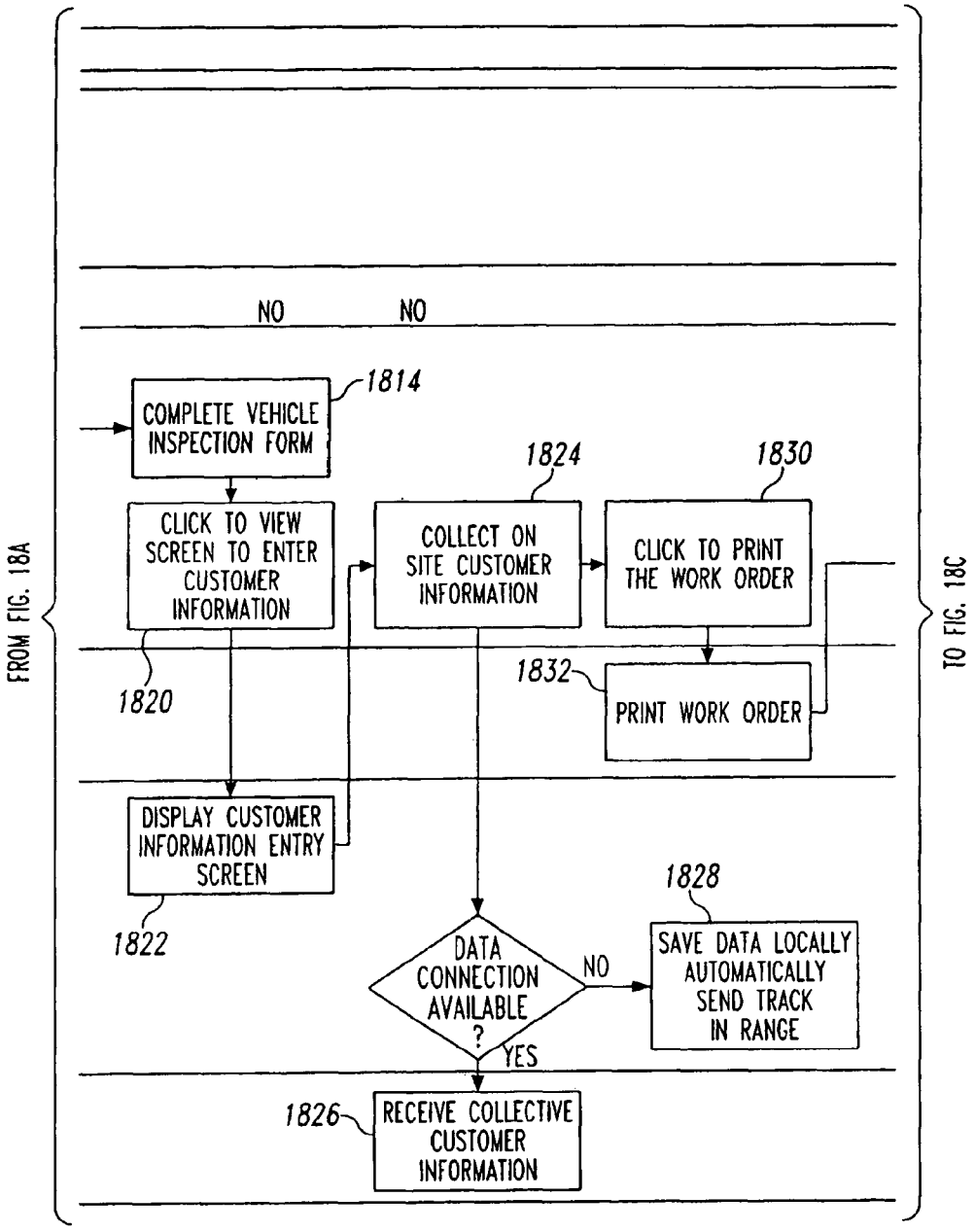

FROM FIG. 18A
TO FIG. 18C
NO
NO
COMPLETE VEHICLE INSPECTION FORM
1814
CLICK TO VIEW SCREEN TO ENTER CUSTOMER INFORMATION
1820
COLLECT ON SITE CUSTOMER INFORMATION
1824
CLICK TO PRINT THE WORK ORDER
1830
PRINT WORK ORDER
1832
DISPLAY CUSTOMER INFORMATION ENTRY SCREEN
1822
DATA CONNECTION AVAILABLE ?
NO
YES
SAVE DATA LOCALLY AUTOMATICALLY SEND TRACK IN RANGE
1828
RECEIVE COLLECTIVE CUSTOMER INFORMATION
1826

Fig. 18B

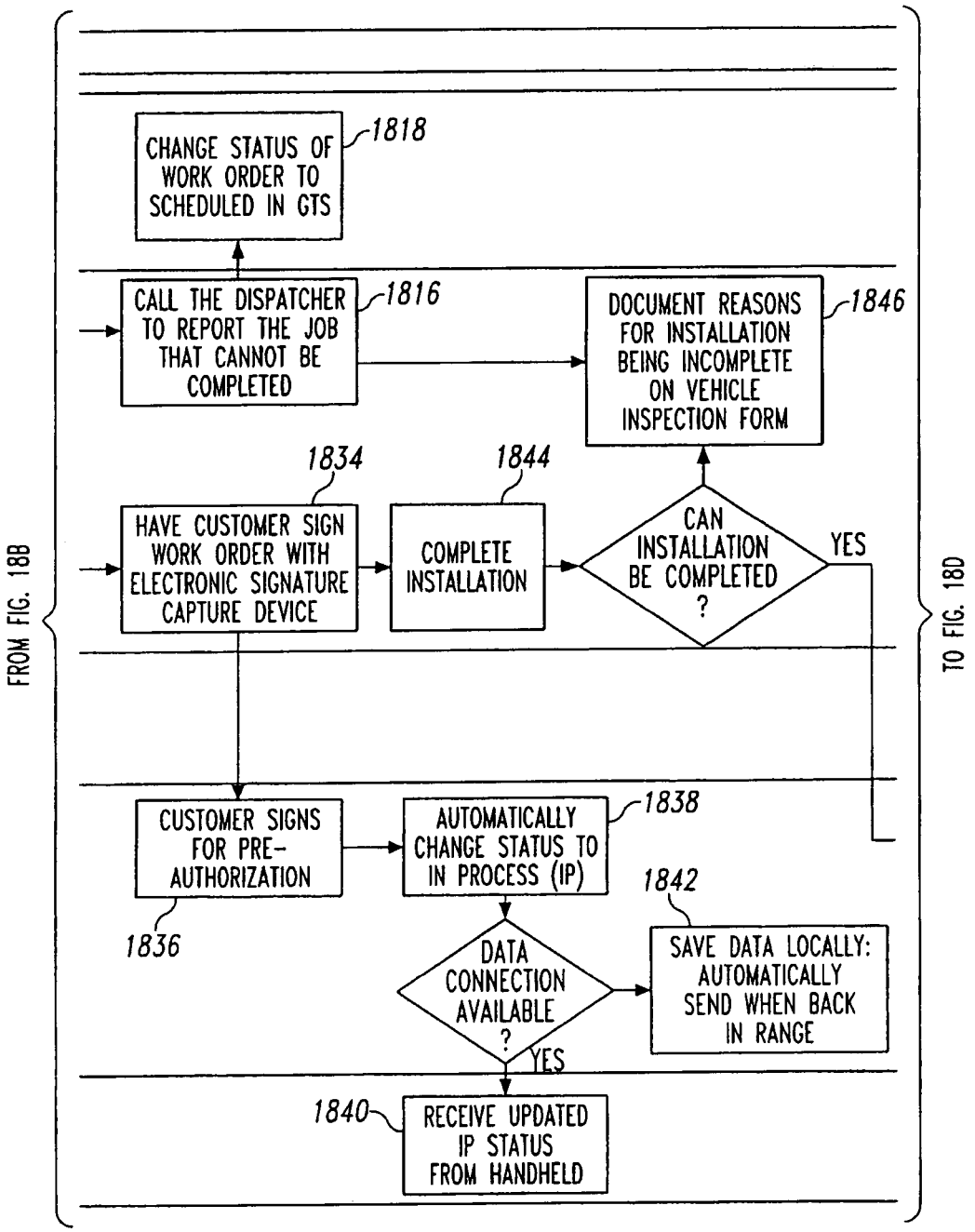

FROM FIG. 18B
CHANGE STATUS OF WORK ORDER TO SCHEDULED IN GTS
1818
CALL THE DISPATCHER TO REPORT THE JOB THAT CANNOT BE COMPLETED
1816
DOCUMENT REASONS FOR INSTALLATION BEING INCOMPLETE ON VEHICLE INSPECTION FORM
1846
1834
1844
HAVE CUSTOMER SIGN WORK ORDER WITH ELECTRONIC SIGNATURE CAPTURE DEVICE
COMPLETE INSTALLATION
CAN INSTALLATION BE COMPLETED ?
YES
CUSTOMER SIGNS FOR PRE-AUTHORIZATION
1836
AUTOMATICALLY CHANGE STATUS TO IN PROCESS (IP)
1838
1842
DATA CONNECTION AVAILABLE ?
SAVE DATA LOCALLY: AUTOMATICALLY SEND WHEN BACK IN RANGE
YES
1840
RECEIVE UPDATED IP STATUS FROM HANDHELD
TO FIG. 18D

Fig. 18C

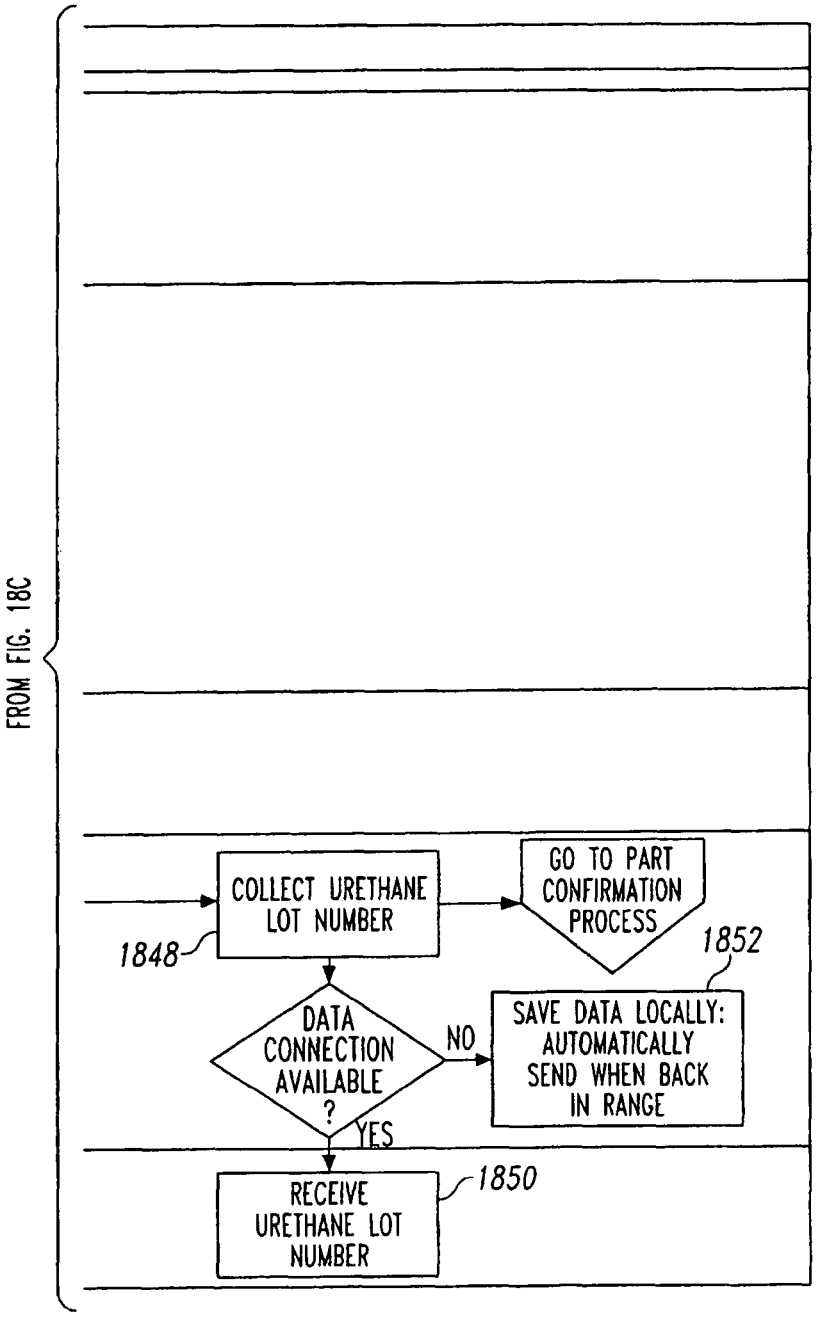

FROM FIG. 18C
COLLECT URETHANE LOT NUMBER
1848
GO TO PART CONFIRMATION PROCESS
1852
DATA CONNECTION AVAILABLE ?
NO
SAVE DATA LOCALLY: AUTOMATICALLY SEND WHEN BACK IN RANGE
YES
RECEIVE URETHANE LOT NUMBER
1850

Fig. 18D

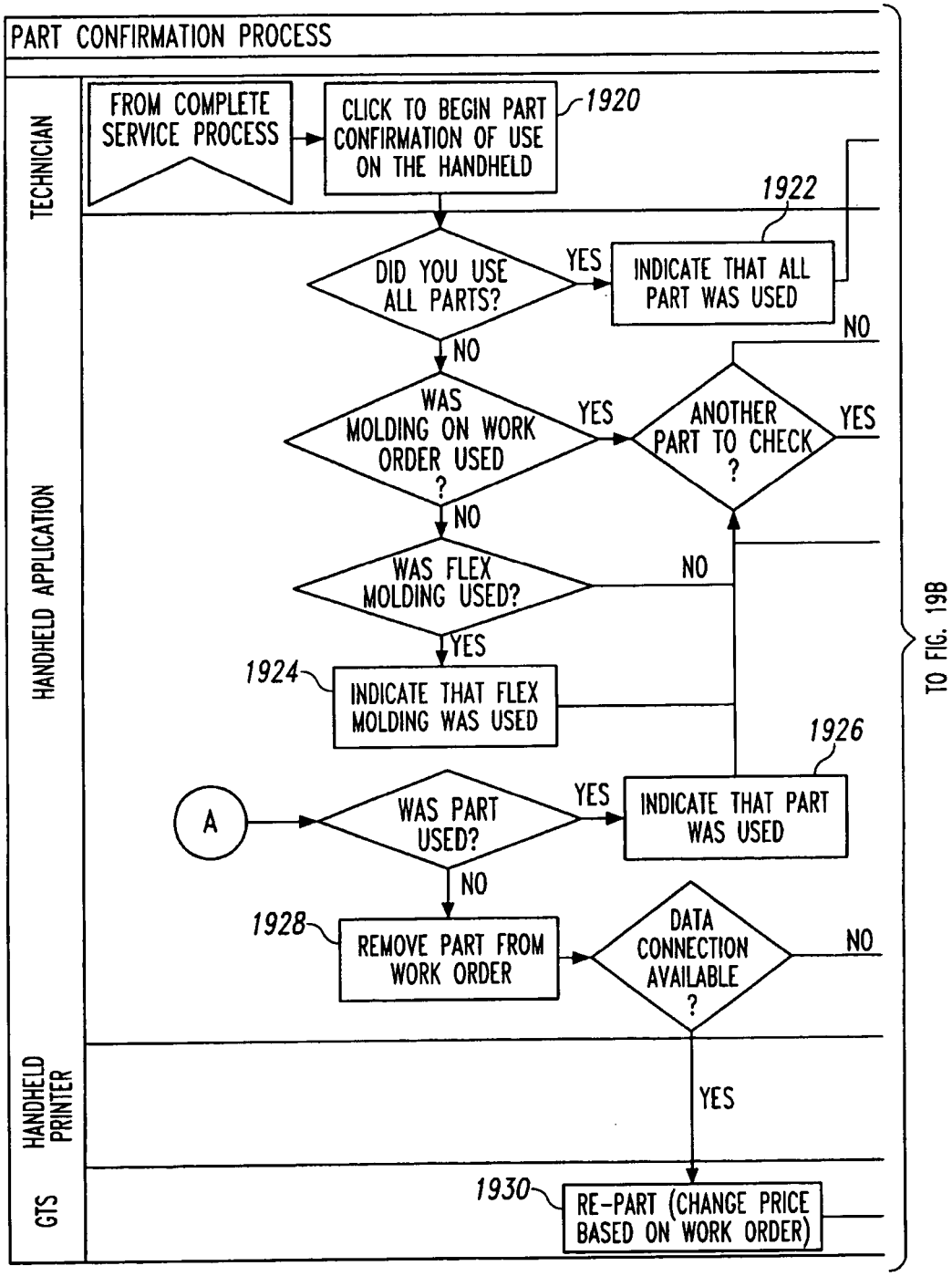

PART CONFIRMATION PROCESS
TECHNICIAN
HANDHELD APPLICATION
HANDHELD PRINTER
GTS
FROM COMPLETE SERVICE PROCESS
CLICK TO BEGIN PART CONFIRMATION OF USE ON THE HANDHELD
1920
1922
DID YOU USE ALL PARTS?
YES
INDICATE THAT ALL PART WAS USED
NO
WAS MOLDING ON WORK ORDER USED ?
YES
ANOTHER PART TO CHECK ?
YES
NO
NO
WAS FLEX MOLDING USED?
NO
YES
1924
INDICATE THAT FLEX MOLDING WAS USED
1926
A
WAS PART USED?
YES
INDICATE THAT PART WAS USED
NO
1928
REMOVE PART FROM WORK ORDER
DATA CONNECTION AVAILABLE ?
NO
YES
1930
RE-PART (CHANGE PRICE BASED ON WORK ORDER)
TO FIG. 19B
1910

Fig. 19A

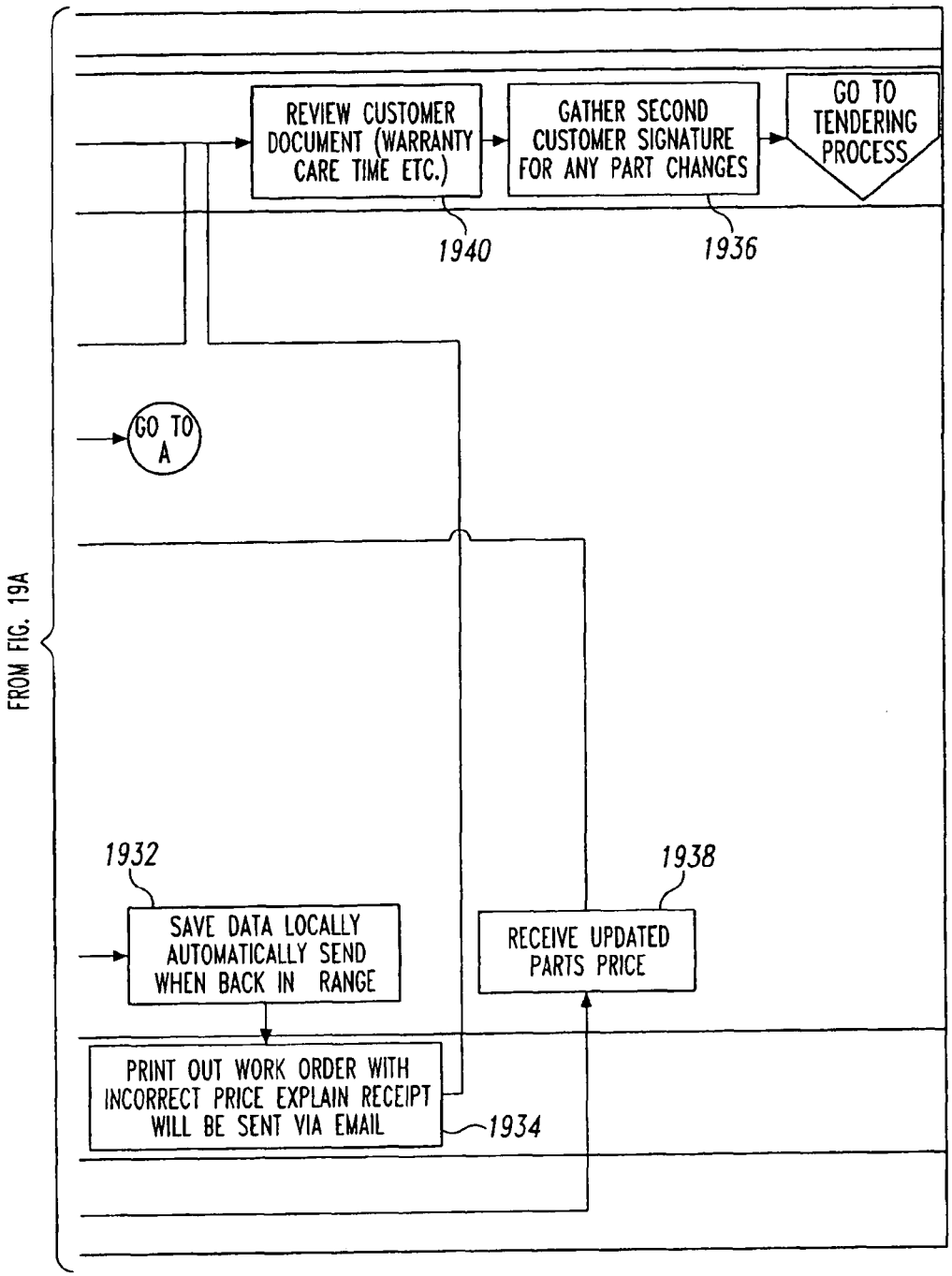

FROM FIG. 19A
REVIEW CUSTOMER DOCUMENT (WARRANTY CARE TIME ETC.)
GATHER SECOND CUSTOMER SIGNATURE FOR ANY PART CHANGES
GO TO TENDERING PROCESS
1940
1936
GO TO A
1932
SAVE DATA LOCALLY AUTOMATICALLY SEND WHEN BACK IN RANGE
1938
RECEIVE UPDATED PARTS PRICE
PRINT OUT WORK ORDER WITH INCORRECT PRICE EXPLAIN RECEIPT WILL BE SENT VIA EMAIL
1934

Fig. 19B

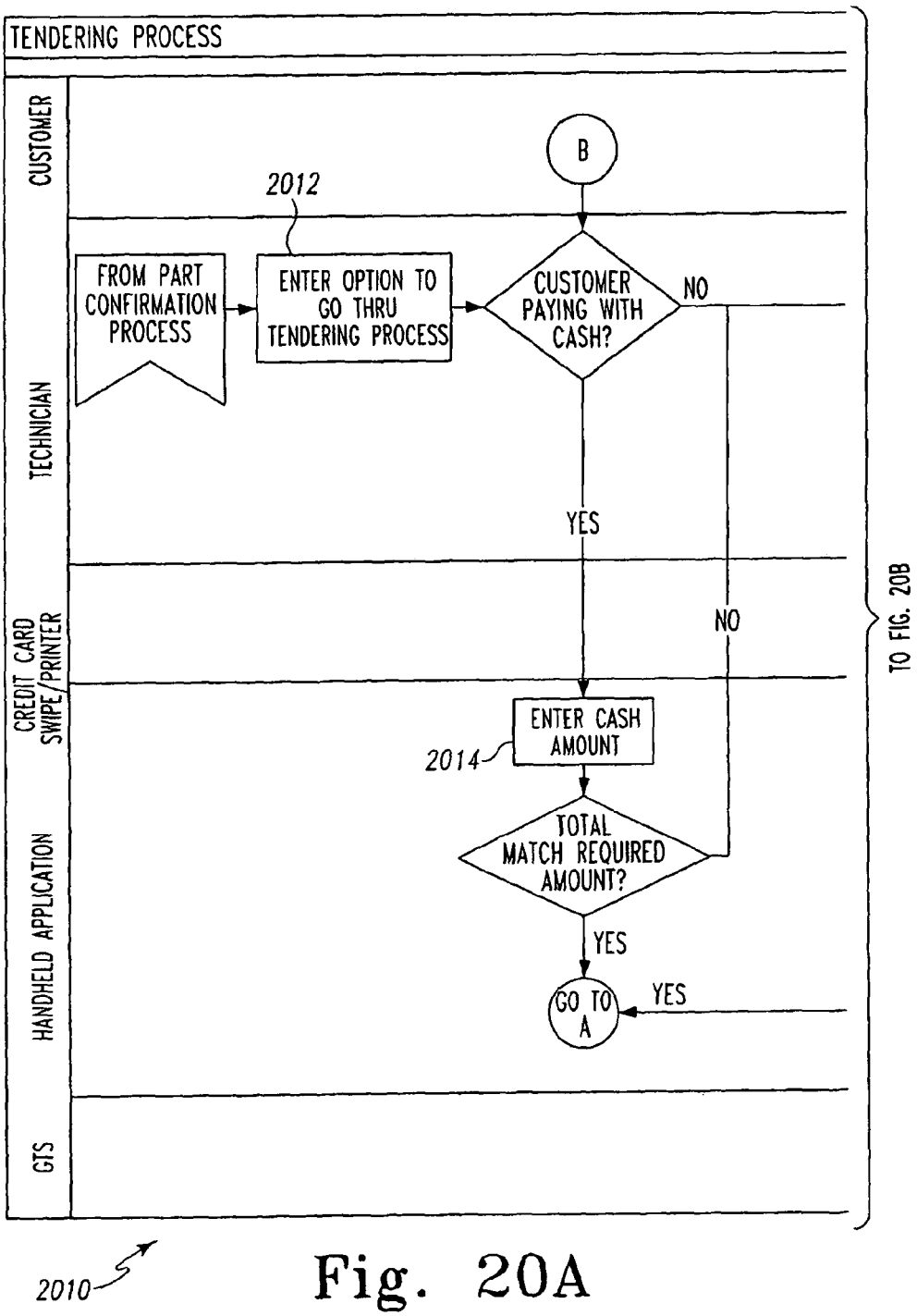

TENDERING PROCESS
CUSTOMER
TECHNICIAN
CREDIT CARD SWIPE/PRINTER
HANDHELD APPLICATION
GTS
B
2012
FROM PART CONFIRMATION PROCESS
ENTER OPTION TO GO THRU TENDERING PROCESS
CUSTOMER PAYING WITH CASH?
NO
YES
NO
ENTER CASH AMOUNT
2014
TOTAL MATCH REQUIRED AMOUNT?
YES
GO TO A
YES
TO FIG. 20B
2010

Fig. 20A

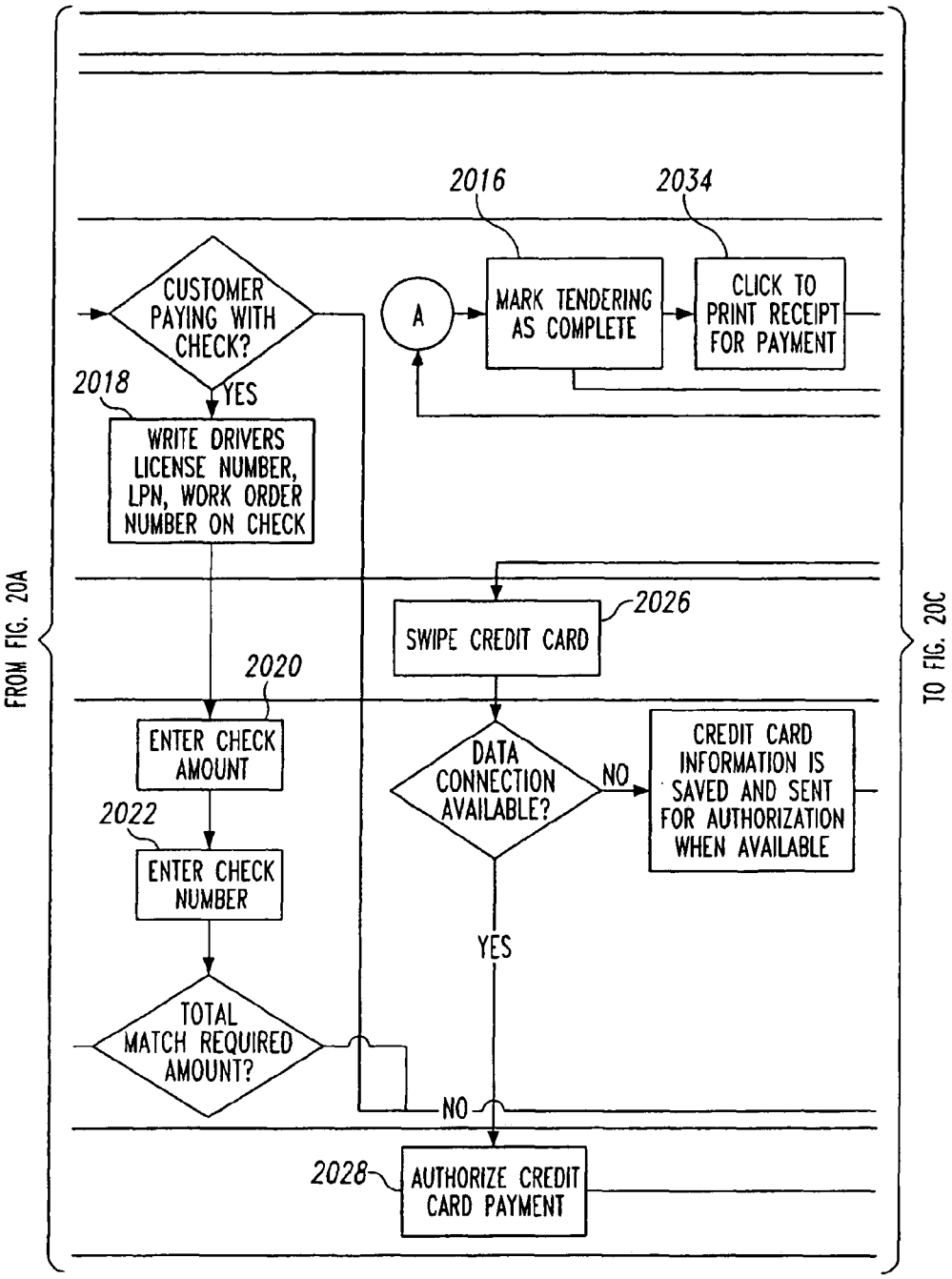

FROM FIG. 20A
TO FIG. 20C
2016
2034
CUSTOMER PAYING WITH CHECK?
A
MARK TENDERING AS COMPLETE
CLICK TO PRINT RECEIPT FOR PAYMENT
2018
YES
WRITE DRIVERS LICENSE NUMBER, LPN, WORK ORDER NUMBER ON CHECK
SWIPE CREDIT CARD
2026
2020
ENTER CHECK AMOUNT
DATA CONNECTION AVAILABLE?
NO
CREDIT CARD INFORMATION IS SAVED AND SENT FOR AUTHORIZATION WHEN AVAILABLE
2022
ENTER CHECK NUMBER
YES
TOTAL MATCH REQUIRED AMOUNT?
NO
2028
AUTHORIZE CREDIT CARD PAYMENT

Fig. 20B

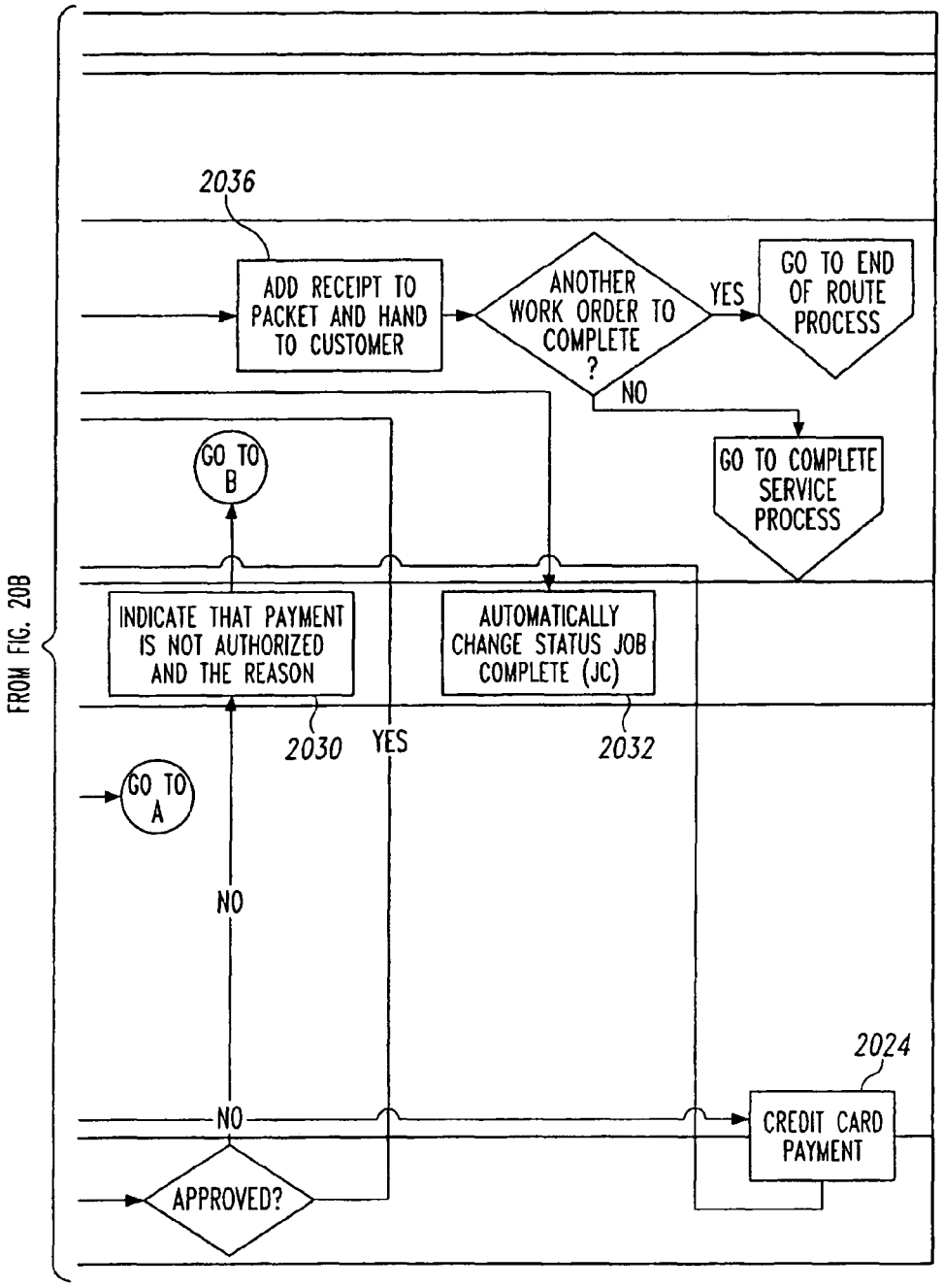

FROM FIG. 20B
2036
ADD RECEIPT TO PACKET AND HAND TO CUSTOMER
ANOTHER WORK ORDER TO COMPLETE ?
YES
GO TO END OF ROUTE PROCESS
NO
GO TO COMPLETE SERVICE PROCESS
GO TO B
INDICATE THAT PAYMENT IS NOT AUTHORIZED AND THE REASON
AUTOMATICALLY CHANGE STATUS JOB COMPLETE (JC)
2030
YES
2032
GO TO A
NO
2024
NO
CREDIT CARD PAYMENT
APPROVED?

Fig. 20C

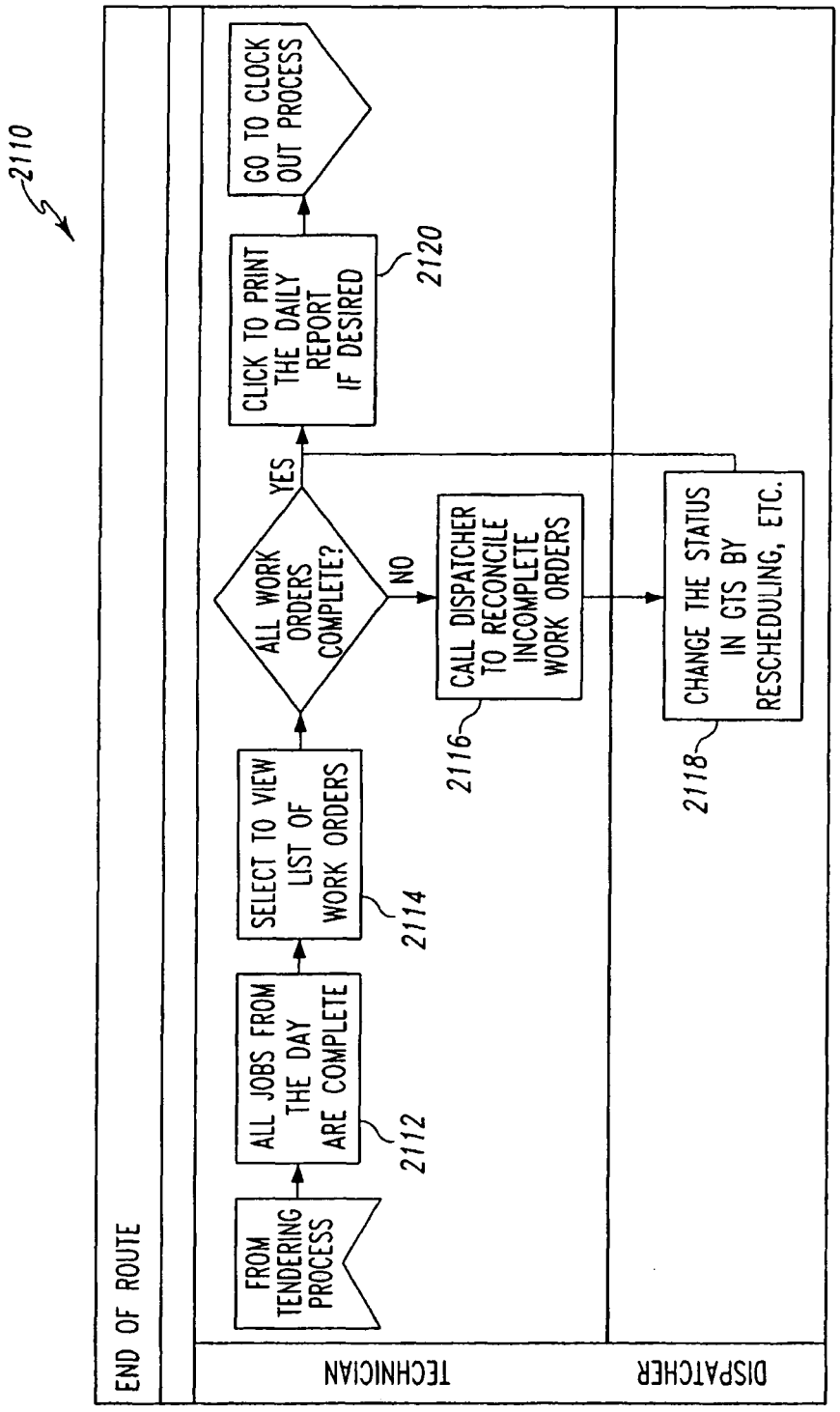

2110
END OF ROUTE
TECHNICIAN
DISPATCHER
FROM TENDERING PROCESS
ALL JOBS FROM THE DAY ARE COMPLETE
2112
SELECT TO VIEW LIST OF WORK ORDERS
2114
ALL WORK ORDERS COMPLETE?
YES
NO
CALL DISPATCHER TO RECONCILE INCOMPLETE WORK ORDERS
2116
CHANGE THE STATUS IN GTS BY RESCHEDULING, ETC.
2118
CLICK TO PRINT THE DAILY REPORT IF DESIRED
2120
GO TO CLOCK OUT PROCESS

Fig. 21

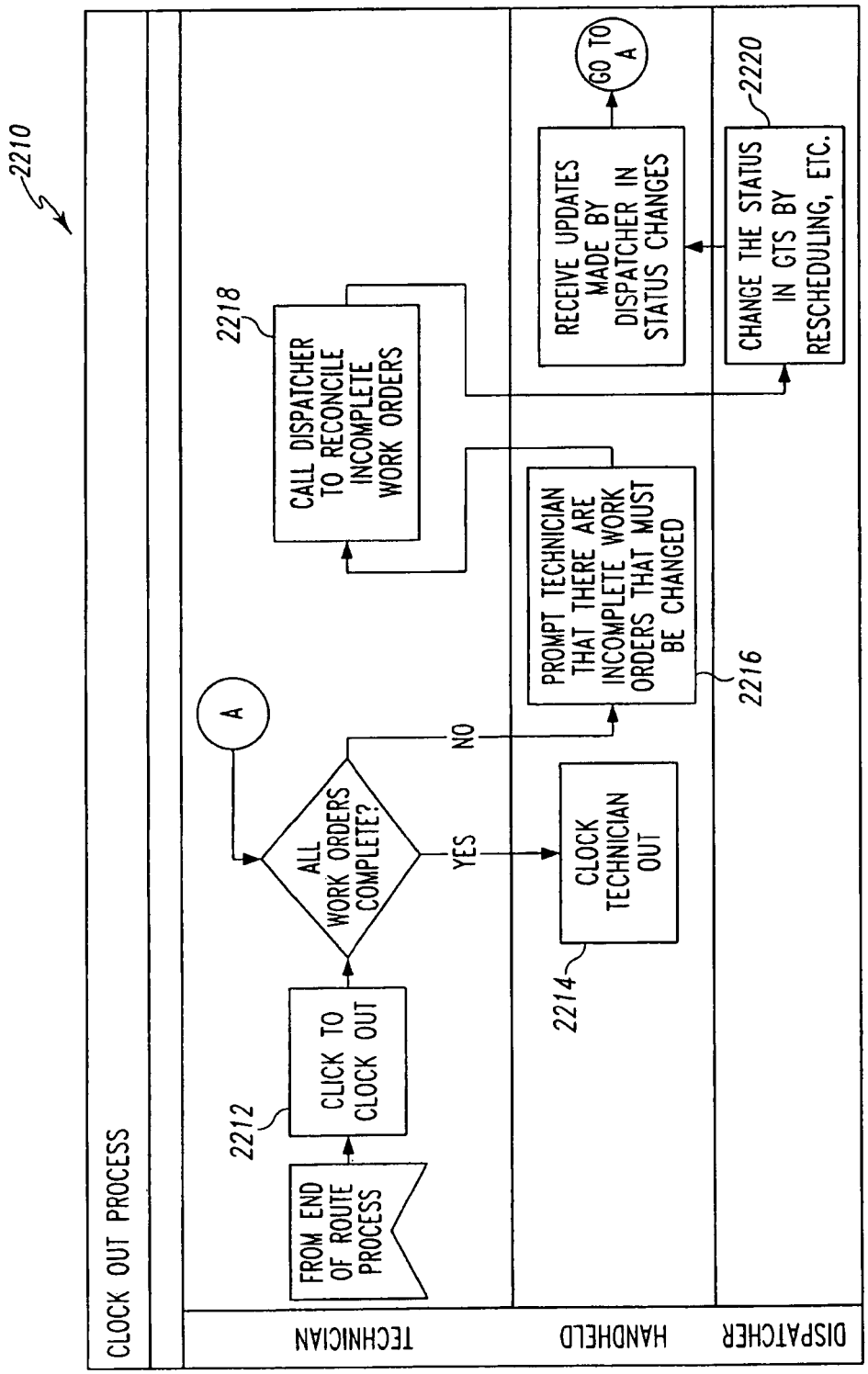

2210
CLOCK OUT PROCESS
TECHNICIAN
HANDHELD
DISPATCHER
FROM END OF ROUTE PROCESS
2212
CLICK TO CLOCK OUT
A
ALL WORK ORDERS COMPLETE?
NO
YES
2214
CLOCK TECHNICIAN OUT
PROMPT TECHNICIAN THAT THERE ARE INCOMPLETE WORK ORDERS THAT MUST BE CHANGED
2216
2218
CALL DISPATCHER TO RECONCILE INCOMPLETE WORK ORDERS
RECEIVE UPDATES MADE BY DISPATCHER IN STATUS CHANGES
GO TO A
2220
CHANGE THE STATUS IN GTS BY RESCHEDULING, ETC.

Fig. 22

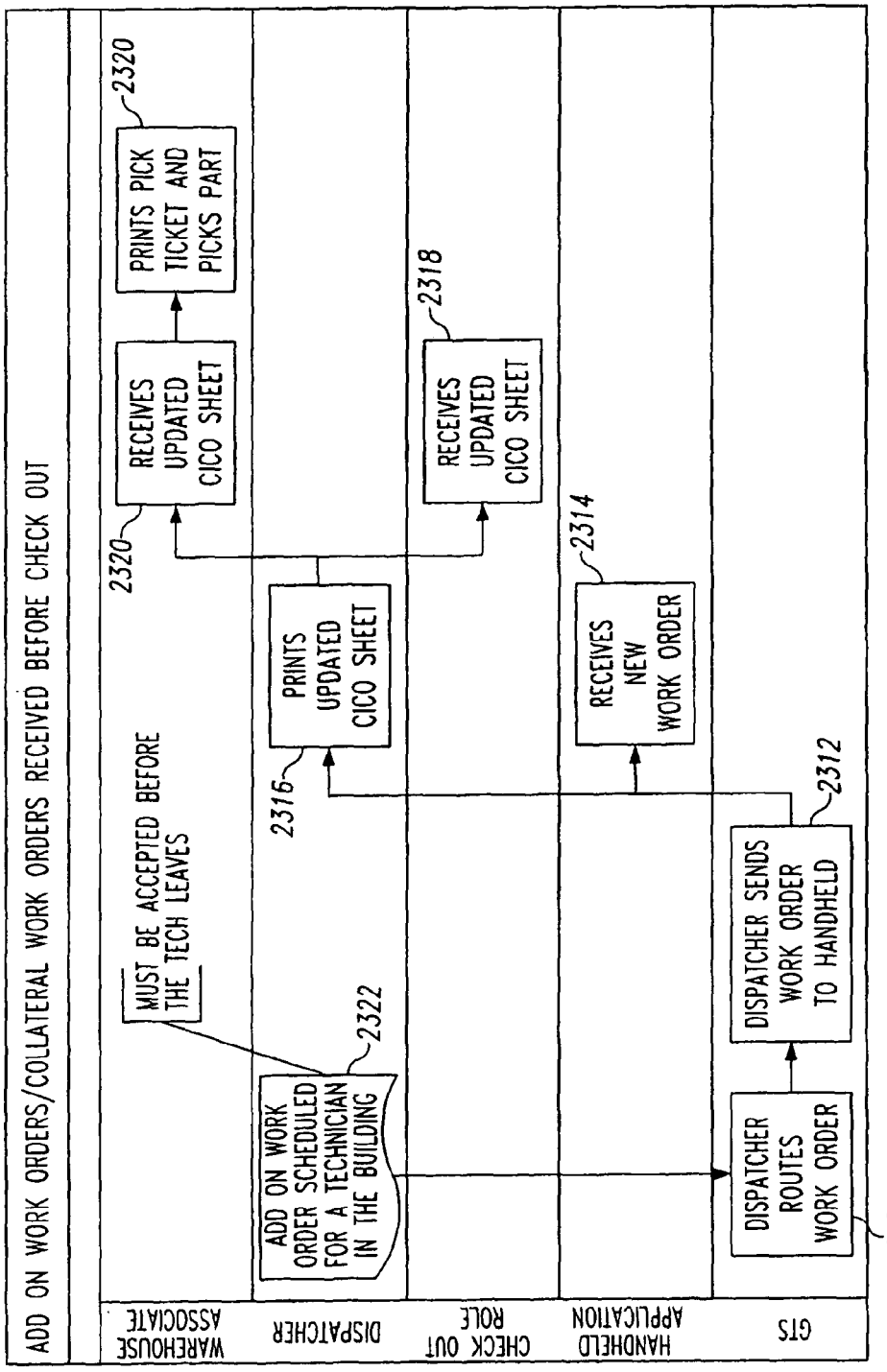

ADD ON WORK ORDERS/COLLATERAL WORK ORDERS RECEIVED BEFORE CHECK OUT
WAREHOUSE ASSOCIATE
DISPATCHER
CHECK OUT ROLE
HANDHELD APPLICATION
GTS
MUST BE ACCEPTED BEFORE THE TECH LEAVES
ADD ON WORK ORDER SCHEDULED FOR A TECHNICIAN IN THE BUILDING
2322
PRINTS UPDATED CICO SHEET
2316
RECEIVES UPDATED CICO SHEET
2320
PRINTS PICK TICKET AND PICKS PART
2320
RECEIVES UPDATED CICO SHEET
2318
RECEIVES NEW WORK ORDER
2314
DISPATCHER ROUTES WORK ORDER
DISPATCHER SENDS WORK ORDER TO HANDHELD
2312
2310

Fig. 23

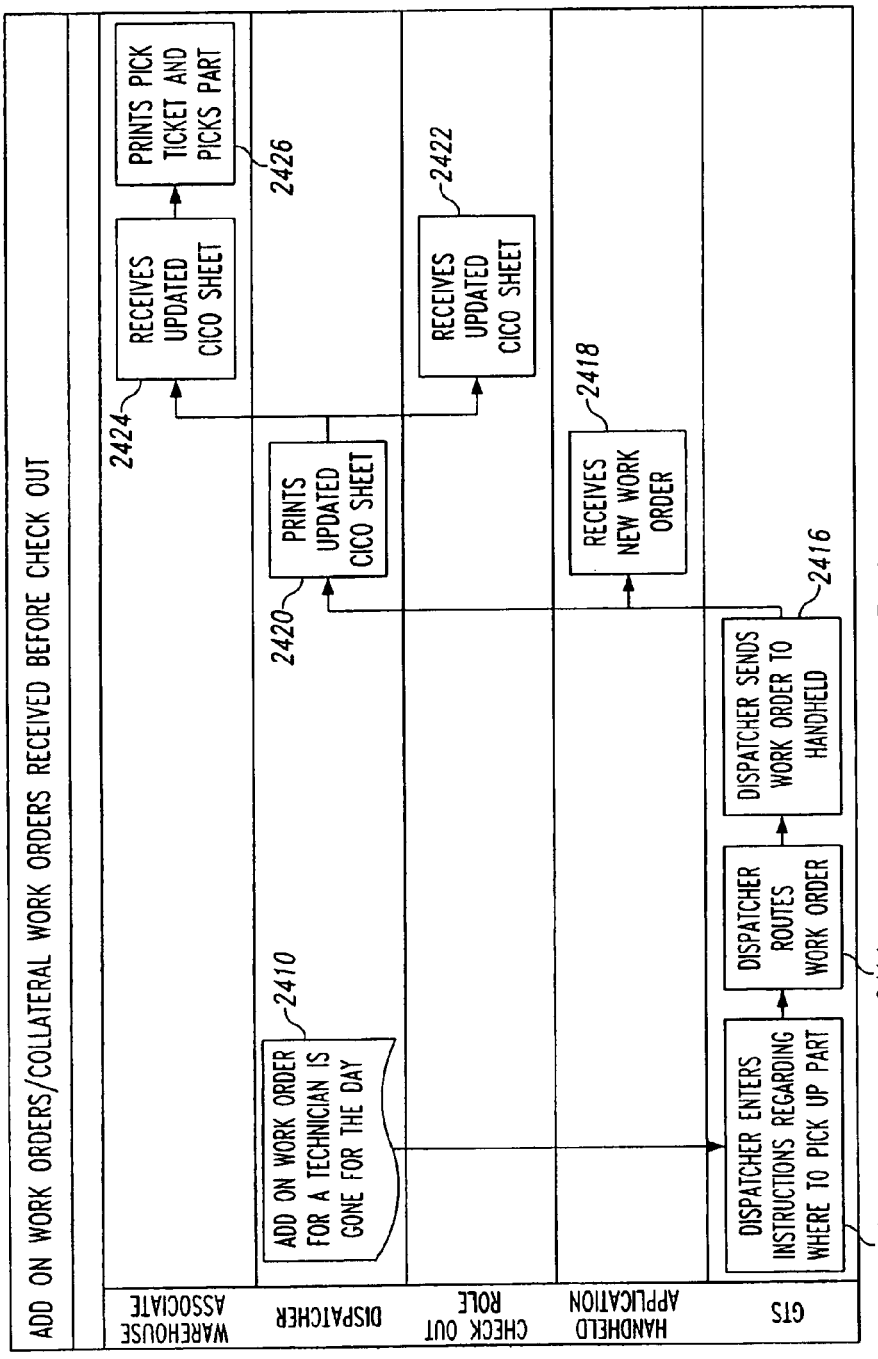

ADD ON WORK ORDERS/COLLATERAL WORK ORDERS RECEIVED BEFORE CHECK OUT
WAREHOUSE ASSOCIATE
DISPATCHER
CHECK OUT ROLE
HANDHELD APPLICATION
GTS
ADD ON WORK ORDER FOR A TECHNICIAN IS GONE FOR THE DAY
2410
PRINTS UPDATED CICO SHEET
2420
RECEIVES UPDATED CICO SHEET
2424
PRINTS PICK TICKET AND PICKS PART
2426
RECEIVES UPDATED CICO SHEET
2422
RECEIVES NEW WORK ORDER
2418
DISPATCHER ENTERS INSTRUCTIONS REGARDING WHERE TO PICK UP PART
2412
DISPATCHER ROUTES WORK ORDER
2414
DISPATCHER SENDS WORK ORDER TO HANDHELD
2416

Fig. 24

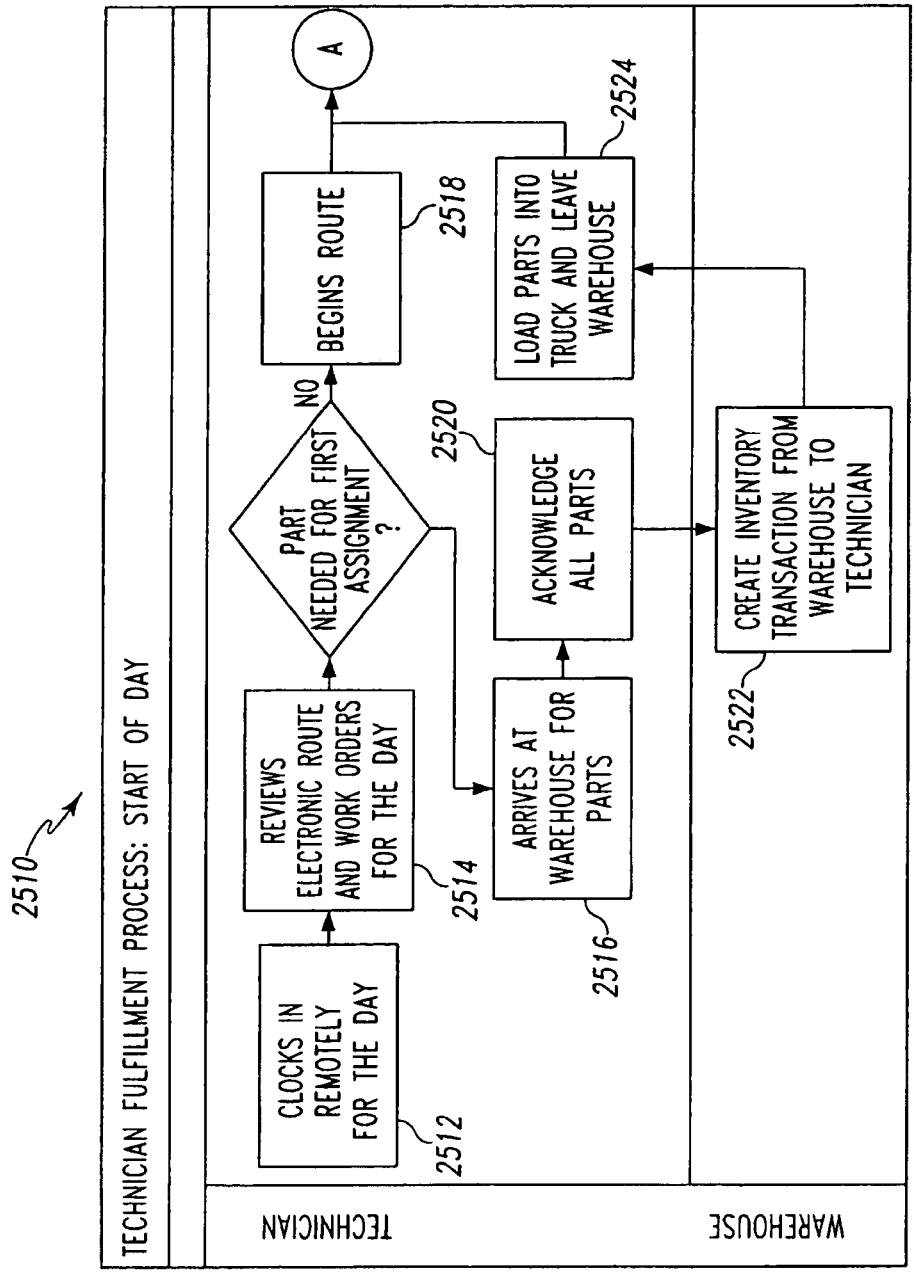

2510
TECHNICIAN FULFILLMENT PROCESS: START OF DAY
TECHNICIAN
WAREHOUSE
CLOCKS IN REMOTELY FOR THE DAY
2512
REVIEWS ELECTRONIC ROUTE AND WORK ORDERS FOR THE DAY
2514
PART NEEDED FOR FIRST ASSIGNMENT ?
NO
BEGINS ROUTE
2518
A
ARRIVES AT WAREHOUSE FOR PARTS
2516
ACKNOWLEDGE ALL PARTS
2520
LOAD PARTS INTO TRUCK AND LEAVE WAREHOUSE
2524
CREATE INVENTORY TRANSACTION FROM WAREHOUSE TO TECHNICIAN
2522

Fig. 25

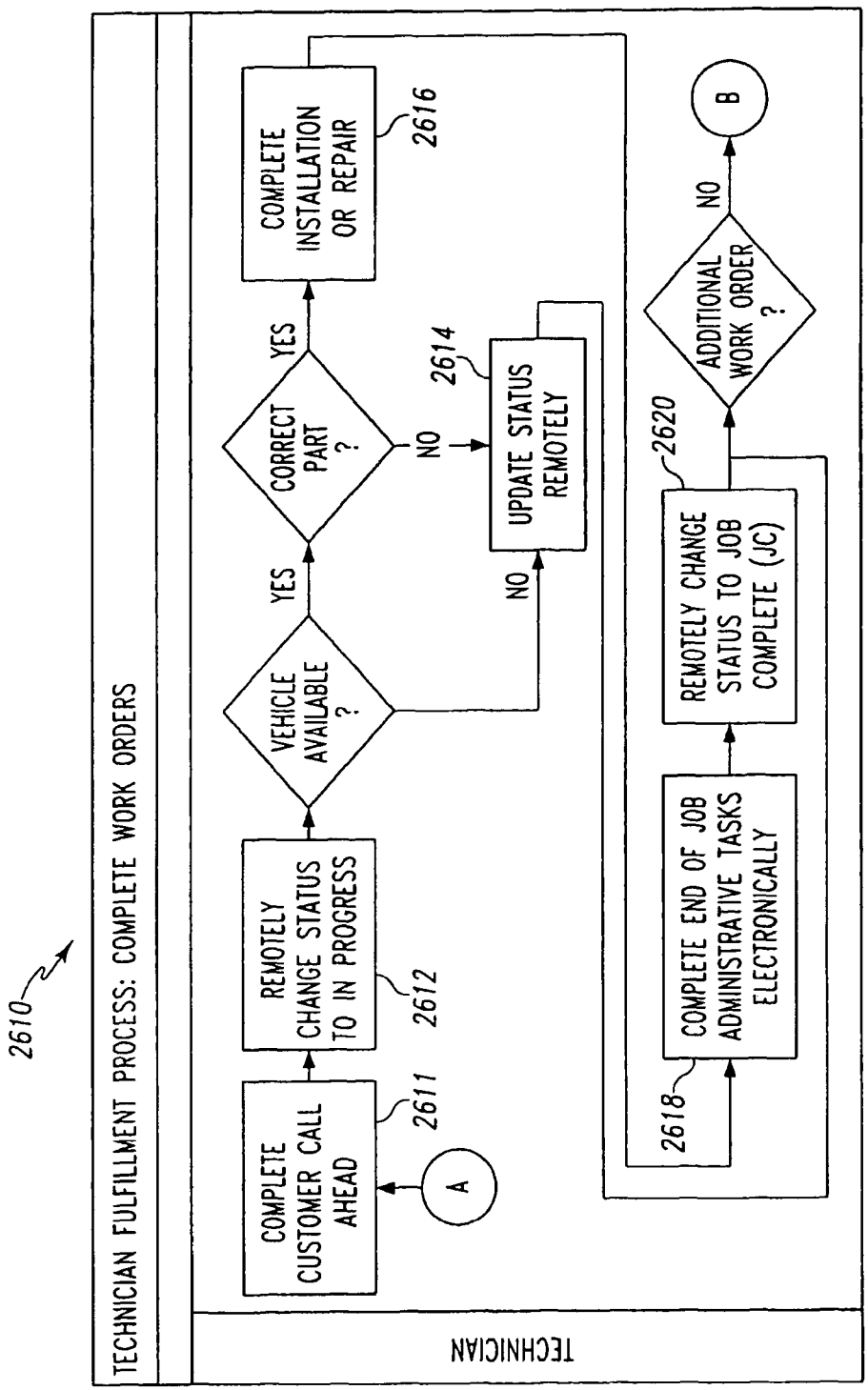

2610
TECHNICIAN FULFILLMENT PROCESS: COMPLETE WORK ORDERS
TECHNICIAN
COMPLETE CUSTOMER CALL AHEAD
2611
A
REMOTELY CHANGE STATUS TO IN PROGRESS
2612
VEHICLE AVAILABLE ?
YES
NO
CORRECT PART ?
YES
NO
2614
UPDATE STATUS REMOTELY
COMPLETE INSTALLATION OR REPAIR
2616
2618
COMPLETE END OF JOB ADMINISTRATIVE TASKS ELECTRONICALLY
REMOTELY CHANGE STATUS TO JOB COMPLETE (JC)
2620
ADDITIONAL WORK ORDER ?
NO
B

Fig. 26

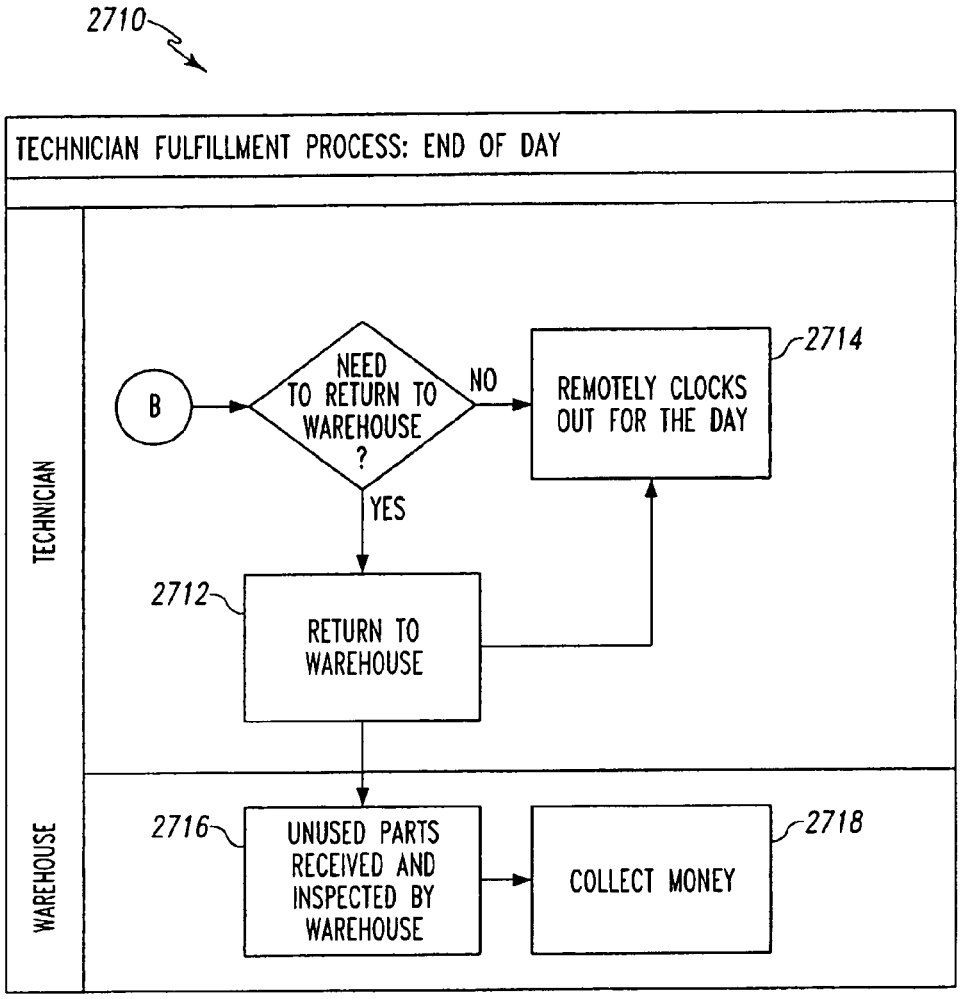

2710
TECHNICIAN FULFILLMENT PROCESS: END OF DAY
TECHNICIAN
WAREHOUSE
B
NEED TO RETURN TO WAREHOUSE ?
NO
YES
2714
REMOTELY CLOCKS OUT FOR THE DAY
2712
RETURN TO WAREHOUSE
2716
UNUSED PARTS RECEIVED AND INSPECTED BY WAREHOUSE
2718
COLLECT MONEY

Fig. 27

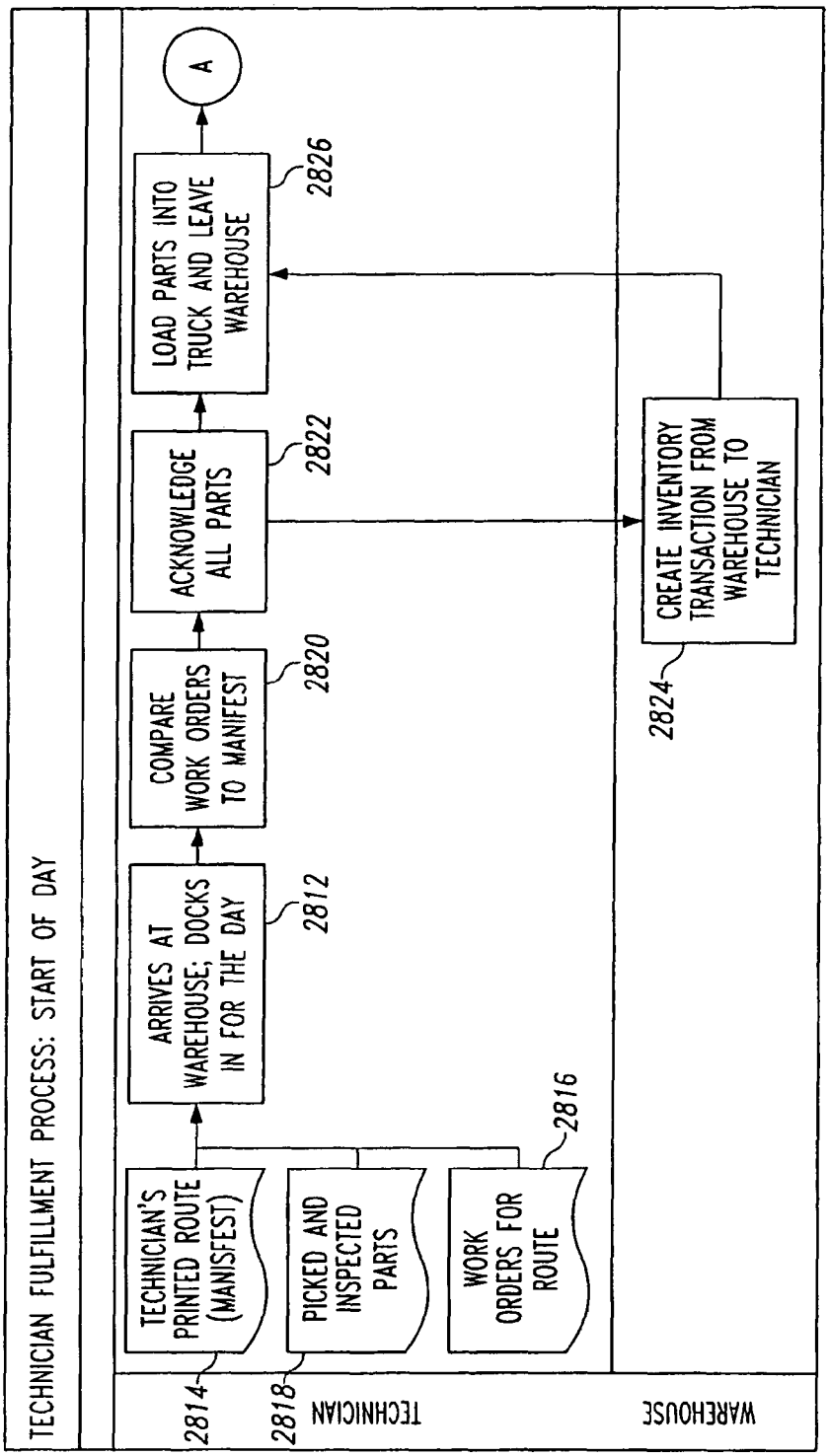


Fig. 28
(Prior Art)

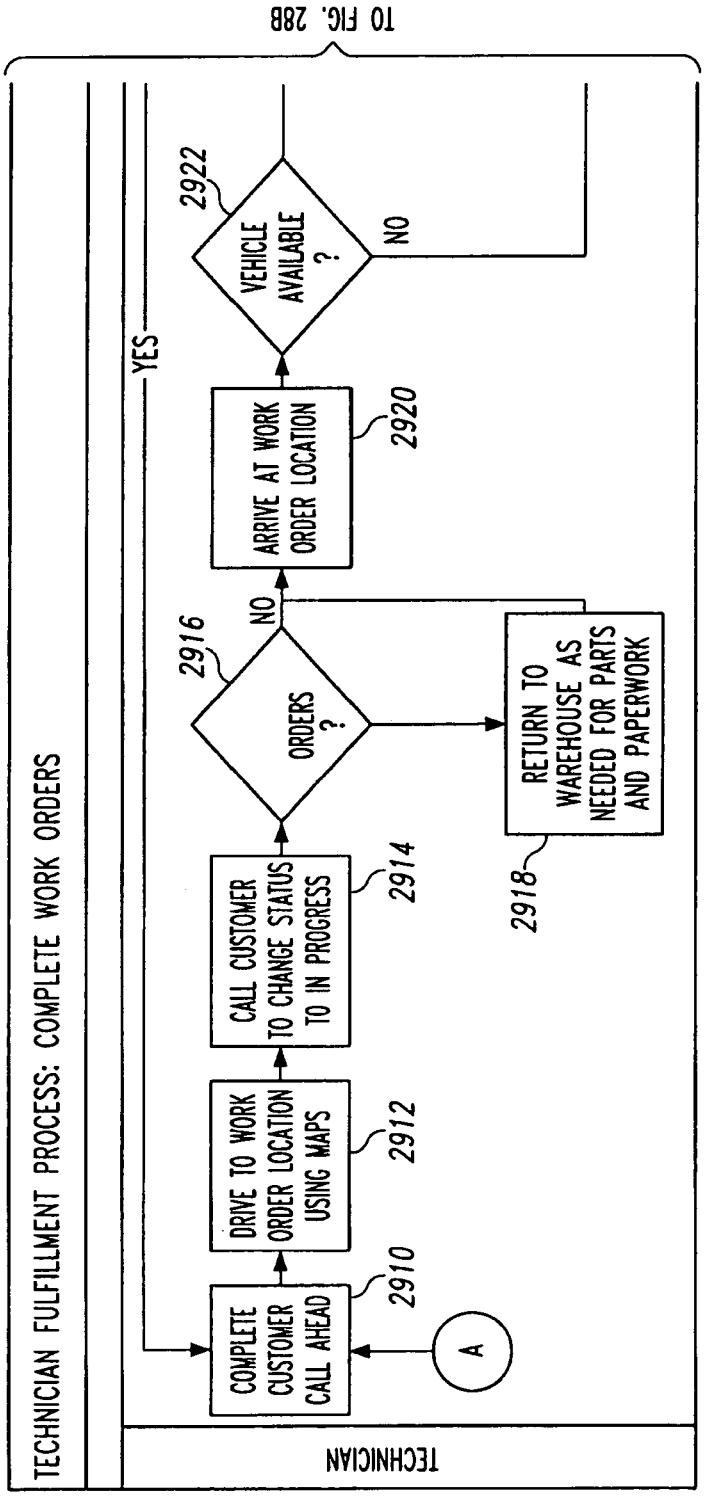


Fig. 29A
(Prior Art)

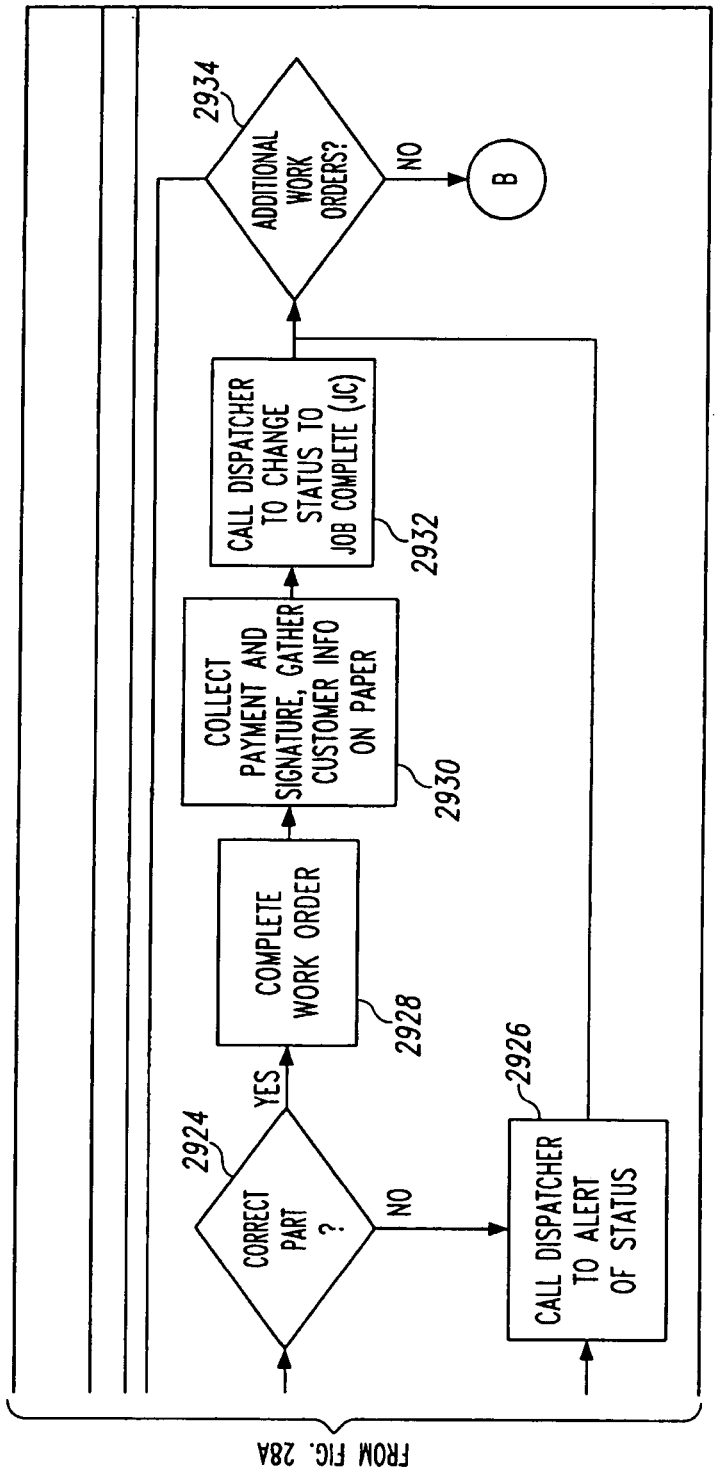


Fig. 29B
(Prior Art)

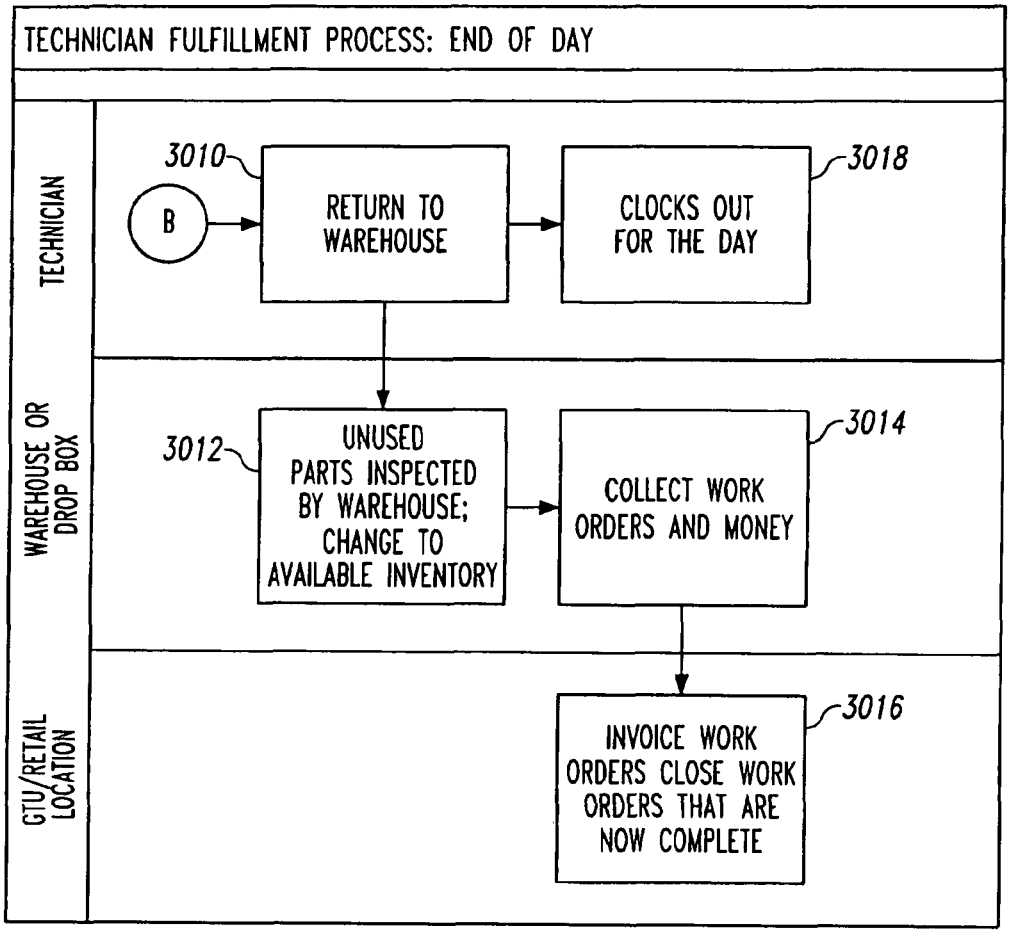


Fig. 30
(Prior Art)

METHOD AND DEVICE FOR PROVIDING LOCATION BASED CONTENT DELIVERY

RELATED APPLICATIONS

The present application is a U.S. National Stage Application of International Patent Application Serial No. PCT/US2007/001701, filed Jan. 19, 2007, the contents of which are herein incorporated by reference, which claims priority to U.S. Provisional Patent Application Ser. No. 60/760,320, filed Jan. 19, 2006, the contents of which are herein incorporated by reference.

BACKGROUND AND SUMMARY

This disclosure relates to the field of providing location based content delivery and more particularly to providing location based content delivery wherein work is performed off-site utilizing products that are stored at centralized locations wherein the products to be utilized during the provision of location based content delivery can typically be determined in advance.

Many businesses provide services to a variety of customers who need services performed at locations other than the locations of the businesses. Often the services provided require products to be consumed during the provision of the services. One example of such a business is the auto glass installation business wherein automotive glass is installed in a customer's vehicle either at the customer's home or at some other location convenient to the customer. In the auto glass installation business, by determining the make, year and model of the automobile, any tinting options and the window which needs replaced, it can be determined in advance precisely what automotive window glass, seals and sealants may be required for servicing each vehicle.

Currently one method of providing off-site automotive glass repair and installation utilizes several steps as shown, for example, in FIGS. **28-30**. Since the disclosed device and method will have the greatest impact on technician fulfillment processes, those processes implemented in the prior art will be described in some detail. As shown, for example, in FIG. **28**, in the described current method of providing location based content delivery, such as auto glass replacement services, at the start of the technician's day, a technician arrives at a warehouse and clocks in for the day in step **2812**. After arrival the technician receives, one or more of the following, work orders for the technician's route **2816**, a preprinted technician's printed route or manifest **2814**, picked and inspected parts **2818** for completing the work orders **2816**. The technician then compares the work orders to the manifest in step **2820** to determine that all of the work orders **2816** are included in the manifest **2814**. The technician then acknowledges that all of the parts required for completion of the work orders are included in the picked and inspected parts list in step **2822**. An inventory transaction is created from the warehouse in step **2824**. The parts are loaded onto the technician's truck prior to the technician leaving the warehouse in step **2826**.

As shown for example in FIG. **29**, after the start of the day the technician completes work orders in the existing technician fulfillment process utilizing various steps. The technician completes a customer call ahead in step **2910** utilizing a cell phone or other mobile telephone. After completing the customer call ahead the technician drives to the work order location utilizing pre-printed maps available in the technician's vehicle in step **2912**. Technician calls the dispatcher utilizing their cell phone to change status of the work order to in progress in step **2914**. If it is determined in step **2916** that there are new work orders then the technician returns to the warehouse as needed for parts and/or paperwork in step **2918**. Otherwise, the technician continues to the work order location until they arrive at the work order location in step **2920**. If it is determined in step **2922** that the vehicle requiring glass repair is available the technician then determines in step **2924** if they have the correct part or parts to perform the required service. If it is determined that either the vehicle is not available in step **2922** or that the correct part is not on the technician's vehicle in step **2924**, then the technician calls the dispatcher to alert the dispatcher of the status of the work order in step **2926**. If the vehicle is available and the correct part is on the technician's vehicle then the technician completes the work order in step **2928**. After completing the work order the technician collects payment and signature and gathers customer information on a hard paper copy in step **2930**. The technician then calls the dispatcher to change the status of the job to job complete in step **2932**. After the job is completed, or after it is determined that the job cannot be completed because the vehicle is unavailable or the correct part is not on the technician's vehicle, the technician determines if there are additional work orders in step **2934**. If there are additional work orders the technician returns to step **2910** and completes steps **2910** through steps **2932** for the additional work order. If there are no additional work orders then the technician begins the existing end of the day technician fulfillment process, as shown, for example, in FIG. **30**.

At the end of the day the technician returns to the warehouse in step **3010** and returns unused parts to the warehouse for inspection in step **3012**. At the warehouse or drop box, money and work orders are collected in step **3014**. After the work orders and money are collected the CTU retail location invoices the work order and closes the work orders that are now complete in step **3016**. After returning to the warehouse in step **3010** and delivering the unused parts to the warehouse in step **3012** the technician clocks out for the day in step **3018**. This clocking out typically involves putting a time card into a manual time clock which time card is utilized to enter the hours worked by the technician during each day.

The current fulfillment process has several drawbacks that result in some inefficiencies. Among a few of the drawbacks of the current process are: high technician dependence on the automotive glass company's infrastructure; high administrative costs associated with technician support; and reduced process efficiency and customer service.

The high technician dependence on the automotive glass company's infrastructure is evidenced by the fact that the daily work functions of the technician require much input and involvement from various components of the automotive glass companies infrastructure. Technicians are tightly coupled with the dispatchers, central telephone units "CTUs", warehouses, and drop-boxes. This is evident in the clocking-in/clocking-out, getting route and work-orders, getting parts, changing work-order status, processing credit card payments, returning unused parts, returning payments, determining job-site location, changing routes, and addressing customer inquiries steps of the current process. Under the current process, the majority of technicians must drive to a warehouse location to clock-in and out at the beginning and end of each day. Also under the current process, at the beginning of the day, all routes and work-orders are picked up from a warehouse or drop-box. All parts are acquired from the automotive glass repair company's warehouse or drop-box. Throughout the day, technicians must call the dispatcher in order to change the status of their work-orders. This can happen as much as three times per work-order or twenty times

per day per technician. In order to take a credit card payment at the job site, under the current process, the technician calls the dispatcher and provides the credit card and payment information. The dispatcher is then responsible for processing the payment. At the end of the day, all unused parts are returned to the automotive glass company's warehouse or drop-box. All payments collected at the job-sites are returned to the automotive glass company's warehouse or drop-box at the end of each day. The majority of questions regarding directions or job-site location are answered by the dispatcher via a phone call from the technician. Any route changes that occur when the technician is in route are communicated through a phone call from the dispatcher. In addition, route changes may require an extra trip to the warehouse to pick-up parts. Service related questions asked by the customer on the day of service are channeled from the customer to the technician through the dispatcher. Any return communications are returned via the same path.

Because of the tight coupling between technicians and the automotive glass company's infrastructure, administrative and support costs associated with service delivery are higher than necessary. These costs manifest themselves in both payroll and operating expense. Under the above described current process of providing automotive glass repair services, much paper is required to support the process. Among the paper work generated by such process are route manifests, check-in/check-out lists, invoices, and multiple work-orders printed for each job. Additionally, management of parts, routes, work-orders, payments, and customer inquiries creates a ratio of support personnel to technician that is overly high. The combination of payroll and operating expense required to support the automotive glass company's technicians creates a higher than desired fixed cost.

Under the above described current process of providing automotive glass repair services, there exists some reduced process efficiency and customer service. The order fulfillment process includes steps that have an overall negative impact on efficiency and customer service. While technician productivity may be high under the above described process, it is believed that there are opportunities for improvement in technician idle time, phone calls required to complete a job, trips to and from a warehouse or drop-box, work-order slippage, inventory loss, service differentiation, and invoice and work-order quality. Thus, providers of location based content delivery would appreciate a process and device that improves the currently available processes in one or more of the following ways, either alone or in combination.

Providers of location based content delivery would appreciate a process and device that reduces technician idle time by increasing the amount of time technicians spend on fulfilling orders. Providers of location based content delivery would appreciate a process and device that reduces the amount of phone calls technicians place to the dispatcher and vice-versa. Providers of location based content delivery would appreciate a process and device that reduces the amount of trips the technician takes to and from a warehouse or drop-box. Providers of location based content delivery would appreciate a process and device that allows technicians to complete a greater percentage of scheduled jobs thereby reducing work-order slippage. Providers of location based content delivery would appreciate a process and device that increase control of inventory checked-out of the warehouse. Providers of location based content delivery would appreciate a process and device that facilitate providing higher quality service in a way that differentiates their services from competitors' services. Providers of location based content delivery would appreciate a process and device that improve compliance with "Write-it-right" regulations and increase quality of customer data collected at the job site.

The method and device disclosed herein decrease technician dependence on the location based content delivery provider's distribution infrastructure, reduce administrative costs associated with technician support and increase process efficiency and customer service. In the disclosed embodiments these improvements are accomplished by implementing broad business process improvements and mobile resource management (MRM) technical solutions.

It is believed that through the use of wireless networks an increase in business agility may be realized through continuous data access and synchronization. Better business decisions may be made when information is more timely and accessible. Wireless solutions deliver customizable data exchange solutions that fit with an location based content delivery provider's mobile business model. Wireless solutions also help office-based employees manage day-to-day workflow and keep up with mobile users in real-time. MRM allows users to track their work as they complete it, whether they are selling, installing, delivering, servicing, or monitoring product delivery.

According to one aspect of the disclosure, a device for providing location based content delivery includes a server, a hand-held electronic device and a processing device. The server is located at a first location and includes memory and data stored in the memory relating to a plurality of requests for service at a location other than the first location. The hand-held electronic device is configured for communication via a wireless network with the server to receive data relating to the plurality of requests for services and to transmit data regarding the provision of services for updating the data stored in the memory. The processing device is communicatively coupled to the hand-held device.

According to another aspect of the disclosure, a method for providing location based content delivery at a job site other than the location of a service provider comprises providing a technician with a hand-held electronic device, clocking the technician in remotely utilizing the hand-held electronic device, clocking the technician out remotely utilizing the hand-held electronic device, transferring work orders to the hand-held electronic device, viewing the work orders on the hand-held electronic device, capturing customer information electronically on the hand-held electronic device at a job site and transferring captured customer information to a server accessible at the location of the service provider.

Additional features and advantages of the disclosed devices and methods will become apparent to those skilled in the art upon consideration of the following detailed description of preferred embodiments exemplifying the best mode of carrying out the invention as presently perceived.

BRIEF DESCRIPTION OF THE DRAWINGS

The illustrative devices and methods will be described hereinafter with reference to the attached drawings which are given as non-limiting examples only, in which:

FIG. **1** is a block diagram of a device for facilitating the provision of location based content delivery services;

FIG. **2** is a diagram of an embodiment of a device for facilitating the provision of location based content delivery services without technician location services or turn by turn direction services;

FIG. **3** is a diagram of an embodiment of a device for facilitating the provision of location based content delivery

services with messaging listening and load balancing capabilities and with fire wall protection of the local area network;

FIG. **4** is a simplified block diagram of some of the major components of device including the eCentral device, the application server, the wireless network Server, the firewall and the hand-held device;

FIG. **5** is a logical layer diagram of the simplified block diagram of FIG. **4**;

FIG. **6** is a logical layer diagram of the simplified block diagram of FIG. **4**;

FIG. **7** is diagram of an embodiment of a device for facilitating the provision of location based content delivery services with technician location services and turn by turn direction services;

FIG. **8** is diagram of an embodiment of a device for facilitating the provision of location based content delivery services with off-site credit card capture;

FIG. **9** is a diagram of an embodiment of the device for facilitating the provision of location based content delivery services utilizing a signature capture pad;

FIG. **10** is a diagram of an embodiment of the device for facilitating the provision of location based content delivery services utilizing a signature capture digital pen;

FIG. **11** is a high level process flow diagram of a method for providing location based content delivery services;

FIG. **12** is a flow diagram of a routing step of the method for providing location based content delivery services of FIG. **11**;

FIG. **13** is a flow diagram of a log-in process step of the method for providing location based content delivery services of FIG. **11**;

FIG. **14** is a flow diagram of a clock-in process step of the method for providing location based content delivery services of FIG. **11**;

FIG. **15** is a flow diagram of a check-in step of the method for providing location based content delivery services of FIG. **11**;

FIG. **16** is a flow diagram of a check-out step of the method for providing location based content delivery services of FIG. **11**;

FIG. **17** is a flow diagram of a technician call ahead step of the method for providing location based content delivery services of FIG. **11**;

FIG. **18** is a flow diagram of a complete service step of the method for providing location based content delivery services of FIG. **11**;

FIG. **19** is a flow diagram of a part confirmation step of the method for providing location based content delivery services of FIG. **11**;

FIG. **20** is a flow diagram of a tendering process step of the method for providing location based content delivery services of FIG. **11**;

FIG. **21** is a flow diagram of an end of route step of the method for providing location based content delivery services of FIG. **11**;

FIG. **22** is a flow diagram of a clock-out step of the method for providing location based content delivery services of FIG. **1**;

FIG. **23** is a flow diagram of a step for adding a work order before a technician performs the check-out step of the method for providing location based content delivery services of FIG. **11**;

FIG. **24** is a flow diagram of a step for adding a work order after a technician performs the check-out step of the method for providing location based content delivery services of FIG. **11**;

FIG. **25** is a flow diagram of a start of day portion of the technician fulfillment process step of the method for providing location based content delivery services;

FIG. **26** is a flow diagram of a complete work orders portion of the technician fulfillment process step of the method for providing location based content delivery services;

FIG. **27** is a flow diagram of a start of day portion of the technician fulfillment process step of the method for providing location based content delivery services;

FIG. **28** is a flow diagram of a start of day portion of the technician fulfillment process step of a prior art method for providing location based content delivery services;

FIG. **29** is a flow diagram of a complete work orders portion of the Technician fulfillment process step of a prior art method for providing location based content delivery services; and,

FIG. **30** is a flow diagram of a start of day portion of the technician fulfillment process step of a prior art method for providing location based content delivery services.

DETAILED DESCRIPTION

The disclosed method and device offer improvements over the current processes implemented to provide location based content delivery wherein inventory located at centralized locations is installed or otherwise utilized at an off-site job site. While the disclosed method and device are described with reference to the provision of off-site automotive glass installation, those skilled in the art will recognize that the method and device disclosed herein can be utilized in providing other types of location based content delivery such as appliance repair and other equipment installation purposes.

The disclosed device and method improve the provision of location based content delivery in one or more of the following manners. The clock-in/clock-out process is improved by eliminating or reducing the number of phone calls and amount of travel required for a technician to clock-in and out. This is accomplished by providing the technician with a hand-held electronic device that allows the technician to clock-in and out remotely. This can result in realization of additional benefits such as reduction in fuel usage and reduction in mileage put on company owned or leased vehicles possibly resulting in extended vehicle life and/or reduced maintenance and repair costs.

Additionally, paperwork is reduced in one or more of the following ways by implementing the disclosed embodiments. Paperwork is reduced by eliminating or substantially reducing the number of printed versions of work orders. In the disclosed embodiments, the technician is allowed to view the work orders on a hand-held electronic device. Paperwork is reduced by eliminating or substantially reducing the printed Check-in/Check-out ("CICO") lists. CICO lists are generated electronically and are viewable on not only the technician's hand-held electronic device but also on devices such as computers work stations and/or hand-held electronic devices in the warehouse. Paperwork is further reduced by eliminating or substantially reducing the number of printed versions of the route manifest provided to each technician. In the disclosed embodiments, technicians are allowed to view the work order sequence on their hand-held electronic device. Work order status changes are simplified by eliminating or reducing the number of telephone calls from the technician to the dispatcher previously required to change work order status and the need to print updated work orders and manifests reflecting the changed or added information.

In the disclosed embodiments, the technician remotely updates the status of work orders through their hand-held

devices on a central computer or server (occasionally referred to herein in specific examples and in the drawings related to off-site glass repair as "GTS") without travel or phone conversations. Technicians are able to obtain parts as needed from centralized inventory locations and/or alternative parts suppliers. Additionally, technicians are provided more opportunities with regard to the disposition of unused parts. The hand-held device allows dispatchers to reprioritize work orders in real-time by sending the reprioritized work order to the technician's hand held device.

In the disclosed embodiment, through the utilization of GPS or other location technology, the dispatcher is able to know the location of the technician in real-time, thereby reducing the number of phone calls between the technician and dispatcher. Additionally, the provision of location technology facilitates delivering turn by turn directions and electronic maps to the hand-held device **14** to provide directions to the technicians next worksite.

Instead of manually notating customer information, such as Vehicle Identification numbers, license numbers and mileage when services are provided on an automobile, the technician is allowed to capture this information electronically on the hand-held electronic device at the job-site. Additionally, the disclosed device facilitates electronic capture of the customers signature at the job site. With the provision of the portable printer, the technician can print necessary receipts and other required documentation at the job site for delivery to the customer. Phone calls to the dispatcher from the technician are substantially reduced by utilizing the hand-held device to process credit card payments at the job-site. The disclosed embodiments also allow the technicians the ability to invoice, record payments and make bank deposits utilizing their hand-held electronic devices.

As shown for example, in FIG. **1**, the disclosed embodiment of a device **10** for providing location based content delivery includes a server **12**, a hand-held electronic device **14**, a signature capture device **16**, a processing device **18** and a location information-receiving device **20**. The device **10** may also include a separate a mobile phone **26** provided for use by the technician. In the illustrated embodiment, hand-held electronic device **14** is a personal data assistant ("PDA") device configured for wireless communication. The PDA **14** and the server **12** are each configured to communicate via a wireless communication network **22**. The signature capture device **16** is communicatively coupled to or implemented by the hand-held electronic device **14**. The processing device **18** is communicatively coupled to the hand-held electronic device **14**. Thus, the components of the device **10** are configured to permit data to be transferred between the various devices.

In the illustrated embodiment, processing device **18** is configured to automatically capture information regarding an account represented by a credit card and to print documents. In one embodiment, the processing device **18** is configured to allow credit cards to be swiped in a credit card capture device **24** and print documentation from a printer **28**. While shown as a single device **18**, it is within the scope of the disclosure for processing device **18** to be implemented by a separate credit card capture device **24** and a separate printer **28** or for either or both of the credit card capture device **24** and printer **28** to be implemented by the hand-held electronic device **14**.

The location information-receiving device **20** is communicatively coupled to or implemented by the PDA **14**. The location information-receiving device **20**, in one disclosed embodiment, is a GPS transceiver **20** configured to receive signals from and transmit signals to the global positioning satellite system. The location information is sent by the hand-held device **14** to the server **12** to permit electronic and real time monitoring of the location of the technician with which the hand-held device **14** is associated. In one embodiment of the disclosed device and method, employee information regarding the technician to which the hand-held device **14** is assigned is associated with the specific hand-held device **14** assigned to the technician. For instance, the server **12** may link the employee identification number of the technician to the MAC address ID of the hand-held electronic device **14** assigned to the technician.

In one exemplary embodiment, the PDA **14** is a Blackberry Model 7520, available from Research In Motion Limited. Those skilled in the art will realize that while a personal data assistant is described it is within the scope of the disclosure for other portable electronic communication devices to be utilized in the device **10**. For example, as the memory capacity of cellular telephones increase they may exhibit the functionality required to act as hand-held electronic device **14**.

While described as a single wireless communication network **22** and a single server **12**, as shown, for example, in FIGS. **1**, **2**, **3**, **7** and **8** multiple wireless communication networks and multiple servers and gateways may be utilized in the implementation of the device **10**. As shown, for example in FIG. **2**, the term server **12** is used broadly to encompass various network and back office components. For example, the server **12** from a development environment perspective may include computers such as mainframes **30**, eCentral **32** and an S3 router **34** computer devices coupled through a network bus **36** to each other and/or to an application server **38**, and/or a shared SQL database server **40** coupled to an SQL database **41**. In the illustrated embodiment, the application server **38** is communicatively coupled to both the network bus **36** and the shared SQL database server **40**. Illustratively, the database server **40** runs an operating system such as Windows 2000 or 2003 server operating system or Boleris The application server **38** is also communicatively coupled to the wireless communication network server **42**, which in the illustrated embodiment is a Blackberry Enterprise server **42** communicatively coupled to a SQL database **44**. In the illustrated embodiment, the wireless communication server **42** is coupled through a local area network **46** and network **48** to the wireless communication network server **50**. Illustratively, network **48** provides bi-directional TCP communication between the wireless communication server **42** and the wireless communication network server **50**. In the illustrated embodiment, the wireless communication network server **50** is the RIM BES network provided by Research In Motion Limited.

The server **12** also includes an HR ID module **52** communicative coupled between the wireless network server **50** and a gateway device **54**. The gateway device **54** in the illustrated embodiment is communicatively coupled to additional gateways **56**, such as those provided by a cellular telephone service, because the communication services with the hand-held device **14** are provided through the cellular telephone wireless network which includes a plurality of communication towers **58**. If the communication with the wireless hand-held device **14** is not provided by a cellular telephone service provider, additional gateways **56** may not be required and wireless communication from the server **12** to the hand-held device **14** may be provided through another wireless communication network. Additionally, other wireless communication networks including proprietary wireless networks may be utilized to provide communication between the server and the hand-held device.

While the application server **38**, SQL database server **40** and wireless communication network server **42** are shown as

separate machines, it is within the scope of the disclosure for each of these servers to be implemented as virtual machines or otherwise implemented on a single computer. Additionally, the web server **70** (FIGS. **7** and **8**) may be implemented on the same computer as the application server **38**, SQL database server **40** and wireless communication network server **42**. It has been found that a single computer such as and IBM x335 with dual 2.8 GHz CPUs, four GB of RAM, two mirrored 36 GB internal disk drives and two GigE network interfaces may be configured to act as web server **70**, application server **38**, SQL database server **40** and wireless communication network server **42**. Other computers may also be utilized to implement the web server **70**, application server **38**, SQL database server **40** and wireless communication network server **42** within the scope of the disclosure.

As shown for example in FIG. **3**, device **10** may also include one or more load balancer devices **60** communicatively coupled to the application server **38** and a firewall **62** protecting the local area network **46**. Additionally, a Messaging listener **64** is coupled between the eCentral device **32** and the load balancer **60** in the communication link with the application server **38**.

When GPS location and turn-by-turn direction functionality are provided, device **10** also includes an ASP map serves application server **66** coupled to a data source **68**, as shown, for example, in FIG. **7**. Additionally, device **10** includes a web server **70** coupled to an ESRI data source **72**. When credit card capture is implemented in device **10**, as shown for example in FIG. **8**, device **10** also includes a transaction manager server **74** communicatively coupled to the web server **70** and the hand-held device **14**.

In one example of the device **10**, the processing device **18** is an Oneil printer Model MF 4T which is configured to interface with the PDA **14**. The disclosed processing device **18** includes a built in credit card swipe slot **24** and a printer **28**. The disclosed processing device **18** is blue-tooth enabled to provide communicative coupling between the processing device **18** and the hand-held device **14**. Preferably, the processing device **18** is lightweight (the disclosed O'Neil printer is under five pounds), has flushable ROM and has multiple power sources which may include battery power, a DC adapter for powering the device from an automobile electrical system and an AC power source for powering the processing device from an electrical outlet.

Alternative signature capture devices **16** include a signature pad **76** or a wireless pen **78** configured to interface with the PDA. Preferably such separate signature capture devices **16** are blue-tooth enabled. Among the currently available signature capture devices **16** that may be utilized within the scope of the disclosure is the digital wireless pen **78** available from Standard Register, 600 Albany St., Dayton, Ohio, and the Siglite Bluetooth enabled pressure-sensitive electronic signature pad **76** available from Topaz Systems, Inc., 650 Cochran Street, Unit 6, Simi Valley, Calif., 93065. Additionally, it is within the scope of the disclosure for the hand-held device **14** to be configured to include a signature capture device **16**, although that feature is not available on the BlackBerry 7520.

In one illustrated embodiment, utilizing a Topaz signature pad as the signature capture device and a Blackberry 7520 as the hand-held device **14**, the hand-held device **14** runs a J2ME application to facilitate capturing signature data provided by the signature capture device **16**. IN such embodiment, RIM's Bluetooth API is utilized to establish a Bluetooth connection between the hand-held device **14** and the signature capture device **16**. The J2ME application listens on a specific port for incoming signature data from the signature capture device **16** and retrieves such signature data when it is present utilizing the Java library associated with the Topaz Signing pad **76**. Upon receiving the signature data, the hand-held device **14** displays the captured signature on the screen of the hand-held device **14**. The hand-held device **14** captures the signature data as a set of coordinates which can be converted to any popular graphics format for storage and wireless transfer to the server **12**. In one illustrated embodiment, the signature data is captured in scalable vector graphics format.

The BlackBerry 7520 hand-held electronic device **14** disclosed in one embodiment of the device **10** includes a color screen, a user input device, illustratively an integrated keyboard, and includes global positioning satellite functionality and is Bluetooth enabled. The communication service to the disclosed hand-held device **14** is covered under a cell phone plan although it is within the scope of the disclosure for the communication service to be provided under some other plan. The disclosed PDA **14** is configured to integrate with common business software products such as software products provided by Microsoft, Inc. including but not limited to Outlook and Office Products such as Word, Excel, Access and Outlook. It is within the scope of the disclosure for the PDA **14** to integrate with other or additional business software products. The BlackBerry 7520 Wireless Hand-held provides Email, phone, browser, SMS, a GPS transceiver **20** and organizer applications in a single, integrated hand-held. The 7520 series is fully compatible with all version of the BlackBerry Enterprise Server as well as many POP3, IMAP and Webmail accounts.

As shown, for example, in FIGS. **4** and **5**, the eCentral computer **32** includes a presentation layer, a database layer, an application bus rules layer and a messaging layer. The eCentral computer **32** communicates through MQ protocol with the Dexterra application server **38**. The Dexterra application server **38** includes a control consol layer, a metadata layer, a service manager layer and a messenger layer. The Dexterra application server **38** communicates with the wireless network server **42**. The wireless server network **42** includes a security layer, a messaging layer, a local database, and transmission services. The wireless network server **42** communicates through SSL protocol with the BlackBerry hand-held device **14**. The hand-held device **14** includes a presentation layer through the screen, a data persistency layer, a provisioning layer, a local database, and a network CDMA.

As shown, for example, in FIG. **6**, the server **12** includes applications for services **610**, transmission and reception services **612**, MQ messaging **614**, security **616**, a server manager **618**, work flow manager **620**, and an integration server **622**. The hand-held electronic device **14** includes a MIDP profile UI tier presentation **624**, an MIDP (ex-BB) TCP IP HTTP UDP networking interface persistence **626**, a profile tier **628**, a profile application tier **630**, a configuration messaging security tier **632** and JBM services metadata configuration tier **634** and a CLDC connect device configuration tier **636**. The application layer **610** of the server **12** communicates with the MIDP profile UI tier presentation layer **624** of the hand-held device **14**. The transmit and receive services layer of the server communicates with the MIDP (ex-BB) TCP IP HTTP UDP networking interface persistence layer **626** of the hand-held device **14**. The MQ messaging layer **614** of server **12** also communicates with the MIDP (ex-BB) TCP/IP HTTPS UDP networking interface persistence layer **626** of the BlackBerry device. The security layer **616** of the server **12** communicates with the CLDC connect device configuration layer **636** of the BlackBerry device **14**. The work flow layer **620** of the server communicates with the CLDC connect device configuration

layer **636** of the BlackBerry device **14**. The integration server layer **622** of the server communicates with the JVM services metadata configuration layer **634** of the BlackBerry device **14**. This communication will be better understood as the description of the device **10** for providing turn-by-turn directions and GPS location shown in FIG. **7** and for providing credit card capture as shown in FIG. **8** as well as the method of providing location-based content delivery with regard to an auto glass replacement service as shown in FIG. **11** are explained.

As shown, for example, in FIG. **7**, the device **10** for providing location-based content delivery is configured to provide the dispatcher with GPS location information regarding the location of the technician who is in possession of the hand-held device **14** and to provide the technician with turn-by-turn directions. In the illustrated embodiment, the hand-held device **14** is GPS-enabled and includes the GPS transceiver device **20**. The hand-held device **14**, in one embodiment is configured to send real-time date and time stamped GPS latitudinal and longitudinal information to the web server **70** of the server **12**. In one embodiment, the hand-held device **14** is configured to send the GPS latitudinal and longitudinal information to the web server **70** once every minute. Those skilled in the art will recognize that GPS latitudinal and longitudinal information can be sent on the hand-held device **14** more or less frequently as desired by the location-based content delivery service within the scope of the disclosure. The hand-held device **14** is configured to send an HTTP message containing the GPS latitudinal and longitudinal information to both the ASP map serves application server **66** and the web server **70**. The ASP map service application server **66** is configured as a web server but can generate JAVA VM (JRE JDK) HTTP post protocol messages through a servlet API, a servlet engine and an ESRI database. Similarly, the web server **70** is configured to generate a JAVA VM (JRE-JDK) X and L content in queue messaging utilizing a servlet engine and an ESRI database. The web server **70** provides web services which include a JAVA connector, a servlet connector, a cold fusion connector, and active ex-connector and a .NET link connector. The web server **70** communicates with the application server **38**.

The ASP map service application server creates a HTTP post message which displays the address location of the technician on the S3 router. The web server **70** generates an XML call with address to the GOGEOCODE database. This XML message is sent to the eCentral service **32** and the presence of the message is detected by the MQ messaging listener **64**. The server **12** is configured to use Baan service to provide turn-by-turn directions that are displayed on the hand-held device **14**. The GPS latitudinal and longitudinal coordinates read from the hand-held device **14** enable the Baan service server to provide turn-by-turn directions to be displayed on the hand-held device **14**. The BlackBerry device is configured so that upon the technician launching the mobile resource management application, the GPS device **20** begins to track their location. GPS events are captured with a date and time stamp throughout the day. Among the occurrences when GPS data and time stamps are recorded are clock-in, clock-out, arrival at job sites and completion of a job task.

As shown, for example, in FIG. **8**, the device **10** allows a technician to capture credit card information and a customer's signature at the job site and receive credit card authorization while still at the job site. Upon completion of the work order, the technician, typically after printing out an invoice, can accept payment in several fashions including cash payments, check payments and credit card payments. When the customer elects to use a credit card to pay for the invoice, the technician swipes the credit card in the credit card slot of the processing apparatus **18**. The processing apparatus **18** communicates the information obtained from the magnetic strip on the credit card to the hand-held device **14**. The hand-held device **14** then places an HTTP call to both the transaction manager's server **74** and the web server **70**. The processing device **18** is equipped with an applet, a JAVA card framework and a JAVA card BM so that the appropriate link protocol can be established between the processing device **18**, the hand-held device **14** and the transaction manager server **74** and web server **70**. The web server **70** includes a connector for receiving wireless communication through HTTP protocol from the hand-held device **14** and a host application and JAVA servlet for receiving credit card information. The transaction manager **74** includes similar connectors, applications and servlets so that the transaction manager server **74** can communicate with the hand-held device **14**. The transaction manager server **74** queries a database to determine whether credit card payment should be authorized and then generates a credit card authorization document that is communicated in extended mark-up language ("XML") format to the web server **70** and the eCentral device **32**. The server **12** then communicates via the wireless network **22** with the hand-held device to advise the technician that credit card payment has been authorized. When a signature is required for credit card authorization, the signature capture device **16** is utilized to obtain an electronic copy of the cardholder's signature which is transferred in scalable vector graphics, bitmap or some other image file form to the hand-held device **14** which communicates with the transaction manager server **74** as described above to obtain authorization.

As shown, for example, in FIGS. **9** and **10**, the signature capture device **16** in one embodiment is a blue tooth enabled signature capture pad **76** (FIG. **9**) and in a second embodiment is a blue tooth enabled digital pen **78** (FIG. **10**). The signature capture device **16** may be utilized not only for submitting a signature for credit card authorization as described with regard to FIG. **8**, but also to obtain a customer's signature on a direct or thermal work order and/or invoice printed on the printer of the processing device **18**. The main computer **32** of the server **12** includes print files **910** which may be communicated to the hand-held device **14** each time a document is to be printed or a single time and stored on the memory of the hand-held device **14**. This print files are editable on the hand-held device **14** so that changes to the work order or invoice may be made by the technician at the job site.

Once the desired work order or invoice is ready for printing, the hand-held device **14** digitally communicates a print file to the printer **28** of the processing device **18**. The printer **28** of the processing device **18** then generates a hard copy **912** of the desired work order or invoice on digital paper. When the signature capture pad **76** is utilized to obtain the customer's signature, as shown, for example, in FIG. **9**, the hard copy work order on digital paper **912** is placed in the signature capture pad **76** so that the signature line is below the signature location on the signature capture pad **76**. The signature is then captured by the signature capture pad **76** and placed in scalable vector graphics, bitmap or other image format

When a digital pen **78** is used to capture the signature of the customer, as shown, for example, in FIG. **10**, the hard copy of the work order on digital paper is provided to the customer along with the digital pen **78**. The digital pen **78** is in communication with the hand-held device **14** and communicates scalable vector graphics, bitmap or other image file of the signature to hand-held device **14**, similar to the manner in

which the signature capture pad **76** communicates the scalable vector graphics, bitmap or other image file of the signature to a hand-held device **14**.

Regardless of which device captures the signature and transmits it to the hand-held device **14**, the hand-held device **14** wirelessly transfers the image of the printed form (e.g. a work order or invoice) along with any variable data and/or signature added to the form and captured by the signature capture device **16**. This information is wirelessly transferred to the central computer **32** of the server **12** which generates and stores a server image of the form with variable data and signature **914**.

In one embodiment of signature capture using a signature capture device **16** print-ready work orders are downloaded to each technician's hand-held device **14** on a daily basis utilizing a wireless network. That same work order file is downloaded to the main computer **32** of the server **12** and stored thereon. The technician electronically receives a work order for each job order in a generated task list for the day. The technician then prints each work order individually at each job site utilizing the processing device **18** rather than receiving a batch of pre-printed work orders every morning. Once the customer's signature is captured with the signature capture device **16**, the signature capture device **16** transmits an image of the signature to the hand-held device **14**. The hand-held device **14** transmits the image of the signature to the central computer **32** of the server. The signature is matched to the appropriate work order downloaded earlier in print file format to the central computer of the server **12**. Central computer **32** then generates an image of the form with variable data and signature **914**. This server image of the form with variable data and signature **914** is then stored on the central computer of the server **12** and is available for viewing or printing as needed. In one embodiment, the central computer **32** on the server **12** runs EXPEDATA service which facilitates the generation and modification of the print files and the server image of the form with variable data and signature **914**. Those skilled in the art will recognize that other software may be running on the central computer **32** of the server **12** to facilitate generation of print files **910** and server image files of forms with variable data and signatures **914**.

The method for providing location-based content delivery **1110** is shown, for example, in FIGS. **11** through **25**. Referring now to FIG. **11**, from a high level field process flow perspective, the method of providing location-based content delivery **1110** may be subdivided into pre-dispatch processes **1112**, core impact processes **1114** and post check-in processes **1116**. Among the pre-dispatch processes **1112** are routing **1210** which includes scheduling orders **1212** and routing scheduling orders **1214**. After the routing step **1210** is accomplished the routes are dispatched to hand-held devices **1118**. While not specifically represented by the high level process flow diagram, the pre-dispatch processes **1112** include the picking process wherein parts required for completion of the order are removed from inventory and prepared for loading onto a technician's vehicle, generating check-in/check-out ("CICO") sheets and other existing processes carried out by the location-based content delivery service provider.

After the routes are dispatched to hand-held devices **14** they may be viewed at any time during the core impacted processes **1114** and thus are routed to hand-held devices **14** prior to the log-in step **1310** or the clock-in step **1410**. When a technician starts his workday, he performs a log-in step **1310** on the hand-held device **14** to access the information wirelessly communicated to the hand-held device **14**. Following log-in **1310**, the technician may perform a clock-in step **1410**. Following clock-in step **1410**, the technician performs the checkout step **1500** which includes a check-in step **1510** and a checkout for current day process **1610**. During the checkout step **1500** parts **1120** that were picked during the pre-dispatch process **1112** in order to fulfill the work orders **1122** that were routed to the hand-held device **14** are provided to the technician. Also, the manifest and check-in/check-out sheet are provided to the technician so that he can perform the various checkout steps **1500**. Following checkout **1500**, the technician performs the complete service step **1700**.

In performing the complete service step **1700** the technician performs the technician call aheads **1710**, the complete service process **1810**, the part confirmation process **1910**, the tendering process **2010** and the end of route process **2110**. During the complete service step **1700**, electronic work orders are remotely printed, electronic signatures are captured and remote part changes are designated. Also, the technician electronically changes the status of work orders, performs electronic clocking-in and clocking-out (such as during a lunch break) and performs remote credit card transactions by capturing credit card information and obtaining authorization while still at the job site. Upon completion of service step **1700**, the check-in step **1510** is performed. During the check-in step **1510**, unused parts **1126** are returned to a warehouse or a drop box and cash and check payments are tendered in step **1128**. Following check-in, the clock-out step **2210** is performed.

Because work orders are not pre-printed and given to the technician at the time he departs the warehouse or drop box of the location-based content delivery service provider, at any time during the core impacted processes **1114** add-on or collateral work orders can be sent to the hand-held device **14**. FIG. **23** represents an add-on work order or collateral work order received before check-out while FIG. **24** represents process flow when an add-on work order or collateral work order is received after check-out. Upon completion of the core impacted processes **1114**, the post check-in processes **1116** are performed. The post check-in processes **1116** include a close of day step **1130**. Close of day step **1130** includes reconciling invoices, paper work in files and making bank deposits, among other processes.

Some of the main steps in the high level field process flow **1110** will be explained in further detail in the following descriptions regarding FIGS. **12** through **24**.

As shown, for example, in FIG. **12**, the routing step **1210** includes a scheduled work order step **1212**. In a specific embodiment of the method of providing location-based content delivery **1110**, related to the provision of auto glass replacement and repair services, the scheduled work orders step **1212** is accomplished by a sales representative or dispatcher entering information regarding the customer, the vehicle type, any insurance coverage, the auto glass that has been damaged, and the location, date and requested time when the service should be performed into a main computer such as eCentral **32** or the application server **38** of the server **12**. This dispatcher then imports the work orders from the central computer (indicated by the initials "GTS" in the drawings) to the router **34** (indicated by the symbol "S3") in the drawings) in step **1214**. The GTS then exports work orders to the router **34** and changes the status of the work order to exported in step **1216**. The router **34** then optimizes and places work orders in routed order and changes the status of the work order to optimized in step **1218**. The router **34** then exports the work orders to the technology system in step **1220**. The technology system changes the status of the work order to routed in step **1222**. The dispatcher accessing the technology system then, if necessary, prints manifest and

check-in/check-out sheets in step **1224**. Thus, work order manifests including a list of work orders are created for each technician which are stored in the technology system. Alternatively, if a router **34** is not available to automatically optimize the distribution of work orders to create routes for the various technicians, scheduled work orders may be sent to the dispatcher who can use cut and drop in the technology system to manually route work orders and then automatically create a manifest for each driver.

Following the routing step, as shown, for example, in FIG. **11**, a manifest including the list of work orders to be carried out by each technician is electronically dispatched to the appropriate technician's hand-held device **14** in step **1118**. Thus, the manifest for each technician is available for viewing on the hand-held device **14** following the log-in step **1310** and throughout the core impacted processes **1114**. Although shown in FIG. **11** as occurring only once during the day prior to clock-in, the log-in step **1310** may be performed at any time throughout the day should the technician not wish to leave the mobile resource management program running. Thus, the technician will log-in to the hand-held device and the mobile resource management software each time he gets into the vehicle during the day including prior to clock-in so that he can track the use of company property at all times. Thus, in step **1320** the technician gets into the vehicle for the work day and turns on the hand-held **14** as shown, for example, in FIG. **13**.

The technician then logs into the hand-held in step **1322**. Log-in step **1322** may include entry of a password associated with the technician. Once a technician has logged into the hand-held **14** in step **1322**, the hand-held application begins to capture GPS information in step **1324**. This GPS information is wirelessly transferred from time to time or continuously to the technology system in step **1326**. By turning off the mobile resource management software, the technician can stop the GPS data from being sent to the technology system. After logging into the hand-held device **14**, in the illustrated embodiment, the technician is offered the opportunity to agree to a usage policy in step **1328**.

If the technician elects to not agree to the usage policy, the hand-held device logs the technician out of the mobile resource management application in step **1327**. Among things that may be included in the usage policy are an agreement whereby the technician states that the hand-held device and any cellular telephone provided to the technician will not be used while driving. This may be particularly valuable when the device **10** is utilized in states in which it is illegal to operate a motor vehicle while utilizing a cellular telephone.

If the technician agrees to the usage policy then the technician may view items on the hand-held screen **14** that are in the To Do list representing the manifest generated during the routing step in step **330**. The To Do list will typically include, in the illustrated embodiment, a list of work orders to be performed in a prioritized order. With regard to each work order to be performed the technician may then utilize the menu screen on the hand-held device to obtain additional information regarding the work order including customer name, customer telephone number, the location of the vehicle and may access turn-by-turn directions to the location of the vehicle based on the current location indicated by the GPS signal sent to the technology system. After logging in the technician then arrives at the next location, which, if he has the appropriate parts on the truck, may be the location of the first installation or may be the warehouse or other drop box in step **1332**.

Upon arriving at the first location for performing work during the day the technician goes through the clock-in process. In one specific embodiment of performing the log-in step **1310**, the technician turns on the hand-held device **14** and enters a password using an alphanumeric keyboard on the hand-held device **14**. The technician clicks the mobile resource management application icon to initiate the mobile resource management software running on memory on the hand-held device **14**. When a usage agreement appears the technician clicks an "OK" icon and then is able to view their To Do list representing the work orders in the manifest assigned to them. After arriving at the first location necessary for carrying out the technician's To Do list, the clock-in step **1410** is performed. As shown, for example, in FIG. **14**, the clock-in step **1410** is performed at the beginning of the day start time in step **1420**. In one specific embodiment, the step **1420** is performed by clicking on an item in the menu displayed on the screen of the hand-held device **14** which includes an icon stating "clock-in" when the technician is currently clocked-out and includes an icon "clock-out" after the technician has clocked-in. Upon clicking the clock-in icon, the hand-held device **14** captures data regarding the time and GPS location when the clock-in icon was clicked in step **1430**. This time and GPS data is then transferred wirelessly to the technology system in step **1440**. The technology system stores this time and GPS data and utilizes it to track the technician's clock-in and clock-out times each day. In one specific embodiment, upon the technician clicking on the clock-in and/or the clock-out option, a dialogue box appears on the hand-held device **14** that asks the technician to confirm the action prior to the clock-in or clock-out event actually occurring.

Following, although typically not immediately after, the clock-in step **1410**, the check-out step **1500** is performed as shown, for example, in FIG. **15**. If the technician still has vehicle inspection forms **1520**, cash and checks **1530** or unused parts **1540** from a previous day, then they select to view the list of work orders that are being checked in step **1550**. In step **1560**, the hand-held device **14** displays a list of work orders for check-in process. As part of the check-in role in step **1570**, the money and parts are reconciled against a check-in/check-out sheet that is printed in step **1565**. If all of the items reconcile then the check-out process **1610** is performed. If the items do not reconcile then comments are added to the check-in/check-out sheet and the technology system is updated with any changes in step **1580**. While shown in FIG. **11** as occurring as a portion of the check-out process **1500**, the check-in step **1510** can be performed at different times of the day. Some technicians can complete check-in step **1510** at the end of the day before clocking out and sometimes the technician will complete check-in step **1510** the next morning after clocking in. Some technicians will also complete the check-in step **1510** remotely at drop boxes or other locations with no check-in associate present.

Check-out step **1610** occurs before the technician leaves the location to begin their work day. Parts of the check-out process **1610** can also occur during the day if the technician needs to return to pick up additional parts. Since the hand-held device **14** has downloaded the work order information in electronic form in the order in which they are supposed to be completed, the hand-held device **14** will be able to display an assumed list of parts based on the existing work orders stored on the hand-held device **14**. If a work order is removed from the hand-held device **14**, the parts associated with the removed work order are removed from the list of parts stored on the hand-held device **14**. Among the information included in the work order information, in one specific example, is the customer name, the job name, the customer phone number,

the vehicle identification information, the parts required for completion of the job, and any other information required to create an invoice.

During the check-out process **1610** if the system is available then a check-in/check-out sheet is printed in step **1620** as shown, for example, in FIG. **16**. In addition to printing the check-in and check-out sheet, if the system is then available a copy of all work orders and manifests are printed out in step **1630** in place of using the information available from the hand-held device **14**. The technician then selects a view of the list the work orders that are being checked-out in step **1640**. The hand-held device **14** displays this list of work orders during the check-out process in step **1650**. The technician acknowledges the work orders displayed on the hand-held device **14** against physical parts in step **1650**. The physical parts are compared to the printed check-in/check-out sheet in step **1660**. Comments are added to the check-in/check-out sheet and the technology system is updated with changes in step **1670**. The parts that were removed from inventory and set aside as part of the pre-dispatch process **1112** are loaded onto the truck in step **1680**. Other materials that are needed for the day are gathered in step **1690**. These may include additional supplies **1691**, warranty/cure time documents **1692**, vehicle inspection forms **1693**, and door hangers **1694** in a glass installation embodiment. Those skilled in the art will recognize that when the method **1100** is utilized to supply other outside services, the list of materials required will be different. If it is time for a weekly inspection then a manager may inspect the vehicle to verify mileage, decals and other information in step **1695**. After completion of the vehicle inspection, or if it is not time for the vehicle inspection the technician then goes to the complete service step **1700**.

One step included in the complete service step **1700** is the technician call ahead step **1710** as shown, for example, in FIG. **17**. Technicians can perform the technician call ahead step **1710** to cause a customer to confirm the appointment and the work that is to be completed. Depending on the technician's day, the technician call ahead step **1710** may be done at different times. Since it may be preferable to perform the technician call ahead step **1710** in the morning before leaving for the first work order, the technician call ahead step **1710** is described herein as a first step towards completed service step **1700**.

The prior descriptions have referred to and the following descriptions will refer to certain items that are displayed on the display screen of the hand-held device **14**. The display on the display screen of the hand-held device **14** is generated by the mobile resource management software running on the hand-held device **14**. A portion of the mobile resource management software running on the hand-held device **14** will cause a mobile resource management icon to be displayed upon the display screen of the hand-held device **14** in additional to other icons that are normally displayed upon the initial screen of the hand-held device **14**. Upon activating the MRM icon, the display screen of the hand-held device **14** prompts the technician to log-in. Upon logging-in, the MRM causes the hand-held device **14** to display a screen including a To Do list and the current clock-in/clock-out status of the technician. The clock-in/clock-out status of the technician can be changed by clicking on the icon indicating their current status. Upon clicking upon the icon, a confirmation dialogue box is displayed requesting confirmation that the technician desires to change their clock-in/clock-out status. The To Do list includes a list of all jobs presented in a prioritized manner indicating a job number, the date on which the job is to be performed, the address at which the job is to be performed and the current status of the job.

By clicking on a specific job in the To Do list, a job detail screen is generated that includes a job detail tab and a parts tab. The job detail tab causes the screen to be displayed with information regarding the job to be performed. In the glass repair embodiment, the information included on the job detail screen may include the name of the customer, the job type, the address, the time and place where the vehicle will be available, the model, color and year of the car, instructions on how access to the vehicle is to be obtained, a contact phone number, a home phone number and a work phone number for the customer, the amount to be collected, the labor costs and material costs, and/or a deductible amount and a menu. The menu includes a start job icon which changes the status of the job from routed out ("RO") to in progress ("IP"), an end job icon which changes the status of the job from IP to job complete ("JC"). A directions icon which activates the turn-by-turn directions functionality of the system and generates a job directions screen, a job detail parts tab which switches to the parts tab screen, a hide menu option, a select option, and a close option. The job detail parts tab screen displays a list of part numbers, price and description for each of the parts to be utilized in completion of the job. The start job screen also allows an authorization copy to be printed out for customer signature prior to beginning the job. The end job icon also activates the processing device **18** to permit printing of receipts and processing of credit card information.

Upon logging into the MRM process the To Do list is displayed on the hand-held device **14**. The To Do list displays a high level overview of the jobs that are expected to be completed by the technician for the day. Pop up messaging will alert the technician that a new work order has been added or modified on their To Do list. When the status of the technician indicates that they are clocked-in, clicking on the To Do list allows work order details to be displayed on the hand-held device **14**.

When the technician performs the technician call ahead step **1710**, the technician utilizes a speaker phone or an ear bud connected to the hand-held device **14** to call the customer **1720**. The customer call information is available on the hand-held device **14** through the job detail screen. Thus, the utilization of the speaker phone or the ear bud allows the technician to view the hand-held screen while talking on the phone. The technician selects to view the list of work orders in step **1730** and displays the list of work orders in step **1740**. After completing one or more call aheads, the technician begins the complete services process **1810** as shown, for example, in FIG. **18**.

The technician arrives at the job in step **1812**. Prior to arriving at the job, the technician may access turn-by-turn directions utilizing the directions menu button displayed on the screen of the hand-held device **14** with regard to the work order they are performing. If the vehicle is available for the job and the correct parts are available then the technician completes the inspection form in step **1814**. If the vehicle or the parts are not available then the technician calls the dispatcher to report the job cannot be completed in step **1816** and the status of the job is changed to schedule in the technology system in step **1818**. When the vehicle is available and correct parts are available and the vehicle inspection form has been completed in step **11814**, the technician then clicks to view the screen to enter customer information in step **1820**. This causes the hand-held application to display a customer information entry screen in step **1822**. The technician then collects onsite customer information in step **1824**.

The onsite customer information collected **1824** may include viewing the vehicle identification number of the vehicle to be serviced and entering that information into the

hand-held device **14**. If there is a data connection, then the customer information entered into the hand-held device **14** is sent to the technology system of server **12** in step **1826**. Otherwise, the data is saved locally on the hand-held device **14** and automatically sent back to the technology system of server **12** when the hand-held device **14** comes into range for transmission in step **1828**. After collecting the onsite customer information the technician clicks to print a work order in step **1830**. The hand-held printer **28** of the processing device **18** then prints a work order in step **1832**. This work order is given to the customer for signature with electronic signature capture device **16** in step **1834**. If the hand-held device **14** detects that the signature capture device **16** has captured a customer's signature indicating pre-authorization in step **1836**, the hand-held device automatically changes the status of the work order to in process (IP) in step **1838**. If a data connection is available then the status of the work order is updated in the technology system in step **1840**. If a data connection is not available the data is saved locally and automatically sent back when the hand-held device **14** is in range in step **1842**.

After receiving the customer's signature on the work order, the technician completes the installation and other work indicated on the work order in step **1844**. This of course assumes that the installation can be completed. If the installation cannot be completed then the technician documents reasons for being unable to complete the work order on a vehicle inspection form in step **1846**. This information is then transmitted to the dispatcher to report that the job cannot be completed in step **1816** and the status of the work order is changed to scheduled in step **1818**. If the work order can be completed, in the glass installation embodiment, the technician utilizes the hand-held application **14** to collect the urethane lot number in step **1848**. If there is a data connection available, the urethane lot number is transmitted by the hand-held device **14** to the technology system of the server **12** in step **1850**. Otherwise, the urethane lot number information data is stored locally on the hand-held device **14** and automatically sent back to the technology system of the server **12** when a data connection becomes available in step **1852**.

In one specific embodiment, the MRM program running on the hand-held device **14** will require the technician to input the vehicle identification number at the job site on all accounts. Or otherwise validate the VIN if that information has already been completed on the work order. Such entry or validation may also be required for license plate numbers and state of license and mileage.

After completing the installation and collecting the information from the customer, the technician performs the part confirmation process step **1910**, as shown, for example, in FIG. **19**, to determine that the parts were used during the installation process. The part confirmation process **1910** is initiated by the technician clicking begin part confirmation of use on the hand-held device **14** in step **1920**. If the technician used all the parts on the work order in completing the job then they indicate that all the parts were used in step **1922**. If some parts, such as molding in a glass installation situation, were not used, then it is determined whether a flexible form of molding was used and if so, in step **1924**, the hand-held application indicates that flex molding was used. If molding was not on the work order and there is another part to check then if that part was used the hand-held application indicates that the part was used in step **1926**. If a part was not used, the part is removed from the work order in step **1928**. Anytime a part was not used if there is a data connection available, then the hand-held device **14** sends that information to the technology system of server **12** to re-part the work order based on a change in price in step **1930**. If a data connection is not available, then that data is saved locally until back in range in step **1932**. When the data connection is not available, a work order must be printed out and an explanation indicated thereon that the price is incorrect and that an updated work order will be provided to the customer in step **1934**. If a connection is available, then the technology system of server **12** wirelessly transfers an updated parts and price list for the work order to the hand-held device **14** in step **1938**.

Once all the parts are confirmed the technician reviews customer documents including such things as warranty and cure times in step **1940**. If there was a parts change, then in step **1936** the technician obtains another customer signature for the parts change. Like the first signature obtained, the signature is obtained on the signature capture device **16** and an image is sent to the hand-held device **14**. The technician then seeks to obtain payment for the services rendered.

After the part confirmation process **1910** has been performed, the technician performs the tendering process **2010**, as shown, for example, in FIG. **20**, in an attempt to obtain payment from the customer. The technician enters the option to go through the tendering process in step **2012**. It is then determined whether the customer is paying with cash. If the customer is paying with cash, then the cash amount is entered into the hand-held device **14** in step **2014**. If the cash amount matches the total amount required then the technician marks the tendering as complete in step **2016**. If the total amount of cash does not match the amount required then it is determined whether the customer will be paying by check. If so, the technician writes the driver's license number, LPN and work order number on the check in step **2018**. The technician then enters the check amount into the hand-held device **14** in step **2020** and also enters the check number into the hand-held device **14** in step **2022**. If the total amount of the check either alone and/or in combination with any cash tendered, is equal to the total amount required, then the technician marks tendering as complete in step **2016**. If the total amount of the check alone or the check in combination with any cash submitted does not match the total, then it is determined whether the customer would like to make a credit card payment in step **2024**.

If the customer selects to make a credit card payment, the processing device **18** is utilized to swipe the credit card through the credit card slot **24** in step **2026**. If there is a data connection available, the information captured by the processing device **18** and transferred to the hand-held device **14** is then wirelessly sent for authorization of the credit card payment in step **2028**. If authorization is approved then the technician marks tendering as complete in step **2016**. If no data connection is available, the credit card information is saved and sent for authorization when a connection becomes available. If authorization is not received, then in step **2030**, the processing device **18** indicates that payment is not authorized. If credit card is not authorized then the technician again asks the customer whether they wish to pay with cash and if not, whether they wish to pay with check. Upon marking the tendering process as complete **2016** the hand-held application automatically changes status of the work order to complete in step **2032**.

The technician then clicks an icon on the hand-held device **14** to print a receipt for payment in step **2034**. The receipt is then added to a packet of printed documents and handed to the customer in step **2036**. That packet of printed documents may contain additional information such as warranty information, cure time information and other details required by law. If after giving the receipt to the customer there is another work order to be completed the complete service step **1810** is then

performed with regard to the next work order on the To Do list. If there is not another work order to be completed then the end of the route process **2110** is begun.

During the end of route step **2110**, as shown, for example, in FIG. **21**, when the technician has completed all the jobs he will complete for the day in step **2112**, the technician then selects to view a list of work orders on the screen of the hand-held device **14** in step **2114**. If all the work orders are completed, the technician then clicks the icon to print out a daily work report if desired in step **2120**. If the technician had not completed all of the work orders when all the jobs are completed for the day, the technician then calls the dispatcher to reconcile incomplete work orders in step **2116**. The dispatcher changes his status in the technology system by rescheduling any incomplete work orders in step **2118**.

Once the end of the route step **2110** is completed the clock-out process **2210** is performed, as shown, for example, in FIG. **22**. The technician clicks on the current clock-in/clock-out status icon to change the status to clock-out in step **2212**. If all the work orders are complete the technician is clocked-out on the hand-held device **14** in step **2214**. If all the work orders are not complete the technician is prompted by a message on the screen of the hand-held device **14** indicating that there are incomplete work orders that must be reconciled in step **2216**. The technician then calls the dispatcher to reconcile incomplete work orders in step **2218**. The dispatcher changes the status in the technology system of the incomplete work orders by rescheduling or canceling the work order in step **2220**. As shown, for example, in FIG. **11**, prior to performing the clock-out process **2210**, the technician may return to the warehouse to complete the check-in process **1510**.

At any time during the work day add-on work orders or collateral work orders may be received by the technician, as shown, for example, in FIG. **23**. If an add-on work order is scheduled by a dispatcher and a technician has not completed the check-out step **1500**, the dispatcher routes the work order to the technology system in step **2310**. The dispatcher then sends the work order to the technician's hand-held device **14** in step **2312**. The hand-held device **14** receives the work order in step **2314**. In step **2316**, the dispatcher prints an updated check-in/check-out sheet. The updated check-in/check-out sheet is sent to the person filling the checkout role in step **2318** and to the warehouse associate in step **2320**. The warehouse associate then prints a pick ticket and picks the parts necessary for completion of the work order in step **2320**. The add-on work order scheduled for a technician that is in the building **2322** who has not completed check-out must be accepted before the technician completes check-out in order for the add-on work order process shown in FIG. **23** to be performed.

If the technician is already in the field and an add-on work order for a technician is received **2410**, the dispatcher enters instructions regarding where to pick up the parts in step **2412**. The dispatcher then routes the work order in step **2414** and sends the work order to the appropriate technician's hand-held device **14** in step **2416**. The technician's hand-held device **14** receives the new work order in step **2418**. The dispatcher prints an updated check-in/check-out sheet in step **2420** and the updated check-in/check-out sheet is sent to the person filling the checkout role in step **2422** and to the warehouse associate in step **2424**. The warehouse associate picks the parts and creates a print ticket in step **2426**. When the new work order is received by the hand-held device **14** in step **2418**, the To Do list is automatically reprioritized and a pop up message may be presented indicating to the technician that a new work order has been assigned to the To Do list. Thus, the disclosed method and device modifies the technician fulfillment process from the prior art technician fulfillment process described in the background and summary section of the application and illustrated in FIGS. **28** through **30**.

Thus, referring to FIG. **25**, the start of the day portion of the Technician fulfillment process step **2510** is performed by the technician clocking in remotely using the hand-held device **14** in step **2512**. After clocking-in remotely, the technician reviews the electronic route and work orders for the day in step **2514**. If it is determined that based on the review of the work orders that parts are needed for the first assignment, the technician arrives at the warehouse for parts in step **2516**. The technician then acknowledges all parts to be received from the warehouse in step **2520**. The warehouse creates an inventory transaction from the warehouse to the technician in step **2522**. Preferably, this inventory transaction is entered into the technology system of the server **12**. The technician then assures that the parts required are loaded onto the truck and leaves the warehouse to begin the route in step **2524**. If after reviewing the route and work orders, the technician determines that no parts are needed for the first assignment, the technician begins the route from the clock-in location in step **2518**. The technician has then completed the start of the day portion of the Technician fulfillment process step **2510** and proceeds to the complete work orders portion of the Technician fulfillment process step **2610**.

Referring to FIG. **26**, the technician initiates the complete work orders portion of the Technician fulfillment process step **2610** by completing the customer call ahead step **2611**. The technician then changes the status of the first work order to be completed, and upon which the call ahead step has been completed, to in progress ("IP") in step **2612** utilizing the hand-held device **14**. If the vehicle or the correct parts required for completion of the work order are not available, the technician remotely updates the status of the work order in step **2614** to indicate that the job is no longer in progress and cannot be completed. The update of the job status is transferred remotely from the hand-held device **14** to the technology system of the server **12** so that the dispatcher can access the technology system to re-route the work order. If additional work orders are available on the technicians To Do list, the technician loops back to the complete customer call ahead step **2611** with regard to the additional work order. If the vehicle and correct parts are available, the technician completes the installation and repair in step **2616**. After completing the installation or repair on the work order, the technician completes the end of the job administrative tasks electronically in step **2618**. Among the tasks included in step **2618** may be such things as gathering a customer signature, tendering, capturing required data to close the work order such as the Vehicle Identification number and credit card information. The technician then remotely changes the status of the work order to "job complete" in step **2620** using the hand-held device **14**. Again, if there are additional work orders, and the technician has not completed the assigned hours for the day, the technician then begins the complete work orders portion of the Technician fulfillment process step **2610** again. Also, the technician is automatically alerted when additional work orders not originally assigned to the To Do list are assigned. The route is optimized to reduce the number of required trips to the warehouse and to minimize the mileage traveled to accomplish the remaining jobs in a timely fashion. The inventory is automatically updated each time a work order is completed.

Referring to FIG. **27**, when the complete work orders portion of the Technician fulfillment process step **2610** is completed, the end of day portion of the Technician fulfillment

process step **2710** is performed. Initially the technician determines if there is a need to return to the warehouse, e.g. to return unused parts or tender payments that could not be electronically reconciled. If so, the technician returns to the warehouse in step **2712**. At the warehouse, the technician returns unused parts which are inspected by the warehouse in step **2716**. Also, assuming the technician collected cash or checks for payment for performance of the work orders, the money is collected in step **2718**. After returning parts or tendering money, the technician remotely clocks-out for the day in step **2714** utilizing the hand-held device **14** while at the warehouse. If the technician does not need to return to the warehouse, then the technician can remotely clock-out for the day in step **2714** using the hand-held device **14** at the last work order location or at some other agreed upon location.

While the present invention has been described in detail with reference to certain exemplary embodiments thereof, such are offered by way of non-limiting example of the invention, as other versions are possible. Moreover, a number of design choices exist within the scope of the present invention, some of which have been discussed above. It is anticipated that a variety of other modifications and changes will be apparent to those having ordinary skill in the art and that such modifications and changes are intended to be encompassed within the spirit and scope of the invention as defined by the following claims.

What is claimed is:

1. A system for providing location based content delivery comprising:

a server located at a first location, the server including memory and data stored in the memory relating to a plurality of requests for services at a corresponding plurality of locations other than the first location;

a hand-held electronic device located at one location of the corresponding plurality of locations configured for

communicating via a wireless network with the server to receive a first data relating to a prioritized list of the plurality of requests for services,

storing the first data in a local database resident on the hand-held electronic device,

receiving updated data regarding a provision of services for updating the data stored in the memory, and

determining if a data connection is available to the server and i) transmitting the updated data to the server substantially immediately when the updated data becomes accessible if a data connection to the server is available, or ii) storing the updated data in the local database and automatically transmitting the updated data to the server at a later time when the data connection becomes available; and

a processing device communicatively coupled to the hand-held electronic device.

2. The system of claim **1** wherein the processing device is configured to automatically capture information regarding an account represented by a credit card and transfer the account information to the hand-held electronic device.

3. A system for providing location based content delivery comprising:

a server located at a first location, the server including memory and a first data stored in the memory relating to a plurality of requests for services at a corresponding plurality of locations other than the first location;

a hand-held electronic device located at one location of the corresponding plurality of locations configured for

communicating via a wireless network with the server to receive the first data relating to a prioritized list of the plurality of requests for services,

storing such the first data in a local database resident on the hand-held electronic device,

receiving updated data regarding a provision of services for updating the data stored in the memory, and

determining if a data connection is available to the server and i) transmitting the updated data to the server substantially immediately when the updated data becomes accessible if a data connection to the server is available, or ii) storing the updated data in the local database and automatically transmitting the updated data to the server at a later time when the data connection becomes available; and

a processing device communicatively coupled to the hand-held electronic device, wherein the processing device is configured to automatically capture information regarding an account represented by a credit card and transfer that information to the hand-held electronic device, and wherein the processing device includes a credit card swipe slot configured to allow swipe of credit cards in a credit card capture device.

4. The system of claim **2** wherein the processing device is configured to print a document from a printer and wherein the document so printed includes information reflective of data stored in the memory.

5. The system of claim **1** wherein the processing device is configured to print a document from a printer and wherein the document so printed includes information reflective of data stored in the memory.

6. The system of claim **1** and further comprising a signature capture device communicatively coupled to the hand-held electronic device and configured to capture an electronic image file of a signature and to convert that electronic image file into a data format for transferring to the hand-held electronic device as part of the updated data for transmission to the server.

7. The system of claim **6** wherein the processing device is configured to automatically capture information regarding an account represented by a credit card and transfer the captured information to the hand-held electronic device.

8. The system of claim **7** wherein the hand-held electronic device is configured to i) wirelessly communicate with a credit authorization server to receive authorization information from the credit authorization server based on transfer to the credit authorization server of the captured information when such wireless communication is available, and ii) store the captured information and automatically and wirelessly communicate with the credit authorization server to receive authorization information from the credit authorization server based on transfer to the credit authorization server of the captured information at a later time when such wireless communication becomes available.

9. The system of claim **8** wherein the processing device includes a credit card swipe slot configured to allow swipe of credit cards in a credit card capture device.

10. The system of claim **6** wherein the processing device is configured to print a document from a printer and wherein the document so printed includes information reflective of data stored in the memory.

11. The system of claim **6** and further comprising a location information-receiving device configured to display location information on a screen of the hand-held electronic device.

12. The system of claim **11** wherein the location information displayed on the screen of the hand-held electronic device includes directions to the plurality of locations other than the first location of the server at which the corresponding plurality of requests for services for which data is stored in the memory are to be performed.

13. The system of claim 12 wherein the hand-held electronic device transfers location data regarding positions of the hand-held electronic device to the server.

14. The system of claim 13 wherein the location information-receiving device is implemented by the hand-held electronic device.

15. The system of claim 14 wherein the location information-receiving device includes a GPS (global positioning system) receiver configured to receive signals from a global positioning satellite system.

16. The system of claim 1 wherein the location data is transferred in an XML (extended mark-up language) format between the server and the hand-held electronic device.

17. The system of claim 6 wherein the signature capture device transfers the electronic image file to the hand-held electronic device utilizing a Bluetooth connection.

18. The system of claim 17 wherein the electronic image file is transferred in an image file format.

19. The system of claim 18 wherein the electronic image file is transferred in a scalable vector graphics format.

20. A method for providing location based content delivery at a job site other than the location of a service provider comprising:

providing a technician with a hand-held electronic device;

clocking the technician in remotely utilizing the hand-held electronic device at location of technician and a computer at the location of the service provider;

clocking the technician out remotely utilizing the hand-held electronic device at location of technician and the computer at the location of the service provider;

communicating a plurality of work orders in a prioritized manner to the hand-held electronic device;

storing the plurality of work orders on a local database on the hand-held electronic device;

viewing the plurality of work orders on the hand-held electronic;

capturing customer information electronically on the hand-held electronic device at a job site; and

determining if a data connection is available to the server and i) transmitting the captured information to a server accessible at the location of the service provider substantially immediately when the captured information becomes accessible if a data connection to the server is available, or ii) storing the captured information in the local database and automatically transmitting the updated data to the server at a later time when the data connection becomes available.

21. The method of claim 20 and further comprising capturing a customer's signature electronically at the job site.

22. The method of claim 21 and further comprising transferring the captured customer's signature to the server as part of the captured information.

23. The method of claim 20 and further comprising remotely updating the status of the plurality of work orders on a server accessible at the location of the service provider wirelessly through the hand-held electronic device.

24. The method of claim 22 and further comprising reprioritizing work orders in real-time by sending the reprioritized work orders to the hand-held electronic device.

25. The method of claim 20 and further comprising obtaining real-time location information of the technician and communicating the real-time location information to the server at the location of the service provider by a location technology device coupled to the hand-held electronic device.

26. The method of claim 20 and further comprising printing documentation regarding work orders at the job site for delivery to customers.

27. The method of claim 20 and further comprising recording payments on the hand-held electronic device at the job site.

28. The method of claim 27 and further comprising wirelessly communicating to transfer information regarding the recorded payments to a server accessible at the location of the service provider when such wireless commotion is available.

29. The method of claim 20 and further comprising invoicing customers utilizing the hand-held electronic device at the job site.

30. The method of claim 24 wherein the work orders are transferred in real-time electronically using CDMA (code division multiple access) or CDPD (cellular digital packet data) or GSM (global system for mobile communications) to the hand-held electronic device prioritized by need for completion.

31. The method of claim 26 wherein the documentation is printed using the hand-held electronic device and a printer wirelessly communicating with the hand-held electronic device.

32. The method of claim 20 and further comprising capturing VIN (Vehicle Identification Number), mileage and license plate information for a customer's vehicle utilizing the hand-held electronic device as part of the captured information.

33. The method of claim 32 and further comprising transmitting wirelessly the captured VIN, mileage and license plate to the server accessible at the location of the service provider without intervention utilizing the hand-held electronic device.

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